

Family Economics Writ Large[†]

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Powerful currents have reshaped the structure of families over the last century. There has been (1) a dramatic drop in fertility and greater parental investment in children; (2) a rise in married female labor-force participation; (3) a significant decline in marriage; (4) a higher degree of positive assortative mating; (5) more children living with a single mother; and (6) shifts in social norms governing premarital sex and married women's roles in the workplace. Macroeconomic models explaining these aggregate trends are surveyed. The relentless flow of technological progress and its role in shaping family life are stressed. (JEL D13, J12, J13, J16, J22, O33, Z13)

1. Introduction

While the economic approach to behavior builds on a theory of individual choice, it is not mainly concerned with individuals. It uses theory at the micro level as a powerful tool to derive implications at the group or macro level. Rational individual choice is combined with assumptions about technologies and other determinants of opportunities,

equilibrium in market and nonmarket situations, and laws, norms, and traditions to obtain results concerning the behavior of groups. It is mainly because the theory derives implications at the macro level that it is of interest to policymakers and those studying differences among countries and cultures.

—Gary S. Becker (1993, p. 402)

One of the functions of theoretical economics is to provide fully articulated, artificial economic systems that can serve as laboratories in which policies that would be prohibitively expensive to experiment with in actual economies can be tested out at much lower cost.

—Robert E. Lucas, Jr. (1980, p. 696)

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Think about the important choices that people make in life. A far-reaching decision concerns whom to marry. Close on the list is how many children to have, and how to raise them. Women, in particular, may be concerned about trading off their time

between working, spending it with their children and husbands, and leisure for themselves. What one chooses may be influenced by social norms. Such norms have prescribed who can work and what is considered illicit behavior by young children and adults. Gary S. Becker taught us that studying this is within the purview of economics.

Aggregating the behavior of individuals to analyze the economy at large is the subject of macroeconomics. Traditionally, macroeconomists have been interested in explaining time trends. Some of the strongest trends concern shifts in the family. Over the course of the past century in the United States, there was a large decline in the prevalence of marriage, a dramatic drop in fertility, an upsurge in educational attainment, and a huge rise in married female labor-force participation. Social norms about married female labor-force participation shifted and attitudes toward premarital sex changed too. These shifts interest policy makers. Many questions arise. Should the tax system be designed to favor, or at least not penalize, marriage and/or married female labor-force participation? Should child care be subsidized? What policies would be beneficial to children growing up with single or divorced mothers? Should policies be designed to encourage (discourage) child bearing in countries with low (high) rates of fertility?

To address these questions, one needs economic models, which can serve as Lucasian laboratories to conduct policy experiments. Some of the macroeconomics models used in family economics are the subject of this review. The review starts in section 2 with a simple model of married female labor-force participation, which is used to analyze the rise in married female labor supply. The model is also used in section 2.2 to study the timing of births along the life cycle and to investigate the taxation of household income at either the level of an individual or a couple (separate or joint taxation). The discussion then

moves onto developing a model of marriage and divorce in section 3. The framework is employed to examine the dramatic fall in marriage and the rise in divorce. It is also used in section 3.2 to explain the recent uptick in positive assortative mating, or the trend of people to marry someone from the same socioeconomic class. The drop in marriage has been associated with a swelling of the fraction of children living with a single mother. The plight of such children is the subject of section 4. Shifts in social norms are considered in section 5; in particular, the shift in norms about married women working and premarital sex. These norms are endogenous in the models presented. Next, models of fertility are reviewed in section 6. The review focuses on the decline in fertility in developed countries and the rise in expenditure by parents on the development of their children. This is Becker's famous quality-quantity tradeoff. Attention is also paid to the enigma of the baby boom and the plunge in fertility during wars. The discussion is restricted to addressing facts and issues in developed countries.

Finally, before using an economic model for policy analysis, it must be quantified. Section 7 shows how a macroeconomic model of the family can be matched with stylized facts from the data, using the baby boom as an illustration. These stylized facts are generally moments—means, standard deviations, and correlations—taken from cross-sectional and time-series data. They also may include facts from nonstructural regression analysis. The model may be matched with the set of stylized facts using a variety of estimation techniques. The example here uses a minimum distance estimation procedure, which incorporates indirect inference.

2. Married Female Labor Supply

Of all historical change in the female labor force, the increased

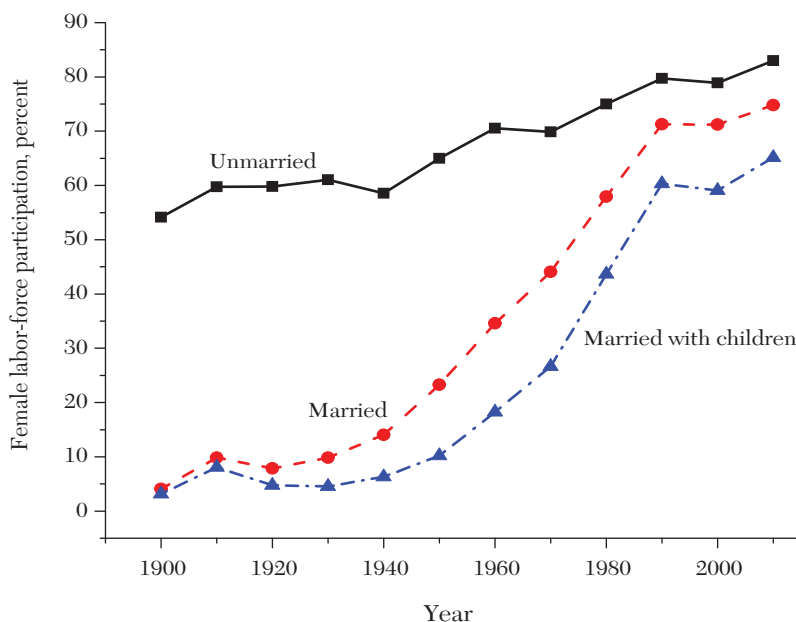


Figure 1. US Female Labor-Force Participation, 1900–2010

Notes: The graph shows the labor-force participation rates for unmarried women, married women, and married women who have at least one child under age six. The sample is restricted to twenty-five to fifty-four-year-old women.

Sources: The US Decennial Censuses, 1900–2000, and 2010 American Community Survey (ACS).

participation of married women has been the most meaningful. ... The movement of married women from the home to the marketplace has been accompanied by social and political change of enormous consequence.
—Claudia Goldin (1990, p. 119).

The entry of married women into the labor force was possibly the most significant change in US labor markets during the twentieth century. Today's households are very far from the stereotypical bread-winner husband and housekeeper wife of the past. The world is different now.

Almost no married women of working age (twenty-five to fifty-four) participated in the

labor market in 1900. Today, about 75 percent of them do (figure 1). The increase in married female labor-force participation displays an S-shaped pattern; the increase was slow initially and followed by a period of rapid growth that eventually leveled off. Large increases in female labor-force participation also occurred in other developed countries (figure 2). In contrast, the labor-force participation rates for unmarried women of the same ages increased by a much smaller amount over this period. The increase was particularly remarkable for married women with young children (under six years old). Their labor-force participation rate more than tripled between 1960 and 2010.

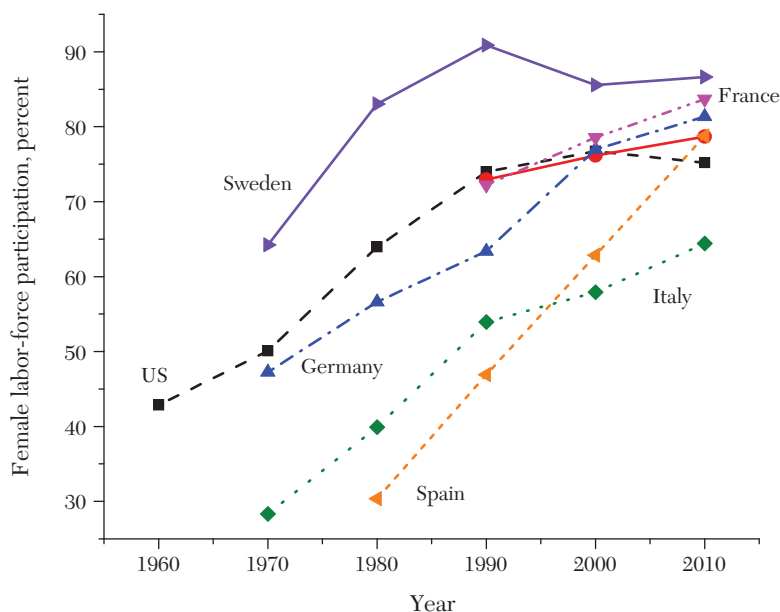


Figure 2. Female Labor-Force Participation, the United States and Western Europe, 1960–2010

Notes: The graph plots female labor-force participation for a select group of countries. The sample is restricted to twenty-five to fifty-four-year-old women.

Source: OECD Labor Force Statistics.

The increase in female employment was accompanied by a decline in the time women spent in household production. Figure 3 shows time use for men and women between 1965 and 2013. Total market work is defined as all time spent working in the market sector on main jobs, second jobs, and overtime, including any time spent working at home, plus commuting and break times. For men, total market work fell from fifty-one to thirty-eight hours per week, a drop of 29 percent. For women, however, the total increased from twenty-three to twenty-five hours per week, an 8 percent increase. Total nonmarket work sums together time spent on meal preparation and cleanup, doing laundry, ironing, dusting, vacuuming, indoor

household cleaning, indoor design and maintenance (including painting and decorating), shopping, and time spent obtaining goods and services, plus time spent on other home production such as home maintenance, outdoor cleaning, vehicle repair, gardening, and pet care. For men, total nonmarket hours increased from ten to thirteen hours, a 26 percent increase, while for women they dropped from thirty-two to twenty-one hours, a 42 percent decline. Total market and nonmarket hours declined for both men and women by about 18 percent. This decline was matched by a rise in leisure and time spent on child care for both sexes (the increase in time spent on child care by mothers is documented in section 4).

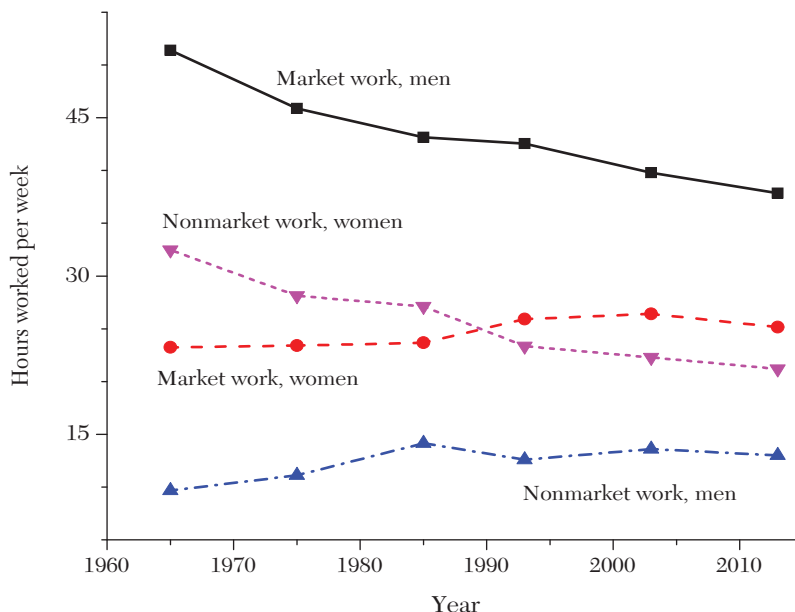


Figure 3. US Market and Nonmarket Hours, 1965–2013

Note: The graph shows total market and nonmarket hours per week for men and women.

Source: Aguiar and Hurst (2007, table II) updated with the 2004–13 American Time Use Survey (ATUS).

2.1 Household Production

A simple model of household production is now presented to understand long-term changes in married female labor supply. Household production treats the home as a small factory. The home uses inputs, intermediate goods and labor, to manufacture home goods. In order to formalize these ideas, imagine an economy populated by married couples. Each person has one unit of time, so the total time endowment of the household is two. Suppose the husband always works and spends the fixed amount I_m in the market at the wage rate w_m . The household collectively decides how much the wife should work. Let l_f denote the wife's hours in the market.

Let w_f represent her wage in the market and $\phi \equiv w_f/w_m$ be the gender wage gap.

Households care about the consumption of market goods, c , and the consumption of home goods, n . All consumption is a public good within the household. The household's preferences are given by

$$(1) \quad U(c, n) = \alpha \ln(c) + (1 - \alpha) \ln(n).$$

The household uses intermediate goods, d , and labor to produce home goods. Think about d as proxying for the wide range of products used at home, such as cell phones, frozen foods, irons, microwaves, computers, refrigerators, washing machines, vacuum cleaners, food storage containers, inter alia. Assume

that the household production function has a constant elasticity of substitution form

$$(2) \quad n = [\kappa d^\sigma + (1 - \kappa)(2 - \mathfrak{l}_m - l_f)^\sigma]^{1/\sigma},$$

where the parameter σ controls the degree of substitution between intermediate goods and labor in household production, which is given by $1/(1 - \sigma)$.

CONDITION 1: (Housework, $2 - \mathfrak{l}_m - l_f$ and intermediate goods, d , are “substitutes” in home production). $0 < \sigma < 1$.

This condition implies that the elasticity of substitution between intermediate goods and housework is high (dubbed substitutes here) in the sense that it exceeds 1 (the Cobb–Douglas case). Intermediate goods can be purchased at the time price $w_f q$; i.e., q measures the goods price in terms of the wife’s time. Hence, the budget constraint of a household is

$$c = w_m \mathfrak{l}_m + w_f l_f - w_f q d.$$

The household’s optimization problem is then given by

$$\max_{c, n, d, l_f} \left\{ \alpha \ln(w_m \mathfrak{l}_m + w_f l_f - w_f q d) \right. \\ \left. + (1 - \alpha) \ln([\kappa d^\sigma + (1 - \kappa)(2 - \mathfrak{l}_m - l_f)^\sigma]^{1/\sigma}) \right\}.$$

Compute the first-order conditions for d and l_f . They are

$$\frac{\alpha w_f q}{c} = \frac{1 - \alpha}{[\kappa d^\sigma + (1 - \kappa)(2 - \mathfrak{l}_m - l_f)^\sigma]} \\ \times \kappa d^{\sigma-1},$$

and

$$\frac{\alpha w_f}{c} \leq \frac{1 - \alpha}{[\kappa d^\sigma + (1 - \kappa)(2 - \mathfrak{l}_m - l_f)^\sigma]} \\ \times (1 - \kappa)(2 - \mathfrak{l}_m - l_f)^{\sigma-1}.$$

The first-order condition for d equates the marginal cost of buying more intermediate goods with the marginal utility from consuming the extra home goods that these intermediate goods generate. Similarly, the first-order condition for l_f equates the benefit of more market hours for the wife with the marginal utility of consuming less home goods. The second first-order condition holds with equality if $l_f > 0$. It is possible, however, that the household finds it optimal to set $l_f = 0$ so that the wife allocates all of her available time to home production. This will happen if the marginal cost of market work is greater than its marginal benefit.

Assume first that $l_f > 0$. Dividing these two first-order conditions and rearranging terms, a relation between d and l_f can be established:

$$d = \left(\frac{(1 - \kappa)q}{\kappa} \right)^{\frac{1}{\sigma-1}} (2 - \mathfrak{l}_m - l_f) \\ \equiv R(q)(2 - \mathfrak{l}_m - l_f).$$

Note that the function $R(q)$ is decreasing in q . By using this equation and the budget constraint, the first-order condition for l_f can be rewritten as

$$(3) \quad \alpha(2 - \mathfrak{l}_m - l_f)[\kappa R(q)^\sigma + (1 - \kappa)] \\ = (1 - \alpha)(1 - \kappa) \\ \times \left[\frac{1}{\phi} \mathfrak{l}_m + l_f - q R(q)(2 - \mathfrak{l}_m - l_f) \right].$$

(See Appendix A.A1 for the details.) The left-hand side of this expression is decreasing in l_f . The right-hand side is increasing in l_f . So, if an interior solution exists, then it will be unique.

It is possible that when $l_f = 0$, the left-hand side of equation (3) is less than the right-hand side. If so, the household will find it optimal to set $l_f = 0$ and the wife will not participate in the labor market.

Now imagine a situation where, due to technological progress, the price of intermediate goods used in household production declines. It is easy to show that households will move away from using labor in household production toward using intermediate goods, a consequence of condition 1. This will increase women's market hours. Likewise, it is also easy to demonstrate that a decline in the gender wage gap (a higher value of ϕ) also leads to an increase in female labor supply. If $l_f = 0$, so that the wife is not participating in the market, then a large enough decline in the price of intermediate goods or the gender wage gap will entice her to enter into the labor market.

PROPOSITION 1: (Married female labor supply). *Married female labor supply, l_f , is decreasing in the price of intermediate inputs, q , and is increasing in the gender gap, ϕ .*

See Appendix A1 for the proof.

2.1.1 U-Shaped Female Labor Supply

The framework described above suggests a monotonic relation between technological advance, which results in lower intermediate goods prices and a lower gender wage gap (a higher ϕ), and female labor supply. There is some evidence, however, that the relation between economic development and female labor supply is U-shaped. This relationship is illustrated in figure 4 for a panel of countries.

Consider a simple extension of the model presented above. Assume that households still care about market consumption, c , and home goods, n . Home goods are produced with the household production technology described earlier. The household, however, faces a fixed cost of household maintenance, denoted by $\mathfrak{c}$. Imagine that $\mathfrak{c}$ captures the bare necessities that a household needs,

such as accommodations or basic furniture. The household's problem is to maximize

$$\mathbf{U}(c, n) = \alpha \ln(c - \mathfrak{c}) + (1 - \alpha) \ln(n),$$

subject to the budget constraint

$$c - \mathfrak{c} = w_m \mathfrak{I}_m + w_f l_f - w_f qd - \mathfrak{c}$$

and the household production technology (2). Following the same steps as above, it can be shown that the first-order condition for an interior solution for l_f is given by

$$(4) \quad \alpha(2 - \mathfrak{I}_m - l_f) [\kappa R(q)^\sigma + (1 - \kappa)]$$

$$= (1 - \alpha)(1 - \kappa)$$

$$\times \left[\frac{w_m \mathfrak{I}_m - \mathfrak{c}}{w_f} + l_f - qR(q)(2 - \mathfrak{I}_m - l_f) \right].$$

If $\mathfrak{c} = 0$, then only the gender wage gap matters for married female labor supply in the sense that a proportional increase in both w_f and w_m does not affect l_f . Suppose that $w_f = aw$ and $w_m = bw$, where w is interpreted as the general level of wages, and $\phi = a/b$ is the gender wage gap. When $\mathfrak{c} = 0$, changes in w will not affect female labor supply, since the income and substitution effects from a change in the general level of wages exactly cancel out. When $\mathfrak{c} > 0$, however, this is not the case anymore. Suppose that $w_m \mathfrak{I}_m > \mathfrak{c}$, so that the husband's income covers the basic necessities of life, a reasonable assumption. Now, a general increase in wages, or a rise in w , will cause l_f to decline because the income effect from a general wage change dominates the substitution effect. A higher value of w increases the term $(w_m \mathfrak{I}_m - \mathfrak{c})/w_f = b \mathfrak{I}_m/a - \mathfrak{c}/(aw) = \mathfrak{I}_m/\phi - \mathfrak{c}/(aw)$. This causes the right-hand side of equation (4) to shift up,

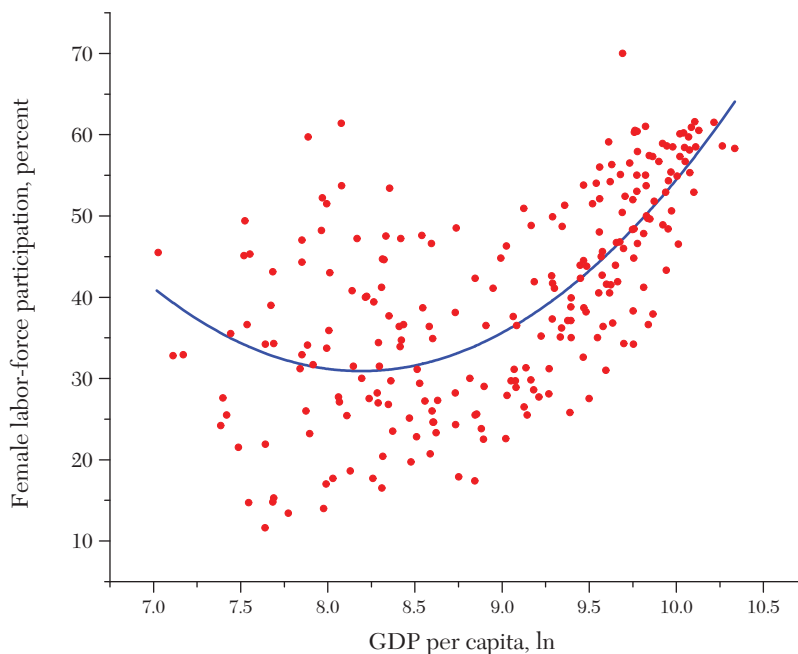


Figure 4. Economic Development and Female Labor-Force Participation for a Panel of Countries, 1890–2005

Notes: The figure plots the relation between log GDP per capita and female labor-force participation for a panel of countries. The fitted line is $FLP = 514.45 - 118.05 \times \ln \text{GDP} + 7.21 \times (\ln \text{GDP})^2$.

Source: Olivetti (2014).

which will result in a smaller value for l_f . A narrowing of the gender gap, due to a lower b or a higher a , reduces $\mathbb{I}_m/\phi - \mathbf{c}/(aw)$, so l_f will move up.

PROPOSITION 2: ($\cup$ -shaped married female labor supply). Suppose $w\mathbb{I}_m > \mathbf{c}$. Married female labor supply, l_f , is decreasing in the general level of wages, w , and the price of household inputs, q , and is increasing in the gender gap, ϕ .

When wages are low, the fixed cost of household maintenance, $\mathbf{c}$, makes the marginal utility of market goods high. As wages increase, the household is better able to cover the cost of

household maintenance and married women can work less. This effect becomes muted as $\mathbf{c}/w$ gets small. This explains the downward portion of the $\cup$. What about the upward portion? To address this, suppose that at some point in time the price of intermediate inputs declines and/or the gender gap narrows (so that ϕ rises.) These forces lead to a rise in married female labor-force participation.

2.1.2 Discussion

The economic analysis of female labor-force participation began with the pioneering works of Mincer (1962) and Cain (1966). The massive rise in female labor-force participation over the course



Figure 5. The US Gender Wage Gap, 1900–2010

Notes: The gender wage gap is defined as the ratio of female to male wages. For the 1940–2010 period, the sample is restricted to females over age eighteen.

Sources: The numbers for 1900, 1920, and 1930 are from Goldin (1990, table 3.1). For the 1940–2010 period, the US Decennial Censuses for 1940–2000 and the 2010 American Community Survey (ACS) are used.

of the twentieth century has interested labor economists. Costa (2000) and Goldin (2006) provide historical perspectives. Much attention has been devoted to examining the extent to which the rise in real wages and the narrowing of the gender gap can account for the rise in female labor-force participation. The classic reference on the gender wage gap is Goldin (1990). Goldin (1995) presents some evidence that female labor supply has a U-shaped pattern over time. She suggests that women have a comparative advantage in service sector jobs. Therefore, the structural transformation from manufacturing to services could be behind the rising part of the U-shaped pattern. Ngai and Petrongolo

(2017), Olivetti and Petrongolo (2014), and Rendall (forthcoming) present models that explicitly link the rise of the service sector to the increase in female labor-force participation and the declining gender wage gap. The declining part of the U-shaped pattern can be linked to the modernization of agriculture and the growing importance of manufacturing. When the level of agricultural technology is low, women need to work hard to cover the subsistence level of consumption, *c.* A recent discussion on the U-shaped female labor supply hypothesis is provided by Olivetti (2014).

Figure 5 shows how the gender wage gap changed during the twentieth century. In

1900, a female worker earned about 50 percent of what a man did, and by 1970 this number had risen to only 60 percent. Today, the observed gender wage gap has shrunk to about 75 percent. A caveat is in order. The gender gap may be due to many things, not just discrimination. It may reflect differences in educational attainment and occupational choice. There is also a tendency for women to experience career interruptions, to take part-time jobs, or to refuse overtime so that they can keep free time for raising children, another form of work. Also, women may prefer to earn lower wages in exchange for better benefits, such as child care, parental leave, and sick leave. Researchers have found that adjusting for such factors reduces significantly the measured gender gap. Blau (1998) and Blau and Kahn (2000, 2017) provide surveys of the literature.

Galor and Weil (1996) present an interesting general equilibrium model in which the increase in women's wages and labor-force participation is a by-product of the process of development. In their analysis, capital accumulation in the market sector raises women's wages relative to men's. The underlying mechanism is that capital in the market sector is more complementary to women's labor than it is to men's labor, since it displaces the need for physical strength. Consequently, capital accumulation will lead to greater increases in women's wages than men's wages. In a similar vein, Jones, Manuelli, and McGrattan (2015) argue that decreases in the gender wage gap can account for increases in average hours worked by married women from 1950 to 1990. The negative relation between the gender wage gap and female labor-force participation also holds in cross-country data. This is shown in the left-hand panel of figure 6 for a set of developed economies.

The economic importance of household production was probably first recognized in a classic book by Reid (1934). She carefully

reported and analyzed the uses of time and capital by households of the era. The data was fragmentary then. Reid (1934) knew in theory that labor-saving household capital could reduce the amount of time spent on housework, but the just-emerging evidence at the time suggested that this effect was modest (see table XIII, p. 91). Important research by Aguiar and Hurst (2007) puts together different time use surveys for the United States to obtain a picture of what has happened to time allocations over the last five decades. Figure 3 updates their numbers. Gimenez-Nadal and Sevilla (2012) document similar trends for other industrialized countries.

In a classic paper, Becker (1965) develops the modern approach to household production: the treatment of the household as a small factory or plant using inputs, such as labor, capital, and raw materials, to produce some sort of home goods. Benhabib, Rogerson, and Wright (1991) and Greenwood and Hercowitz (1991) introduce household production theory into dynamic general equilibrium models in order to study the movement of labor over the business cycle. The idea is that in favorable economic times, households may temporarily move labor out of the home sector to take advantage of goods market opportunities, thereby increasing the elasticity of married female labor supply. Parente, Rogerson, and Wright (2000) use a similar framework to investigate whether household production can explain cross-country income differentials. This is very much in the spirit of Reid (1934).

Greenwood, Seshadri, and Yorukoglu (2005) illustrate how labor-saving appliances and intermediate goods can encourage married female labor-force participation. They embed a Becker-Reid household production model into a dynamic general equilibrium model. They also document the decline in the prices of household durables during the twentieth century. The Second Industrial Revolution marked the beginning of the

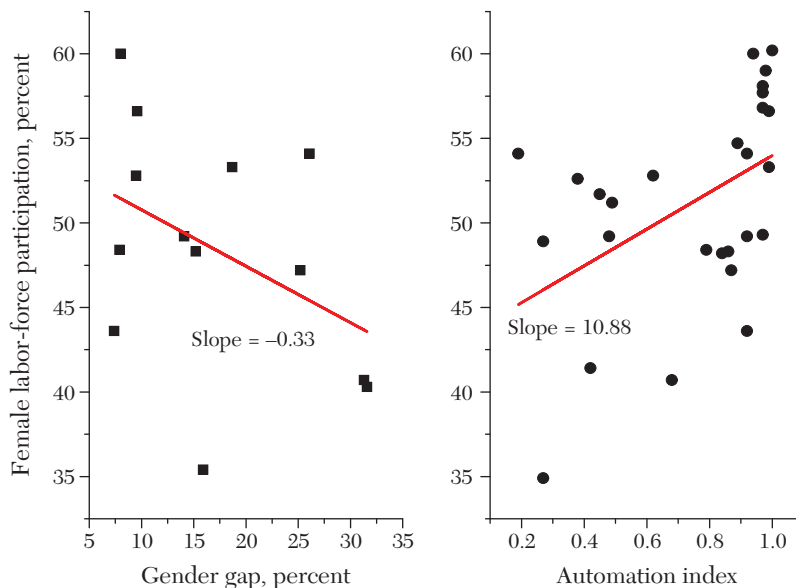


Figure 6. Gender Gap and Household Technologies versus Female Labor-Force Participation for a Cross-Section of Countries

Notes: The left-hand panel shows the cross-country relation between female labor-force participation (ages fifteen+) and the gender wage gap, the ratio of female to male wages. The right-hand panel shows the cross-country relation between the diffusion of household appliances and female labor-force participation (ages fifteen+).

Sources: The automation index is taken from Heisig (2011). The gender wage gap numbers are from Pissarides et al. (2005) and female labor-force participation is based on World Bank Development Indicators.

twentieth century. This era is most often associated with the spread of electricity, the automobile, and the petrochemical industry. But, perhaps just as important was the rise of central heating, dryers, electric irons, frozen foods, indoor plumbing, refrigerators, sewing machines, washing machines, vacuum cleaners, and other appliances now considered fixtures of everyday life. The adoption of appliances was spurred on by a rapid drop in their prices. Quality-adjusted time prices declined, for example, by about 8 percent per annum for refrigerators and 6 percent for dishwashers between the 1950s

and the 1980s. The decline for other appliances, such as microwaves, was even larger.

Empirically, female labor supply is negatively associated with the price of household appliances, as shown by Cavalcanti and Tavares (2008) in a panel of OECD countries. Heisig (2011) looks at a wider set of countries and finds, using a direct measure of appliance diffusion, that household technologies are an important force behind increasing female labor-force participation. The right-hand panel of figure 6 shows the relation between this measure of appliance diffusion (termed an automation index) and

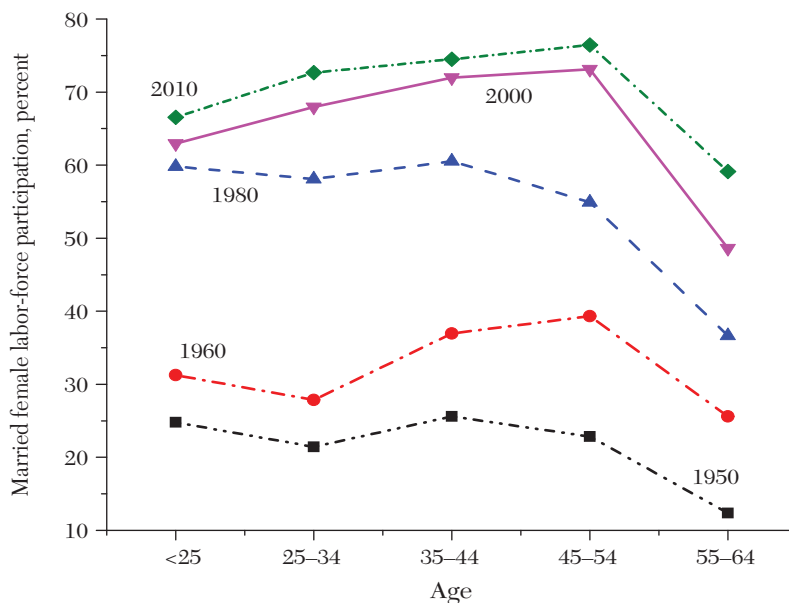


Figure 7. The US Female Labor-Force Participation along the Life Cycle, 1950–2010

Notes: The figure charts the labor-force participation rates of married women at different ages for 1950, 1960, 1980, 2000, and 2010.

Sources: The US Decennial Censuses, 1950–2000, and the 2010 American Community Survey (ACS).

female labor-force participation for a group of developed countries. Female labor-force participation is much higher in countries with widespread use of household appliances. Likewise, Coen-Pirani, Leon, and Lugauer (2010) document, using US census micro data, that a significant portion of the rise in married female labor-force participation during the 1960s can be attributed to the diffusion of household appliances. Similar evidence is provided by Dinkelman (2011), who studies the effect of rural electrification in South Africa.

2.2 Female Labor Supply over the Life Cycle

As married women entered the labor market in greater numbers, their life-cycle

pattern of labor-force participation also changed. In the 1950s and 60s, women tended to participate in the labor force before their childbearing years, reduce their labor-force participation once they had children, and then perhaps return to the labor force later (see figure 7). The dip in female labor-force participation associated with child bearing has disappeared in recent years. Today it follows the typical $\cap$ -shaped pattern over the life cycle. What factors can explain these changes? A simple life-cycle model of female labor supply along the extensive margin will be presented. To fix ideas, a static model of the extensive margin will be developed first.

The economy is populated by married couples. Each household member has one

unit of time. Suppose again that the husband always works a fixed amount of hours, $\mathbf{I}_m$, at the wage rate w_m . The household collectively decides whether the wife should work or not. If the wife enters the labor force, she has to work $\mathbf{I}_f$ hours. Hence, this decision is discrete. The household cares about the consumption of market goods, c , and each party's leisure. Consumption, including leisure, is a public good. The household's utility function is

$$\begin{aligned} \mathbf{U}(c, n, \lambda) = & \alpha \ln(c) + (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ & + (1 - \alpha) \lambda \ln(1 - l_f), \end{aligned}$$

where $1 - \mathbf{I}_m$ and $1 - l_f$ denote the leisure enjoyed by the husband and the wife, respectively. (One could also think about $1 - \mathbf{I}_m$ and $1 - l_f$ as time spent in household production.) The variable λ governs the value that a couple places on the female's time spent at home. It differs across households. Some households value the female's time at home more (a higher λ) than others. In particular, assume that λ is distributed across households in the society according to some cumulative distribution function $\Lambda(\lambda)$.

The household will compare the utility associated with each option, namely $l_f = 0$ and $l_f = \mathbf{I}_f$, and decide which path to take. Suppose first that the wife does not work. Let $\mathbf{V}^{\sim \text{WORK}}(w_f, w_m)$ be the utility for a household in this scenario. It is given by

$$\begin{aligned} \mathbf{V}^{\sim \text{WORK}}(w_f, w_m) = & \alpha \ln(w_m \mathbf{I}_m) \\ & + (1 - \alpha) \ln(1 - \mathbf{I}_m). \end{aligned}$$

If the woman does not work, then the household enjoys $c = w_m \mathbf{I}_m$ in consumption. The husband's leisure is $1 - \mathbf{I}_m$, while the wife spends all her time at home.

Similarly, let $\mathbf{V}^{\text{WORK}}(w_f, w_m, \lambda)$ be the utility that the household will realize if the woman works. It reads

$$\begin{aligned} \mathbf{V}^{\text{WORK}}(w_f, w_m, \lambda) = & \alpha \ln(w_m \mathbf{I}_m + w_f \mathbf{I}_f) \\ & + (1 - \alpha) \lambda \ln(1 - \mathbf{I}_f) \\ & + (1 - \alpha) \ln(1 - \mathbf{I}_m). \end{aligned}$$

A working wife brings more income into the household, so consumption is higher: but the household suffers a utility loss in terms of leisure since $(1 - \alpha) \lambda \ln(1 - \mathbf{I}_f) < 0$. The higher the value of λ , the greater the utility cost of the woman working. As a result, while the indirect utility function $\mathbf{V}^{\sim \text{WORK}}(w_f, w_m)$ does not depend on λ (since $(1 - \alpha) \lambda \ln(1) = 0$), $\mathbf{V}^{\text{WORK}}(w_f, w_m, \lambda)$ is decreasing in λ . These indirect utility functions are portrayed in the upper panel of figure 8. The threshold value for λ that makes a household indifferent between the woman working or not is

$$\hat{\lambda} = \alpha [\ln(\mathbf{I}_m + \phi \mathbf{I}_f) - \ln(\mathbf{I}_m)],$$

where $\alpha \equiv -[\alpha/(1 - \alpha)]/\ln(1 - \mathbf{I}_f) > 0$ and ϕ is the gender wage gap.

The woman will work in a household with a value of λ below $\hat{\lambda}$. The fraction of women working in the society is given by $\Lambda(\hat{\lambda})$, as illustrated in the lower panel of figure 8. It is immediate that the threshold, $\hat{\lambda}$, and married female labor-force participation, $\Lambda(\hat{\lambda})$, are both increasing in w_f and decreasing in w_m , while an equal percentage change in both has no effect. Indeed, any factor that increases the benefits of joint work, $\mathbf{V}^{\text{WORK}}$ relative to $\mathbf{V}^{\sim \text{WORK}}$, as shown by an upward shift in $\mathbf{V}^{\text{WORK}}$ in figure 8, will increase $\hat{\lambda}$ and the female labor-force participation rate. Exactly how much depends on the shape of $\Lambda(\lambda)$.

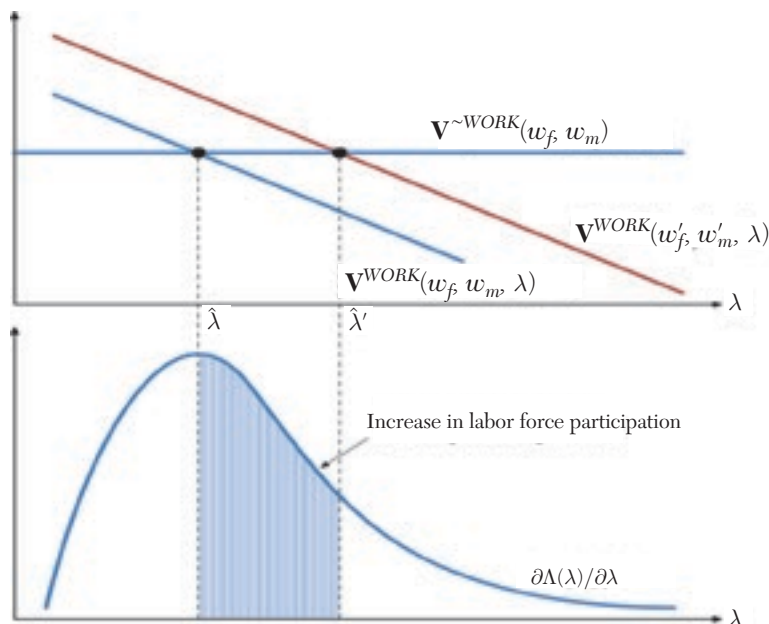


Figure 8. The Determination of Married Female Labor-Force Participation along the Extensive Margin

Notes: The upper panel illustrates $V^{\sim WORK}(w_f, w_m)$ and $V^{WORK}(w'_f, w'_m, \lambda)$ and the determination of the threshold, $\hat{\lambda}$. The lower panel illustrates how the threshold, $\hat{\lambda}$, determines the fraction of married women that work. It is assumed that $w'_f > w_f$ and/or $w'_m < w_m$.

Now, suppose each household lives for two periods. Assume that in the second period both the man and woman work. In the first period, each household has to decide whether or not the woman should work. Introduce two new elements into this environment. First, each household has a child attached to them in the first period. Children are costly, both in terms of time and money. All women need to spend a fixed amount of their *own* time, I_c , for child care. This time cost comes out of the mother's leisure and captures both the cost of childbearing and child care. Women cannot work for some period of time after birthing a child. Pregnancies can also result in temporary or permanent debilities, much more so when medical technology is

not advanced. Child care also involves a fixed time cost for mothers, especially if they are breastfeeding. Additionally, if the household decides that the woman will work, then the family also has to buy child care at the price p_c per unit of time the mother works. This cost simply reflects the fact that someone has to take care of the child when both parents are working. Assume that a mother who stays at home can take care of her child while enjoying her leisure.

Second, there are returns to experience for women. If a woman decides to work in the first period, then her second-period wage is greater than her first-period wage by a factor of χ , so that second-period wage is $w_f(1 + \chi)$. If she decides not to work, then

her second-period wage is simply w_f . There are no returns to experience for a man, so his wage is w_m in both periods. Finally, households discount the future at rate $\beta \in (0, 1)$.

Take a household in which the wife does not work. The household's utility will be

$$\begin{aligned} \mathbf{V}^{\sim WORK}(w_f, w_m, \lambda) &= \alpha \ln(w_m \mathbf{I}_m) + (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &+ (1 - \alpha) \lambda \ln(1 - \mathbf{I}_c) \\ &+ \beta [\alpha \ln(w_m \mathbf{I}_m + w_f \mathbf{I}_f) \\ &+ (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &+ (1 - \alpha) \lambda \ln(1 - \mathbf{I}_f)]. \end{aligned}$$

The woman has to incur the time cost $\mathbf{I}_c$ for childbearing and child care, independent of whether she works or not. Consider now the case where she works. In this case, the household will enjoy

$$\begin{aligned} \mathbf{V}^{WORK}(w_f, w_m, \lambda) &= \alpha \ln(w_m \mathbf{I}_m + w_f \mathbf{I}_f - \mathbf{I}_f p_c) + (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &+ (1 - \alpha) \lambda \ln(1 - \mathbf{I}_c - \mathbf{I}_f) \\ &+ \beta [\alpha \ln(w_m \mathbf{I}_m + w_f(1 + \chi) \mathbf{I}_f) \\ &+ (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &+ (1 - \alpha) \lambda \ln(1 - \mathbf{I}_f)]. \end{aligned}$$

On the one hand, since the wife is working, the household has to pay $\mathbf{I}_f p_c$ for child care. On the other hand, a two-earner household enjoys a higher income in the first period. Furthermore, by working in the first period, the wife earns a higher wage next period.

Again, there will be a threshold value for λ that separates households with working and nonworking wives (in the first period). Note that now both $\mathbf{V}^{\sim WORK}(w_f, w_m, \lambda)$ and $\mathbf{V}^{WORK}(w_f, w_m, \lambda)$ are decreasing functions of λ . The indirect utility function $\mathbf{V}^{WORK}(w_f, w_m, \lambda)$, however, has a steeper slope than $\mathbf{V}^{\sim WORK}(w_f, w_m, \lambda)$, since $\ln(1 - \mathbf{I}_c - \mathbf{I}_f) < \ln(1 - \mathbf{I}_c)$. As a result, a unique solution for the threshold value of λ still exists. The threshold value for λ is given by

$$\begin{aligned} \hat{\lambda} &= \mathfrak{b} [\ln(\mathbf{I}_m w_m + \mathbf{I}_f(w_f - p_c)) - \ln(\mathbf{I}_m w_m)] \\ &+ \mathfrak{b} \beta [\ln(w_m \mathbf{I}_m + w_f(1 + \chi) \mathbf{I}_f) \\ &- \ln(w_m \mathbf{I}_m + w_f \mathbf{I}_f)], \end{aligned}$$

where $\mathfrak{b} \equiv [\alpha / (1 - \alpha)] / [\ln(1 - \mathbf{I}_c) - \ln(1 - \mathbf{I}_c - \mathbf{I}_f)] > 0$. Note that $\partial \mathfrak{b} / \partial \mathbf{I}_c < 0$.

PROPOSITION 3: (Married female labor-force participation). *Married female labor-force participation, $\mu = \Lambda(\hat{\lambda})$, is (1) increasing in the return to labor-force experience, χ , and (2) is decreasing in the personal time required by a mother for child care, $\mathbf{I}_c$, and the price of child care provided by others, p_c .*

In an economy in which children are costly and the returns to work are low, women will tend to stop working during their childbearing years. They will reenter the labor market once their children are older. This pattern will start to change as the cost of children declines or the returns to experience increase. First, it will be easier for women to combine work and children. Second, it will be too costly for women to stay out of the labor market. Both forces will operate to entice women to work more during their childbearing years.

2.2.1 Timing of Births

The same forces that affect married women's incentives to participate in the labor market will also shape their incentives about whether or not to have children and when to have them. Consider the latter problem here. Let married couples live for two periods. Assume that both the husband and wife work for both periods, so there is no labor-force participation decision for the wife. Each household can have one child. They decide whether to have this child in the first or (if possible) second period.

Childhood lasts for one period. Children give the household a utility level ξ . Some households value children more than others. Let ξ be distributed among households according to the cumulative distribution function $\Xi(\xi)$. Suppose further that fecundity declines with the age of the household. All households who want to have a child in the first period can have one with certainty. If the household delays fertility until the second period, then they can have a child with probability π . Beyond biological factors, the probability π is affected by the state of medical technology. Hence, parents have an incentive due to declining fecundity to have their children sooner, rather than later. Having a child early, however, is costly for mothers. If a household has a child in the first period, the wage of the mother is w_f in both periods. If the household decides to wait, the second-period wage of the mother is $(1 + \chi)w_f$. Here, again, χ captures the returns to experience. The basic idea is that mothers, even if they participate in the labor market, might have to take leave from work or reduce effort at their jobs. This hurts the growth in their wages. The lower wage growth of mothers can also be due to discrimination. Set all other costs associated to childbearing to 0; i.e., $p_c = \mathbf{I}_c = 0$. Then the lifetime utility

associated with early childbearing is given by

$$\begin{aligned} \mathbf{V}^{\text{EARLY}}(w_f, w_m, \xi) &= \alpha \ln(w_m \mathbf{I}_m + w_f \mathbf{I}_f) + (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &\quad + (1 - \alpha) \ln(1 - \mathbf{I}_f) + \xi \\ &\quad + \beta [\alpha \ln(w_m \mathbf{I}_m + w_f \mathbf{I}_f) \\ &\quad \quad + (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &\quad \quad + (1 - \alpha) \ln(1 - \mathbf{I}_f)], \end{aligned}$$

while if the household decides to have their children in the second period, they enjoy

$$\begin{aligned} \mathbf{V}^{\text{LATE}}(w_f, w_m, \xi) &= \alpha \ln(w_m \mathbf{I}_m + w_f \mathbf{I}_f) + (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &\quad + (1 - \alpha) \ln(1 - \mathbf{I}_f) \\ &\quad + \beta [\alpha \ln(w_m \mathbf{I}_m + w_f(1 + \chi)\mathbf{I}_f) \\ &\quad \quad + (1 - \alpha) \ln(1 - \mathbf{I}_m) \\ &\quad \quad + (1 - \alpha) \ln(1 - \mathbf{I}_f) + \pi \xi]. \end{aligned}$$

There will be a threshold level of joy from children, $\hat{\xi}$, such that all households with $\xi > \hat{\xi}$ will have their children in the first period. Suppose, for simplicity, that $\mathbf{I}_m = \mathbf{I}_f$, so both husbands and wives work the same hours in the market. Then, the threshold level $\hat{\xi}$ is given by

$$\hat{\xi} = \frac{\beta \alpha [\ln(1 + (1 + \chi)\phi) - \ln(1 + \phi)]}{1 - \beta \pi},$$

where again $\phi = w_f/w_m$ represents the gender wage gap. This threshold is increasing in both π and χ . It is immediate that if $\chi = 0$,

then all children will be born in the first period. If $\chi > 0$, however, a positive fraction of children will be born in the second period. The number of children arising from period-one and period-two births are given by $1 - \Xi(\hat{\xi})$ and $\pi\Xi(\hat{\xi})$, with the total number of children being $1 - \Xi(\hat{\xi}) + \pi\Xi(\hat{\xi})$.

PROPOSITION 4: (Timing of births). *The ratio of late to early births, $\pi\Xi(\hat{\xi})/[1 - \Xi(\hat{\xi})]$, is increasing in fecundity, π , and the returns to experience, χ . The total number of children born, $1 - (1 - \pi)\Xi(\hat{\xi})$, is decreasing in the returns to experience, χ .*

2.2.2 Taxes

The analysis so far has abstracted from taxes. Taxes on the extra income that a wife generates, however, can be an important deterrent to female labor-force participation. To illustrate this, consider again the static, one-period model of female labor-force participation with the following twist. Suppose the husband faces a proportional income tax rate of τ_1 , while the wife's income tax rate is τ_2 . If $\tau_1 = \tau_2$, then both the primary earner (the husband) and the secondary earner (the wife) face the same tax rate. Alternatively, if $\tau_2 > \tau_1$, then the secondary earner is taxed at a higher rate than the primary earner. This would be the case, for example, in the United States, where husbands and wives are taxed jointly. The unit of taxation is the household, and tax liabilities are determined by total household income. As a result, the extra income that the secondary earner brings home is taxed at a higher marginal tax rate than the primary earner. Since taxes are progressive, the secondary earner faces a higher tax rate than she would in a world with separate taxation.

The utilities connected with one and two-earner households are given by

$$\alpha \ln(w_m \mathbf{l}_m(1 - \tau_1)) + (1 - \alpha) \ln(1 - \mathbf{l}_m)$$

and

$$\begin{aligned} & \alpha \ln(w_m \mathbf{l}_m(1 - \tau_1) + w_f \mathbf{l}_f(1 - \tau_2)) \\ & + (1 - \alpha) \lambda \ln(1 - \mathbf{l}_f) + (1 - \alpha) \ln(1 - \mathbf{l}_m). \end{aligned}$$

The threshold value for λ that determines a married female's labor-force participation is now

$$\begin{aligned} \hat{\lambda}(\tau_1, \tau_2) = & \mathbf{a}[\ln(\mathbf{l}_m(1 - \tau_1) + \phi \mathbf{l}_f(1 - \tau_2)) \\ & - \ln(\mathbf{l}_m(1 - \tau_1))], \end{aligned}$$

where again $\mathbf{a} \equiv -[\alpha/(1 - \alpha)]/\ln(1 - \mathbf{l}_f) > 0$. It is trivial to see that $\partial \hat{\lambda}(\tau_1, \tau_2)/\partial \tau_2 < 0$. Thus, higher taxes on the secondary earner lower married female labor-force participation. In contrast, $\partial \hat{\lambda}(\tau_1, \tau_2)/\partial \tau_1 > 0$, so higher taxes on the primary earner will encourage women to work due to the negative income effect.

Like taxes, the transfers that households receive can also affect married female labor-force participation. Many governments, for example, subsidize households' child-care expenditures. Recall that in section 2.2, households with a working mother incurred child-care expenses in the amount p_c , which was subtracted from the household's income. With a subsidy, p_c would be replaced with $(1 - s)p_c$, where s is the subsidy rate. Higher subsidies would increase the threshold level of λ and married female labor participation.

2.2.3 Household Bargaining

The above analysis assumes that all goods are public. Furthermore, husbands work a fixed amount of time, $\mathbf{l}_m$. Consider now a more general setup where the household maximizes a weighted sum of the husband's and wife's utilities. Also, let the household decide on the hours worked both by the

husband and the wife. In particular, consider the optimization problem given by

$$(5) \quad \max_{c, l_m, l_f} \mu [\alpha \ln(c) + (1 - \alpha) \ln(1 - l_m)] \\ + (1 - \mu) [\alpha \ln(c) + (1 - \alpha) \ln(1 - l_f)],$$

where

$$c = w_m l_m + w_f l_f,$$

and μ is the husband's weight in the objective function. Consumption, c , is still a public good. (Note that the weights on $\alpha \ln(c)$ sum to one.) Now, $1 - l_m$ and $1 - l_f$ represent the leisures enjoyed by the husband and the wife, respectively, and these are evaluated according to the weight placed on each party's utility function.

The first-order conditions for l_m and l_f are given by

$$\alpha \frac{w_m}{c} = (1 - \alpha) \mu \frac{1}{1 - l_m}$$

and

$$\alpha \frac{w_f}{c} = (1 - \alpha)(1 - \mu) \frac{1}{1 - l_f}.$$

These two conditions imply

$$(6) \quad \frac{1 - l_m}{1 - l_f} = \frac{w_f}{w_m} \frac{\mu}{1 - \mu}.$$

Suppose that μ is constant and the gender wage gap declines; i.e., w_f/w_m increases. Then, the relative leisure of the husband, $(1 - l_m)/(1 - l_f)$, will rise. The household, which is maximizing the sum of utilities, finds it optimal to allocate more leisure to the husband since the wife's work in the market is now more productive. As a result, the higher gender wage gap implies more market hours for the wife relative to her husband.

The wage gap between men and women has narrowed significantly in the United States during recent decades. Husbands, however, do not appear to have seen their leisure increase relative to their wives. How can this be reconciled with the above framework? One possible answer is that $\mu/(1 - \mu)$ might shift over time. In particular, if μ declines over time, then $(1 - l_m)/(1 - l_f)$ could stay the same or drop, even if the gender wage gap shrinks. How could μ change?

Imagine a world in which the husband and the wife use a cooperative bargaining solution, in particular Nash, to determine l_m and l_f with divorce (or say single life) as their threat points. Let B and G denote these threat points for the husband and the wife, respectively. Then, the Nash bargaining problem is given by

$$\max_{c, l_m, l_f} \left\{ [\alpha \ln(c) + (1 - \alpha) \ln(1 - l_m) - B] \right. \\ \left. \times [\alpha \ln(c) + (1 - \alpha) \ln(1 - l_f) - G] \right\}.$$

The Nash bargaining solution is Pareto optimal. Therefore, there exists a value of μ such that solving the Nash bargaining problem is equivalent to maximizing a weighted sum of the couples' utilities, as given by (5).

Let c^* , l_f^* , and l_m^* represent the optimal decisions associated with the Nash bargaining problem. As shown in Appendix A.2, the weight, μ , is given by

$$\mu = \frac{W - G}{(H - B) + (W - G)},$$

where $W \equiv \alpha \ln(c^*) + (1 - \alpha) \ln(1 - l_f^*)$ and $H \equiv \alpha \ln(c^*) + (1 - \alpha) \ln(1 - l_m^*)$ denote the value of being married for the wife and the husband. A higher outside option for the wife, due to, for example, a smaller gender wage gap (a bigger w_f/w_m), will imply smaller values for μ . This will, following equation (6), lead to a fall in

$(1 - l_m)/(1 - l_f)$. As a result, $(1 - l_m)/(1 - l_f)$ can remain constant even if the gender wage gap declines. More generally, different modes of interaction between husbands and wives and assumptions about the marriage market will lead to different solutions for μ . A reasonable presumption is that if a marriage is formed, the value for μ used in the Pareto problem (5) yields a level of utility for each party that (weakly) dominates what they could obtain in single life.

2.2.4 Discussion

Different features of the simple two-period model of female labor supply developed in section 2.2 have been analyzed in the literature within more realistic multi-period life-cycle models. The importance of labor market experience for female labor-force participation and wages were emphasized by, among others, Altug and Miller (1998), Eckstein and Wolpin (1989), Eckstein and Lifshitz (2011), and Gayle and Golan (2012). Miller (2011) estimates that a year of delay in motherhood is associated with a 9 percent increase in the total earnings of women between ages twenty-one and thirty-four. Olivetti (2006) documents that between the 1970s and 1990s, there was a significant increase in the returns to experience for women, the χ term above, in the United States. She also studies how changes in the returns to experience can account for the disappearance of the dent in female labor supply profiles during childbearing years.

Attanasio, Low, and Sanchez-Marcos (2008) build a model that combines several factors: the returns to experience, χ , child-care costs, p_c , and the gender wage gap, ϕ . They document that child-care costs relative to female earnings declined significantly in the data between the 1980s and 1990s. They argue that this decline, together with a rise in the returns to experience for women and a lower gender wage gap, contributed to both higher female labor-force

participation and its changing shape over the life cycle. Eckstein and Lifshitz (2011) estimate a dynamic life-cycle model of female labor supply to analyze different forces behind the rise in married female labor-force participation and employment during the last fifty years. They find that the rise in education levels accounts for 33 percent of the increase in wages and the narrowing of the gender wage gap accounts for another 20 percent, while about 40 percent remains unexplained by observed household characteristics. They attribute the unexplained portion to cohort-specific changes in preferences or the costs of child rearing and household maintenance.

Family labor supply, the joint behavior of husbands' and wives' labor supplies along both the extensive and the intensive margins, also plays a key role in determining earnings inequality across households, as well as the ability of households to smooth idiosyncratic income shocks. Married female labor supply is highly elastic. As a result, a given increase in wage inequality among married women will translate into an even higher level of earnings inequality among married households. This magnification of inequality occurs because the married women who receive a positive wage shock will work more than those who don't. The increase in household inequality will be even larger if husbands and wives face shocks that are positively correlated.

Hyslop (2001) studies the link between rising wage inequality and family earnings inequality for two-earner families in the period from 1979 to 1985. He shows that labor supply responses by women can account for 20 percent of the rise in family earnings inequality. The insurance provided by family labor supply is studied by Blundell, Pistaferri, and Saporta-Eksten (2016). They estimate a life-cycle model with two earners making consumption and labor supply decisions: most of the insurance against

permanent wage shocks is provided by family labor supply, while savings and transfers play a relatively secondary role.

Uncertainty in finding a new job or losing an existing one, in the face of labor-market frictions, constitute an important part of the shocks that households face. Guler, Guvenen, and Violante (2012) analyze the joint search problem of couples and how it differs from the single-agent search problem, which dominates the literature on search and matching in the labor market. Since families pool their resources, an unemployed married person with an employed partner can be more picky about labor market opportunities than a single person. On the other hand, a married person might decline attractive job offers that require changing locations, which might not be feasible for the other party in the household.

Child-care costs still constitute an important barrier for female labor-force participation. Table 1 documents total direct expenditure on child care as a fraction of a working mother's pretax income for those families that make child-care payments. For a college-educated mother, child-care costs for children under age five claim more than 20 percent of her income. For households with fewer resources, the picture is more bleak. A family with less than \$1,500 in monthly income spends almost half of the mother's income on child care.

The total cost of children, of course, far exceeds what households spend on child care. Table 2 shows total spending on a child as a fraction of household income by household structure, household income, and the age of the child. The expenses are calculated for the younger child in a two-child family.¹ The total expenditure on a child was

about 24 percent of household income for a two-parent family whose yearly income is less than \$61,530 in 2013. Expenses increase only slightly with the age of the child. Independent of the child's age, housing constitutes about 30 percent of these expenses (Lino 2014). As a child ages, parents spend less on child care and education (e.g., about 23 percent for children ages zero to two versus 9.4 percent for the ages between fifteen and seventeen), but more on food, transportation, clothing, and health care (e.g., about 40 percent for children between the ages zero and two, and 55 percent for children between the ages fifteen and seventeen). Single-parent households, led mostly by mothers, spend more, percentagewise, on a child. Those with a yearly income below \$61,530 spend between 30 to 36 percent of total household income on a child.

Public policies can play a role in mitigating the cost of children for families. Countries differ greatly in the amount of benefits they provide to families and the forms these benefits take, as shown in table 3. Family benefits consist of direct, often means-tested, cash transfers to families, direct financing or subsidies for child care, early education and residential facilities, and tax benefits targeted to families, such as child tax credits. Both Denmark and the United Kingdom, for example, spend about 4 percent of their GDPs on family benefits. The amounts are much smaller in Italy, Spain, and the United States. While most of these benefits are in-kind services in Denmark, they mainly consist of cash transfers in the United Kingdom. In other countries, such as Germany and the United States, tax benefits are relatively significant. The last column in

¹The amount spent on the younger child in a family depends on family size due to economies of scale. To calculate the expenses for two children, the figures in table 2 should be summed for the appropriate age categories. To

estimate the expenses for an only child, multiply the total expense for the appropriate age category by 1.25. To estimate the expenses for a family with three or more children, sum the expenses for each child, using the appropriate age category, and multiply by 0.78. For further details on the methodology, see Lino (2014).

TABLE 1
US CHILD-CARE EXPENDITURE AS A PERCENTAGE OF THE MOTHER'S INCOME,
2011

Household type	Age of child	
	< 5	5 to 14
<i>Marital status</i>		
Married	22.2	14.4
Divorced	19.9	11.6
Never married	26.1	15.3
<i>Education of mother</i>		
Less than high school	34.2	22.3
High school	29.0	18.5
Some college	23.2	15.9
College+	21.1	12.3
<i>Employment of mother</i>		
Employed, full	20.9	13.3
Employed, part	30.5	19.3
Self-employed	35.8	17.7
<i>Family income</i>		
Less than \$1,500	46.6	39.3
\$1,500–\$2,999	27.9	20.2
\$3,000–\$4,499	29.7	18.2
\$4,500 and over	20.8	12.6

Notes: The sample comprises families with at least one child under the age of fifteen and with employed mothers that make child-care payments for at least one of their children. The numbers refer to total direct child-care expenditures and include expenditures on organized care (e.g., day care centers or nurseries) and other nonrelative care (e.g., family day care or child care in the child's home). For further detail, see Laughlin (2013).

Source: US Census Bureau, *Who's Minding the Kids? Child Care Arrangements: 2011—Detailed Tables*, table 6. The calculations are based on the Survey of Income and Program Participation (SIPP), 2008 Panel.

table 3 shows how much governments spend on child care and early education for children under age five. Scandinavian countries (Finland, Norway, and Sweden), for example, devote close to 1 percent of their GDPs for child care, which is more than 25 percent of total family benefits. In contrast, public spending on child care in Germany and the United States is much smaller, less than 0.1 percent of GDP. Rogerson (2007), among others, attributes the high levels of female

labor supply in Scandinavia to the scope and magnitude of child-care subsidies there. What would be the effects of more generous, Scandinavian-style, child-care subsidies on female labor-force participation in countries such as Germany or the United States? Domeij and Klein (2013) and Bick (2016) try to answer this question for Germany, while Guner, Kaygusuz, and Ventura (2016) do the same for the United States. There are of course other policies, besides child-care

TABLE 2
TOTAL SPENDING ON A CHILD AS A FRACTION OF HOUSEHOLD INCOME IN THE
UNITED STATES, 2013

	Two-parent families household income, \$		
	< 61,530	61,530–106,540	> 106,540
Age	(39,360)	(82,790)	(186,460)
0–2	0.24	0.16	0.11
3–5	0.24	0.16	0.11
6–8	0.23	0.15	0.11
9–11	0.25	0.17	0.12
12–14	0.26	0.17	0.13
15–17	0.26	0.18	0.14

	Single-parent families household income, \$	
	< 61,530	> 61,530
Age	(27,290)	(111,660)
0–2	0.30	0.16
3–5	0.33	0.17
6–8	0.32	0.17
9–11	0.35	0.17
12–14	0.36	0.18
15–17	0.35	0.19

Notes: The calculations are based on the 2005–06 Consumer Expenditure Survey and the amounts are updated to 2013 dollars using the consumer price index. The sample consists of households with at least one child age seventeen or younger. The figures represent estimated expenses on the younger child in a two-child family. Household expenditures include housing, food, transportation, clothing, health care, child care and education, personal care, entertainment, and reading materials. The numbers in parentheses are mean household income for the indicated income class.

Source: Lino (2014, tables 1 and 7).

subsidies, that affect female labor supply. Many developed countries, for example, have mandatory parental leave policies, which specify a minimum amount of leave time that a person is entitled to in order to care for a newborn child, with a guaranteed job after the leave. Erosa, Fuster, and Restuccia (2010) study how these policies affect female labor supply and fertility decisions.

Albanesi and Olivetti (2016) document that there was a significant reduction in maternal mortality and morbidity between 1930 and 1960. These reductions lower the

fixed cost of childbearing, I_c . At the same time, infant formula became available and its price declined dramatically, about 6.6 percent per year between 1935 and 1960. Infant formula provided working mothers an alternative to breast feeding, again lowering I_c . The authors show that these forces played an important role in the rise of married females' labor-force participation.

Guner, Kaygusuz, and Ventura (2014) document how federal income tax liabilities vary with income, marital status, and the number of dependents. The analysis of taxes and

TABLE 3
PUBLIC EXPENDITURE ON FAMILY BENEFITS IN OECD COUNTRIES, PERCENT OF
GDP, 2011

Country	Cash	Serv.	Tax	Total	Child care
Denmark	1.63	2.42	0.00	4.05	0.68
France	1.24	1.69	0.00	3.64	0.58
Finland	1.58	1.65	0.00	3.22	0.84
Germany	1.21	0.97	0.87	3.05	0.06
Italy	0.74	0.76	0.52	2.01	0.18
Norway	1.29	1.79	0.11	3.20	0.88
Spain	0.54	0.84	0.13	1.51	0.56
Sweden	1.49	2.14	0.00	3.64	1.06
United Kingdom	2.60	1.37	0.29	4.26	0.41
United States	0.09	0.63	0.47	1.19	0.06

Notes: The table reports, as a fraction of GDP, public expenditure on child-related cash transfers to families, services (in-kind transfers) for families with children, financial support for families provided through the tax system, and child care. Public spending includes financial support that is exclusively for families with children. Spending recorded in other social policy areas such as health and housing also assist families with children, but not exclusively so, and is not included. The last column is total public spending on child care and early (under age five) education. Public expenditure on child care and early educational services is all public financial support (in cash, in-kind, or through the tax system) for families with children participating in formal day-care services (e.g., crèches, day-care centers, and family day care for children under three) and preschool institutions (including kindergartens and day-care centers, which usually provide an educational content as well as traditional care for children aged from three to five, inclusive).

Source: OECD Family Database; (i) PF1.1, Public Spending on Family Benefits (Chart PF1.1A); and (ii) PF3.1, Public Spending on Childcare and Early Education (Chart PF3.1.A).

married female labor-force participation follows Kaygusuz (2010) and Guner, Kaygusuz, and Ventura (2012). Kaygusuz (2010) shows that the 1981 and 1986 tax reforms in the United States, which lowered the progressivity of income taxes, account for about one-fourth of the increase in married female labor-force participation between 1980 and 1990. Guner, Kaygusuz, and Ventura (2012) study how a hypothetical tax reform, which moves the US economy from joint to separate taxation, would affect married female labor-force participation. They model female labor supply along both the extensive and the intensive margins and allow for endogenous human capital accumulation

for married women along their life cycle. As a result, a move to separate filing, which effectively lowers taxes for the secondary earner, not only leads to significant increases in married female labor supply, but also results in a smaller gender wage gap. Bick and Fuchs-Schundeln (forthcoming) analyze how much cross-country differences in taxes on the secondary earner contribute to cross-country differences in married female labor supply.

Women in all developed countries today have fewer children than they did decades ago. In 1970, the total fertility rate was 2.5 in the United States. Today it is less than 2. Parents also have children later. In 1970,

22.0 percent of all births in the United States occurred to women older than thirty. This figure almost doubled and increased to 43.3 percent by 2012, as documented by Martin et al. (2013). Caucutt, Guner, and Knowles (2002) study a model of marriage, female labor-force participation, and fertility and show that the increase in the returns to experience for women may be responsible for this shift in the timing of births. Section 2.2 captures the key forces in their analysis. When the wage penalty associated with childbearing is high, women prefer to postpone their fertility and instead first build their human capital. The parameter π in the analysis captures the fact that female fecundity declines with age. Fecundity declines more slowly for males. The implications of differential fecundity between men and women for labor and marriage markets are studied by Siow (1998). Diaz-Gimenez and Giolito (2013) show that differential fecundity is also important for accounting for the age gap between men and women at first marriage. The parameter π also reflects the state of contraceptive technology. The introduction of the pill in the 1960s, for example, allowed women to control their fertility more effectively. The implications of the pill for the labor supply and the marriage behavior of women is studied by Goldin and Katz (2002).

Career interruptions will be even longer and more costly in economies with high unemployment, since women will have a harder time finding jobs. Furthermore, unemployed mothers will have a harder time covering the cost of children. Both forces will compell women to postpone their fertility and lower the overall fertility rate. Da Rocha and Fuster (2006) document that there is a positive relation between female employment and the total fertility rate across developed countries today. The right-hand panel of figure 9 replicates their findings. They build a model of fertility and female

labor-force participation with labor-market search that generates the observed positive relation between fertility and female labor-force participation. Both in the data and their model, there is a negative relation between female unemployment and female labor-force participation. As a result, in countries such as Spain and Italy, high unemployment induces women both to stay out of the labor force and to have fewer children. The relation between female labor-force participation and fertility was not always positive, however, a fact noted by demographers—see Kohler, Billari, and Ortega (2006). In the 1970s, countries with high fertility rates were the ones in which female labor-force participation was low. This is shown in the left-hand panel of figure 9.

The analysis of household bargaining in section 2.2 follows Knowles (2013). He documents how the relative leisure of men has not increased over time. Given equation (6) and the declining gender wage gap, this suggests that some sort of bargaining model may be called for. Different models of household decision making are analyzed in Browning, Chiappori, and Weiss (2014).

The utility value of a woman staying at home, as reflected by λ , is treated as a constant in the current analysis. This does not have to be the case. Fernandez, Fogli, and Olivetti (2004) present evidence suggesting that a man is more likely to have a working wife if his own mother worked than if she didn't. In particular, men who had mothers who worked during World War II had a higher likelihood of marrying working women than those who didn't. They develop a model where attitudes toward working women become more receptive over time. Such forces can be captured by changes in λ . A man whose mother worked is modeled as having a lower λ than a man whose mother did not work. As more women start to work, say due to technological progress, changes in attitudes will have a reinforcing effect.

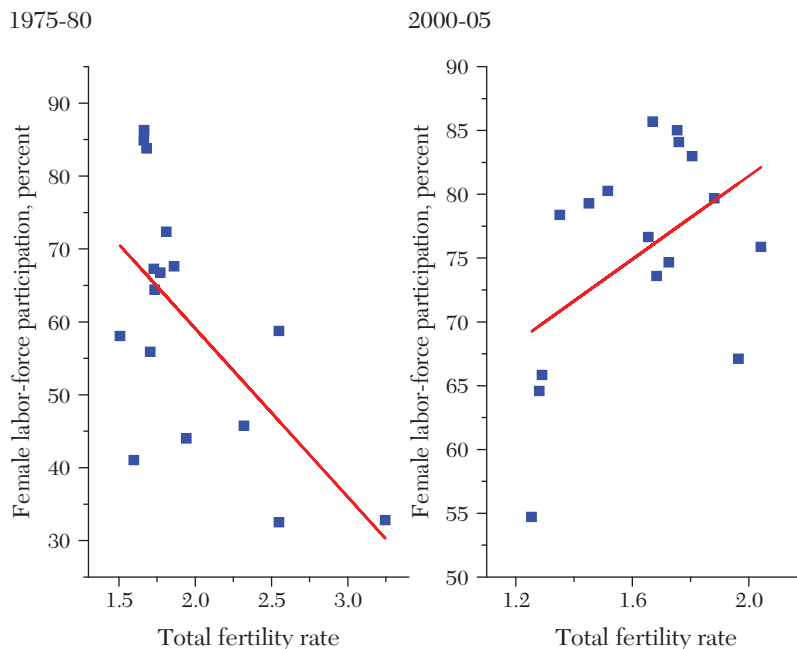


Figure 9. The Cross-Country Relation Between the Total Fertility Rate and the Female Labor-Force Participation Rate

Notes: The left-hand panel shows the relation between female labor-force participation (between ages twenty-five and fifty-four) and the total fertility rate for 1975–80. The right-hand panel shows the same relation for 2000–2005.

Sources: Female labor-force participation, OECD Labor Force Statistics. Total fertility rate, United Nations (2013).

One can also imagine that households have some priors over λ and update their beliefs. Beliefs can change as more women enter the labor force and information grows about the labor-market experience of working women. Fogli and Veldkamp (2011) and Fernandez (2013) show that such learning models can generate the S-shaped increase in married female labor supply observed in the data.

3. Marriage and Divorce

To understand the importance of economic factors in determining marriage and

divorce, consider the facts displayed in figure 10. A much smaller proportion of the adult population is married now compared with fifty years ago. In 1950, 75 percent of women were married (out of non-widows age fifteen or older). By 2016, this had declined to 56 percent. Two factors underlie these observations. First, the crude marriage rate declined. In 1950, there were eleven marriages per 1,000 people compared with just seven in 2016. Second, over the period 1950 to 2016 the crude divorce rate spiked, which led to a smaller fraction of the population being married, although the divorce rate has

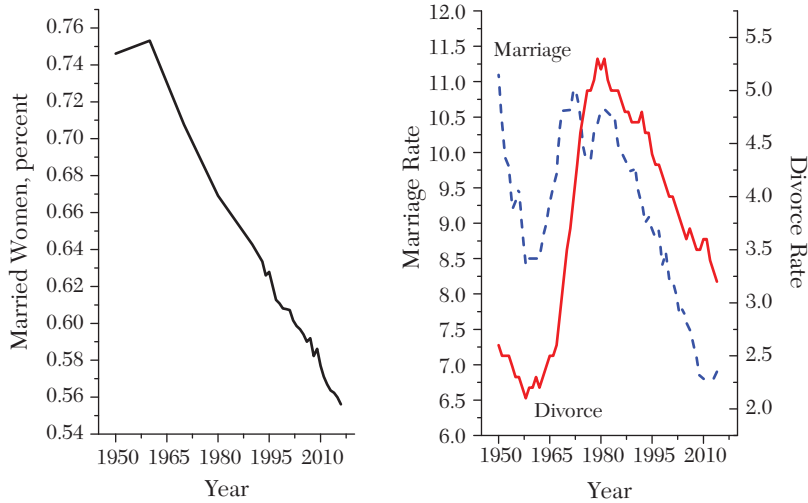


Figure 10. US Trends in Marriage and Divorce, 1950–2016

Notes: The left-hand panel shows the percentage of non-widowed women (ages fifteen and above) who were married. The right-hand panel plots the crude marriage and divorce rates (the number of marriages and divorces per 1,000 people).

Sources: The data for marriage comes from the US Census Bureau. The data on marriage and divorce rates is from the Centers for Disease Control and Prevention (CDC); viz, Monthly Vital Statistics Reports for the pre-1996 period and from the National Vital Statistics Report for the post-1996 period.

been falling more recently. Even today the crude divorce rate of 3.2 per 1,000 people exceeds the 1950 value of 2.6.

These trends had noticeable consequences for the living arrangements of US households. As seen in figure 11, the fraction of married US households has decreased continuously, while alternative living arrangements, singles with or without children, or persons living with a partner that is not a spouse, have increased substantially.

At the same time, there has been an increased tendency for people to marry within a similar socioeconomic class. To see this, consider estimating a regression (using US Census Bureau data) between a wife's educational level and her husband's. In

particular, run a regression of the following form for the years $y = 1960, 1970, 1980, 1990, 2000, 2005$:

$$\begin{aligned} Edu_{my}^w &= \alpha + \beta Edu_{my}^h \\ &+ \sum_{t \in \mathcal{T}} \gamma_t \times Edu_{my}^h \times Year_{ty} \\ &+ \sum_{t \in \mathcal{T}} \theta_t \times Year_{ty} + \varepsilon_{my}, \end{aligned}$$

with $\varepsilon_{my} \sim N(0, \sigma)$.

Here Edu_{my}^h and Edu_{my}^w represent the years of education for the husband and wife in marriage m for year y . The variable $Year_{ty}$ is a time dummy. It is set up so that $Year_{ty} = 1$,

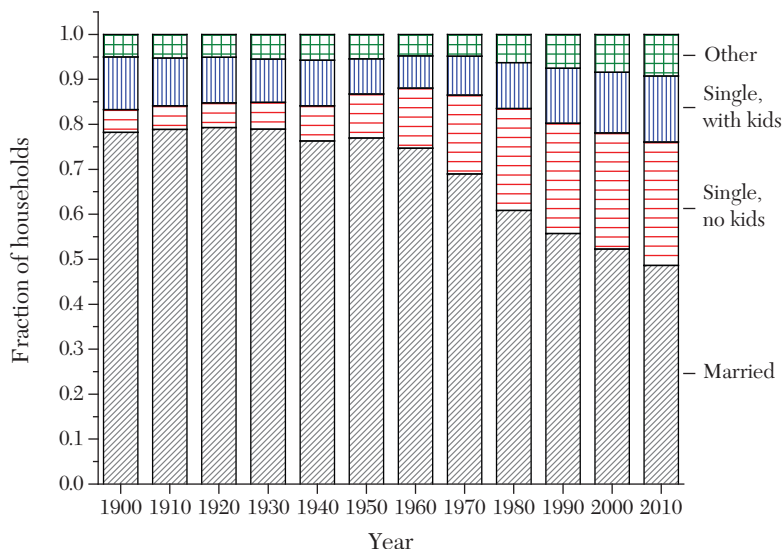


Figure 11. US Trends in Living Arrangements, 1900–2010

Notes: The graph shows fractions of different household types for each year: married couples, singles, singles with children (single mothers and single fathers), and other types of households (unrelated individuals living together).

Sources: The US Decennial Censuses, 1900–2000, and the 2010 American Community Survey (ACS).

if $t = y$, and $Year_{ty} = 0$, if $t \neq y$, where $t \in \mathcal{T} \equiv \{1970, 1980, 1990, 2000, 2005\}$. The coefficient β measures the impact of a husband's education on his wife's education for the baseline year 1960, since $Year_{ty} = 0$, for all t , when $y = 1960$. The coefficient γ_t gives the additional impact of a husband's education on his wife's relative to the baseline year, 1960. The evolution of γ_t over time reflects changes in the degree of assortative mating. The regression also includes a fixed effect for each year as measured by the constants α and θ_t . The θ_t 's control for the secular rise in the educational levels of the married population. Figure 12 plots the upshot of the regression analysis. As can be seen in the figure, γ_t rises over time, implying that the degree of assortative mating has increased.

3.1 A Model of Marriage and Divorce

What determines whether a single person will marry or not? Likewise, whether a married couple will divorce? Two motives for marriage are stressed here: (1) love and companionship and (2) economic motives. The Irish poet Samuel Lover wrote "Come live in my heart and pay no rent." It is hard for an economist to improve on what a poet can write about love and companionship. The love between two people will be modeled here in clinical fashion, as a term in tastes. Two people living together may have a higher level of material well being than if they both lived alone. This could happen because there are economies of scale both in the consumption of market goods and in the consumption/production of nonmarket

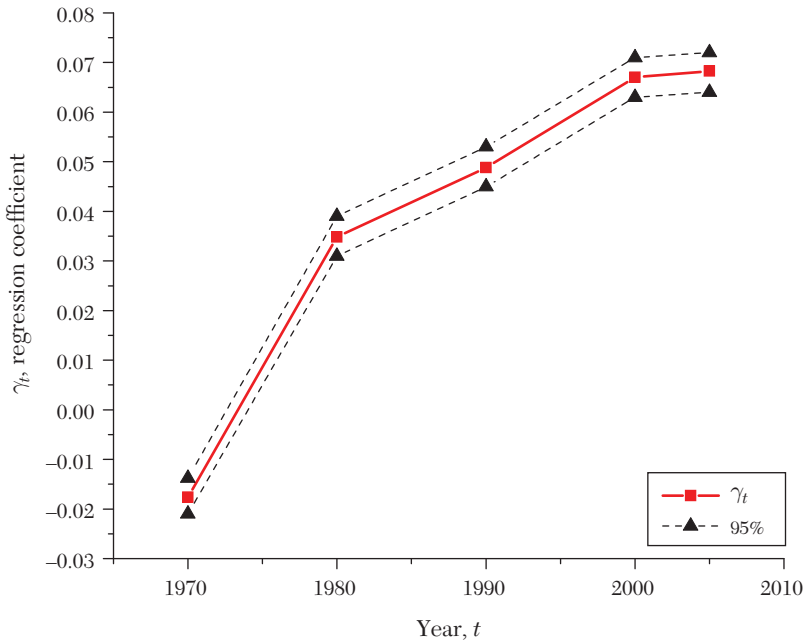


Figure 12. The Rise of Positive Assortative Mating in the United States, 1960–2005

Notes: The variable γ_t is a measure of assortative mating for the years $t = 1960, 1970, \dots, 2000, 2005$. A higher value for γ_t indicates a higher degree of positive assortative mating. See the text for a description.

Source: Computed from data that is freely available from the Integrated Public Use Microdata Series (IPUMS) website.

ones. Additionally, public policies such as family assistance or taxation may favor married or single life.

Households can be composed of $z \in \{1, 2\}$ adults. Suppose that each adult has one unit of time. This can be split between either working in the market or at home. A z -adult household will have a total of z units of time. Let the market wage for men and women be represented by w . Household production is undertaken according to the production function,

$$(7) \quad n_z = [\kappa(d_z)^\sigma + (1 - \kappa)(h_z)^\sigma]^{1/\sigma},$$

where h_z is the household's labor at home and d_z is the inputs of durable goods into home production. Suppose that d_z can be purchased at the price wq , where q again measures the goods price in terms of time. Assume that the household production function satisfies condition 1; i.e., $0 < \sigma < 1$. This implies that durables and labor are quite substitutable in household production in that the elasticity of substitution exceeds unity (the Cobb–Douglas case).

Preferences are represented by the utility function

$$(8) \quad \alpha \mathbf{U}_z\left(\frac{c_z}{z}\right) + (1 - \alpha) \mathbf{V}\left(\frac{n_z}{z}\right) + \mathbf{M}_z \tilde{b},$$

where c_z and n_z denote the household's consumptions of market and nonmarket goods and with $\mathbf{M}_z \equiv I\{z = 2\}$. The utility indexes $\mathbf{U}_z$ and $\mathbf{V}$ are

$$\mathbf{U}_z(x) = \ln(x - \mathbf{c}/z) \quad \text{and} \quad \mathbf{V}(x) = x^\zeta/\zeta.$$

With this specification, the constant $\mathbf{c}$, as in section 2, is a fixed cost associated with maintaining a household and is the source of scale economies here. The parameter ζ determines the curvature of the utility function for the home good. Finally, to incorporate love and companionship, suppose that upon meeting a couple draws a bliss variable, $\tilde{b}$, which is added to the household's utility. The variable $\tilde{b}$ can be positive or negative and measures their degree of compatibility. Assume that this is a random variable drawn from the cumulative distribution function $B(\tilde{b})$.

CONDITION 2: (Strong diminishing marginal utility for home goods). $\zeta < 0$.

The above condition implies that the utility function for nonmarket goods is more concave than the natural logarithm function ($\zeta = 0$). Thus, marginal utility diminishes more rapidly for household goods than market goods. As a consequence, as a household becomes richer it will tend to move consumption toward market goods (in a relative sense). The optimization problem of a z -adult household is

$$\begin{aligned} \mathbf{W}_z(w, q) = \max_{c_z, d_z, h_z} & \alpha \mathbf{U}_z\left(\frac{c_z}{z}\right) \\ & + (1 - \alpha) \mathbf{V}\left(\frac{n_z}{z}\right) + \mathbf{M}_z \tilde{b} \end{aligned}$$

subject to

$$c_z = w(z - h_z - qd_z)$$

and the household production function (7).

It is useful, for the sake of exposition, to define the indirect utility associated with the maximization problems of single and married households net of the effect of the bliss $\tilde{b}$. Define then $\mathbf{S}(w, q) \equiv \mathbf{W}_1(w, q)$ and $\mathbf{M}(w, q) \equiv \mathbf{W}_2(w, q) - \tilde{b}$. The indirect utility functions $\mathbf{S}$ and $\mathbf{M}$ play an important role in the analysis. By using the envelope theorem, it is easy to calculate for future use that

$$(9) \quad \frac{d\mathbf{S}(w, q)}{dq} = -\alpha w \frac{d_1}{c_1 - \mathbf{c}} < 0$$

and

$$\begin{aligned} (10) \quad \frac{d\mathbf{S}(w, q)}{dw} &= \alpha \left(\frac{1 - h_1 - qd_1}{c_1 - \mathbf{c}} \right) \\ &= \frac{\alpha}{w} \left(\frac{1}{1 - \mathbf{c}/c_1} \right) > 0. \end{aligned}$$

For a married household, the analogous analysis yields

$$(11) \quad \frac{d\mathbf{M}(w, q)}{dq} = -\alpha w \frac{d_2}{c_2 - \mathbf{c}} < 0,$$

and

$$(12) \quad \frac{d\mathbf{M}(w, q)}{dw} = \frac{\alpha}{w} \left(\frac{1}{1 - \mathbf{c}/c_2} \right) > 0.$$

The allocations of single and married households, $\{c_z, h_z, d_z\}_{z=1,2}$ relate to each other in an intuitive fashion. The proof for the lemma below is in Appendix A.3.

LEMMA 1: (Household allocations). *The allocations in married and single households have the following relationships:*

$$(i) \quad (c_2 - \mathbf{c}) > [(2 - \mathbf{c}/w)/(1 - \mathbf{c}/w)] \times (c_1 - \mathbf{c});$$

$$(ii) \ d_2 < [(2 - \mathbf{c}/w)/(1 - \mathbf{c}/w)]d_1;$$

$$(iii) \ h_2 < [(2 - \mathbf{c}/w)/(1 - \mathbf{c}/w)]h_1;$$

$$(iv) \ (c_2 - \mathbf{c})/2 > (c_1 - \mathbf{c}) \text{ and } c_2 > c_1;$$

$$(v) \ \mathbf{S}(w, q) < \mathbf{M}(w, q).$$

Now, a married household has $2 - \mathbf{c}/w$ units of disposable time, after netting out the fixed cost of household maintenance, to spend on various things. A single household has $1 - \mathbf{c}/w$ units of disposable time. So, the ratio $(2 - \mathbf{c}/w)/(1 - \mathbf{c}/w)$ reflects how much richer in time a married household is, due to the economies of scale from marriage; i.e., a married household spends less time working per person in order to cover the fixed cost of the household. Lemma 1, part (i), states that a married household will spend a larger fraction of their adjusted time endowment on the consumption of market goods than will a single household. Parts (ii) and (iii) of lemma 1 imply that married households spend less than single households on household inputs, relative to market goods. That is, $qd_2/(c_2 - \mathbf{c}) < qd_1/(c_1 - \mathbf{c})$ and $wh_2/(c_2 - \mathbf{c}) < wh_1/(c_1 - \mathbf{c})$ so that $[qd_2 + wh_2]/(c_2 - \mathbf{c}) < [qd_1 + wh_1]/(c_1 - \mathbf{c})$. When nonmarket goods exhibit strong diminishing marginal utility, bigger households will favor (relative to the consumption patterns of smaller ones) market consumption for their larger adjusted endowment of time. Part (iv) of lemma 1 states that after paying the fixed cost of household maintenance, market consumption per person is effectively higher in a married household than a single one. Also, married households spend more on market goods than do single households. Last, as a consequence of (ii), (iii), and (iv), a married household is better off than a single one, at least on economic grounds. This result is due to the economies of scale from marriage.

A prospective couple will use the following criteria to determine whether or not to marry:

$$\begin{cases} \text{Marry} & \text{if } \mathbf{M}(w, q) + \tilde{b} > \mathbf{S}(w, q), \\ \text{Single} & \text{if } \mathbf{M}(w, q) + \tilde{b} < \mathbf{S}(w, q). \end{cases}$$

The threshold level of bliss, b^* , that equates the value of married and single life is given by

$$(13) \quad b^* = \mathbf{S}(w, q) - \mathbf{M}(w, q) < 0.$$

Interestingly, some people will marry for economic reasons even though they do not love each other (because $b^* < 0$). Note that $1 - B(b^*)$ gives the fraction of the population that gets married.

3.1.1 From Economics to Romance

Is the above framework useful for explaining the decline in marriage since the 1950s? The answer is yes. To see this, break the economic development process up into two underlying forces: a rise in wages and a decline in the price for home inputs. Using (9) to (12) and (13), it is easy to see that

$$(14) \quad \frac{db^*}{dq} = -\alpha w \left[\frac{d_1}{c_1 - \mathbf{c}} - \frac{d_2}{c_2 - \mathbf{c}} \right] < 0$$

and

$$(15) \quad \frac{db^*}{dw} = \frac{\alpha}{w} \left[\frac{1}{1 - \mathbf{c}/c_1} - \frac{1}{1 - \mathbf{c}/c_2} \right] > 0,$$

where the signs of the above expressions follow from Lemma 1.

Technological advance in the form of either a *falling* price for purchased household inputs or rising real wages reduces the economic gain from marriage. This leads to an increase in the threshold value for bliss, b^* . A fall in the price of purchased household inputs leads to the substitution of purchased household inputs for labor in household production.

Single households use labor-saving products the most intensively, so they realize the greatest gain [i.e., $d_2/(c_2 - c) < d_1/(c_1 - c)$ in (14)]. The assumption of strong diminishing marginal utility for nonmarket goods ($\zeta < 0$) is important for a drop in the price of purchased household inputs to reduce the economic return to marriage. As wages increase, the fixed cost for household maintenance matters less. This fixed cost bites the most for single households [i.e., $c/c_2 < c/c_1$ in (15)]. Therefore, single households benefit the most from a rise in wages. Equation (15) shows that in the absence of a fixed cost ($c = 0$), a change in wages will have no impact on the utility differential between married and single life. This leads to the following lemma.

LEMMA 2: (The decline in marriage). *The fraction of the population that is married, $1 - B(b^*)$, is increasing in the time price of durables, q , and decreasing in wages, w .*

3.1.2 Divorce

The decision to divorce or not is analogous to the decision to get married. Imagine a couple who is married. Their household utility is $\mathbf{M}(w, q) + \tilde{b}$. Suppose that the couple gets a new draw for bliss, $\tilde{b}'$, from the distribution $B(\tilde{b}')$. Should they remain married? The answer is yes only if $\mathbf{M}(w, q) + \tilde{b}' > \mathbf{S}(w, q)$. It is easy to add a cost of divorce into the analysis. Let a divorce cost δ in units of time per person. The divorcee's budget constraint would appear as $c = w(1 - h) - wq - w\delta$. The indirect utility function for a divorcee should be rewritten as $\mathbf{S}(w, q, \delta)$, with $d\mathbf{S}(w, q, \delta)/d\delta = -\alpha w/(c - c) < 0$. Now the couple will remain married if $\mathbf{M}(w, q) + \tilde{b}' > \mathbf{S}(w, q, \delta)$. The threshold value for a divorce, $b^{*'}$, is defined by $b^{*'} = \mathbf{S}(w, q, \delta) - \mathbf{M}(w, q)$. Clearly a drop in the cost of divorce, δ , will promote divorce, since it raises the value of single life for a divorcee.

LEMMA 3: (The rise in divorce). *The rate of divorce, $B(b^{*'})$, is declining in the cost of divorce, δ , decreasing in the time price of durables, q , and increasing in wages, w .*

3.2 Assortative Mating

To have assortative mating, men and women need to differ along some dimensions. Assume that there are two types of men, those with low productivity and those with high productivity, working in the labor market. Denote a man's productivity level by μ_i for $i = 1, 2$, with $\mu_2 > \mu_1$. Let there be π_i men who have a productivity level of μ_i . Similarly, suppose that women differ in their market productivity as well. Represent a woman's productivity in the market by ϕ_j for $j = 1, 2$, with $\phi_2 > \phi_1$. Additionally, let women also differ in their productivity at home, η_h , for $h = 1, 2$, with $\eta_2 > \eta_1$. Assume that there are χ_{jh} women with the productivity combination (ϕ_j, η_h) . Thus, there are two types of men and four types of women. Each sex has one unit of time. To keep things simple, suppose that a man spends all of his time working in the market while a married woman divides her time between market work and household production. Normalize the total number of people of each sex to one; i.e., let $\pi_1 + \pi_2 = 1$ and $\chi_{11} + \dots + \chi_{22} = 1$. For simplicity, assume that $\chi_{22} < \pi_2$ and $\chi_{11} < \pi_1$. Who will marry whom in this economy? Assume that all matches are based solely on economic considerations. Clearly, not all men can marry a woman of the ideal type, (ϕ_2, η_2) , and not all women can wed a man of the best type, μ_2 . How will matching be done in this economy?

Let the household's preferences be given once again by (1). Then the maximization problem for a marriage between a type- μ_i man and a type- (ϕ_j, η_h) woman is

$$\mathbf{M}(\mu_i, \phi_j, \eta_h; w, q) = \max_{d, h} \{ \alpha \ln(c) + (1 - \alpha) \ln(n) \},$$

subject to

$$n = [\kappa d^\sigma + (1 - \kappa)(\eta_h h)^\sigma]^{1/\sigma}$$

and

$$c = \mu_i w + w \phi_j (1 - h) - w q d,$$

where q is the time price of durables (per efficiency unit of time). It is easy to calculate that

$$\frac{d\mathbf{M}}{d\mu_i} = \alpha \frac{w}{c} > 0,$$

$$(16) \quad \frac{d\mathbf{M}}{d\phi_j} = \alpha \frac{w}{c} (1 - h) > 0,$$

and

$$(17) \quad \frac{d\mathbf{M}}{d\eta_h} = (1 - \alpha) \frac{(1 - \kappa)(\eta_h)^\sigma}{n^\sigma} h^\sigma > 0.$$

An interesting feature to note is that in a marriage, the value of a woman's productivity in the market, ϕ_j , depends on how much she will work there, $1 - h$. Likewise, the value of her productivity at home, η_h , is a function of how much time she labors there, h .

3.2.1 The Gale–Shapley Matching Algorithm

To characterize the implied matching process, simply make a list of utilities from the pairings, starting from the top and going to the bottom. The best women will be matched with the best men. Now, suppose that there are more of these men than women. Then, some of the men will have to match with the next-best women on the list. The matching process continues down the list in this fashion. At each stage, the remaining best men are matched with the remaining best women. If there is an excess supply of one of the sexes, the overflow of this sex must find a match on the next line(s) of the list.

Now, suppose that the k th position on the list is represented by a match of type (μ_i, ϕ_j, η_h) . Some type- μ_i men may have already been allocated to women that are higher on the list; i.e., to women that have a better combination of ϕ_j and η_h . Let $R_m^k(\mu_i)$ be the amount of remaining type- μ_i men that can be allocated at the k th position on the list. Similarly, let $R_f^k(\phi_j, \eta_h)$ be the number of available type- (ϕ_j, η_h) women. The number of matches is given by $\min\{R_m^k(\mu_i), R_f^k(\phi_j, \eta_h)\}$. Recall that the number of people of each sex is one. Thus, the odds of a match are $\Pr(\mu_i, \phi_j, \eta_h) = \min\{R_m^k(\mu_i), R_f^k(\phi_j, \eta_h)\}$. Any type- μ_i men that are not assigned a mate at position k will be available for position $k + 1$, and similarly so for type- (ϕ_j, η_h) women. Thus, the number of type- μ_i men that will be available for the next position, $k + 1$, will be given by $R_m^{k+1}(\mu_i) = R_m^k(\mu_i) - \min\{R_m^k(\mu_i), R_f^k(\phi_j, \eta_h)\}$, while the number of type- (ϕ_j, η_h) women is $R_f^{k+1}(\phi_j, \eta_h) = R_f^k(\phi_j, \eta_h) - \min\{R_m^k(\mu_i), R_f^k(\phi_j, \eta_h)\}$. To start things off, $R_m^1(\mu_i) = \mu_i$ and $R_f^1(\phi_j, \eta_h) = \chi_{jh}$. The matching process is described in table 4. The odds of some of the matches happening in the table will be zero. For example, consider a marriage of type (μ_2, ϕ_1, η_1) . There will be no type- μ_2 men left by the time the algorithm reaches a type- (ϕ_1, η_1) woman.

3.2.2 The Rise in Assortative Mating

What can explain the rise in assortative mating? Two things come to mind. First, technological progress in the home. Second, technological progress in the market or reductions in discrimination that favor market work by women. If these drivers reshuffle the entries in table 4, then a change in the pattern of assortative mating will occur. The discussion here is heuristic. Take yesteryear as the starting point in time. Suppose that women do little work in the market. A high value for η implies that a woman will have a

TABLE 4
MATCHING PROCESS

Rank	Match	Odds
1	(2, 2, 2)	χ_{22}
$\vdots$	$\vdots$	$\vdots$
k	(i, j, h)	$\min\{R_m^k(\mu_i), R_f^k(\phi_j, \eta_h)\}$
$\vdots$	$\vdots$	$\vdots$
8	(1, 1, 1)	$\min\{R_m^k(\mu_1), R_f^k(\phi_1, \eta_1)\}$ $= \chi_{11}$

Notes: In the second column, (i, j, h) is shorthand for the match (μ_i, ϕ_j, η_h) .

high ranking in the table. From (16) it can be seen that, if women work little in the market ($h \simeq 1$), then their value in marriage will not be affected much by the value of their market productivity. But, their productivity at home will matter, as (17) illustrates. Assume that

$$\mathbf{M}(\mu_i, \phi_1, \eta_2, w, q) > \mathbf{M}(\mu_i, \phi_2, \eta_1, w, q).$$

Here a man of type μ_i will prefer a woman who is good at home production and not good at market work to a woman who is not good at home production and good at market work.

As the economy changes due to technological progress in the home and the market, so will the ordering in the table. Take each form of technological progress in turn.

(i) *Technological Progress in the Home.*

Let the price of home durables, q , fall. This releases labor from the home. As h falls, the value that a women's market productivity has in a marriage will increase. The position of $\mathbf{M}(\mu_i, \phi_1, \eta_2, w, q)$ vis à vis $\mathbf{M}(\mu_i, \phi_2, \eta_1, w, q)$ in the ranking would be reversed. Now, a man would prefer a wife who is not good at home production and good at

market work to the wife who is good at home production and not good at market work.

(ii) *Technological Progress in the Market.*

Suppose that market forces favor a shift from brawn to brain. This could be thought of in two ways. First, a woman's productivity in the market, ϕ , may rise relative to her productivity at home, η . Second, it could be viewed as an increase in the market productivity of women, ϕ , relative to men, μ . Both of these forces will result in a higher level of labor supply, $1 - h$, by women—see Appendix A.4. As a consequence, a woman's ϕ will matter more relative to her η .

The above illustration assumes that a woman's productivity at home and in the market are uncorrelated. Presumably, they are positively correlated. To the extent that this is true, the reordering in the table may be more muted.

3.2.3 Discussion

Stevenson and Wolfers (2007) document some key facts about marriage and divorce over the last 150 years. A nice survey of the early theoretical literature on marriage

and divorce is provided in Weiss (1997). Browning, Chiappori, and Weiss (2014) provide an introduction to the modern theory of marriage and divorce. The first search model of marriage and divorce was developed by Mortensen (1988). Greenwood and Guner (2009) develop a dynamic search model of marriage and divorce and fit it to match the trends in these variables since World War II. (A subsection of the paper examines the model's prediction over the entire twentieth century.) A model of marriage and divorce is used in Regalia and Rios-Rull (2001) to account for the rise in the number of single mothers. They stress market forces, such as a movement in the gender gap, as explaining this rise. Jacquemet and Robin (2012) estimate a search and matching model of the marriage market for the United States. Their analysis focuses on how female and male wages affect marriage probabilities and the share of the marital surplus received by partners.

Fernandez and Wong (2017) examine the implications of a move toward unilateral divorce in the 1970s. They find that this switch promoted divorce and increased married female labor-force participation. In the current analysis, think about this as a drop in δ . In an econometric investigation, Wolfers (2006) finds that unilateral divorce laws explain very little of the long-run rise in divorce rates. Thus, whether unilateral divorce laws promoted a rise in divorce is an open question. Fernandez and Wong (2017) also find that a reduction in the gender gap barely changed divorce and led to a small reduction in the number of married women. It did have a significant impact on married female labor-force participation. The impact of such laws on savings is studied in Voena (2015). She finds that the unilateral divorce combined with an equal division of property encourages savings and discourages female labor-force participation. Voena (2015) formulates the allocations

within a marriage from a dynamic contracting perspective. Under the unilateral divorce regime, the Pareto weights in the married household's problem [a dynamic analogue to (5) in her analysis] may evolve over time as one party in the marriage attempts to keep the other within the marriage by transferring goods and/or leisure to them. Guvenen and Rendall (2015) suggest that changes in divorce laws also contributed to growing female educational attainment, since education acts as insurance against bad marriages for women. Other public policies affect the value of single versus married life as well. Since the unit of taxation in the US tax code is the household, it penalizes married households with two earners. Chade and Ventura (2002) build a search model of marriage and divorce to study the differential tax treatment of single and married individuals and show that a move to individual filing would increase the number of marriages.

The matching algorithm presented here was developed by Gale and Shapley (1962), who characterize stable matching without transfers. Every potential marriage is associated with a given utility value for each partner. This utility cannot be transferred between parties. The algorithm matches people together. The allocation is stable when there is no situation where two people would prefer to be matched with each other as opposed to with their current partners. As a special case, suppose men and women differ by a single characteristic, say their labor market productivities, μ and ϕ . Let $M(\mu, \phi)$ be the common value of a marriage enjoyed by both parties. It is easy to show that if M is strictly increasing in both arguments, then the stable Gale–Shapley algorithm will imply positive assortative mating; that is, high- μ men will match with high- ϕ women and low- μ men will match with low- ϕ women.

An alternative would be to assume that the utility is transferable; that is, the total utility from a match can be split up between the

partners. Shapley and Shubik (1971) and Becker (1973) characterize stable matches when utility is transferable. They show that stable matching requires efficiency in that it must maximize total utility summed across all matches. Becker (1973) presented some conditions under which there will be positive assortative mating. Again, if agents differ by a single dimension, then positive assortative mating requires that the function M be supermodular. In other words, if $\mu' > \mu$ and $\phi' > \phi$, then $M(\mu', \phi') + M(\mu, \phi) \geq M(\mu', \phi) + M(\mu, \phi')$. A more thorough analysis is provided by Browning, Chiappori, and Weiss (2014). It is easy to put randomness into the matching process, such as marital bliss, so that there will not be perfect assortative mating. Choo and Siow (2006) estimate a static transferable utility model of the US marriage market and show that the gains to marriage for young adults fell substantially between the early 1970s and early 1980s.

Burdett and Coles (1997) study assortative mating within a search framework. They use a simple framework where agents differ along a single dimension. In particular, the utility from a given match for a person is the type of his/her match. They show that a class structure will arise: individuals of a given type will in equilibrium only marry people whose types fall within a certain range around the individual's type. As a result, while there is still a tendency for assortative mating, the cost of search forces agents to be less picky compared with a frictionless world. Shimer and Smith (2000) study a search model with transferable utility and show that the household utility function, $M(\mu, \phi)$, needs to satisfy more stringent conditions than supermodularity for positive assortative mating to emerge in equilibrium.

Couples may sort on many dimensions. Chiappori, Orefice, and Quintana-Domeque (2012) show that body mass is one such factor. Fernandez and Rogerson (2001) and Fernandez, Guner, and Knowles (2005)

suggest that positive assortative mating has implications for income inequality and vice versa. Greenwood et al. (2016) illustrate within the context of a structural model how shifts in family formation (changes in educational attainment, shifts in the pattern of marriage, and movements in married female labor-force participation) can significantly amplify the impact of shifts in the wage structure on inequality. A distinguishing feature of the analysis is that the matching probabilities are determined endogenously. Positive assortative mating provides a marriage-market return for female educational investment, in addition to the traditional labor-market return. This is analyzed in Chiappori, Iyigun, and Weiss (2009). Eckstein, Keane, and Lifshitz (2016) estimate a model of family formation, similar in many respects to Greenwood et al. (2016), but they include a fertility decision. In their analysis, the matching probabilities are exogenously imposed using the empirical ones from the data.

The existing empirical literature measures assortative mating by the correlation, or other measures of association, between the educational attainments, as in figure 12, or the earnings of husbands and wives. In addition to household income, educational attainment of individuals can affect assortative mating both by shaping partner preferences as well as by influencing the circle of potential partners that one circulates within. One can imagine that assortative mating along a measure of socioeconomic class that is independent of one's education can possibly shed light on these two forces. Assortative mating along the lines of parental education or along distinctive first or last names could provide such a measure of socioeconomic class (see Guell, Rodriguez Mora, and Telmer 2015 and Olivetti and Paserman 2015).

As an aside, the model above suggests that a married woman will do more housework the higher her productivity is at home relative

TABLE 5
US CHILD WELL-BEING BY FAMILY STRUCTURE

	1 Parent, %	2 Parent, %
Relative income	51.4	100
<High school	14.6	9.2
College	15.0	23.6
Teen births	32.8	21.3
Idle, at age 19–21	25.9	12.1

Notes: See the main text for details.

Source: Panel Study of Income Dynamics (PSID), 1968–2011

to in the market; i.e., the higher is η_h/ϕ_j . It also suggests she will work more in the market the higher her productivity is relative to her husband's in the market; i.e., the higher is ϕ_j/μ_i . On the first point, Rios-Rull (1993) inserts household production into an overlapping-generations model to examine its impact on the time allocations of skilled versus unskilled labor. In his framework, skilled labor (relative to unskilled labor) tends to substitute market goods or services for labor in household production.

4. Growing Up with a Single Mother

There is no longer the need for never-married adults to find a mate and marry for economic reasons, as in the past. Likewise, people no longer have to remain, for economic reasons, in an unhappy marriage. They can divorce and live alone or find someone else. There is a downside, though. Children from single-parent families are less likely to be successful than kids living with both natural parents. This is shown in table 5. The table examines various outcomes for a child based on where they lived from ages zero to sixteen. A two-parent family is defined as a family with both natural parents present. A single-parent family is defined as a family

with just one parent present or a family with one natural parent and one step parent present. A child is assigned to the category where s/he lived for the majority of her or his childhood. As shown in table 5, children living in single-parent households are more likely than children from two-parent families to drop out of high school (14.6 percent versus 9.2 percent), be idle (25.9 versus 12.1 percent), experience a birth before age twenty (32.8 versus 21.3 percent), and are less likely to complete college (15.0 percent versus 23.6).

What economic factors might be important in accounting for these differences? Single-parent families have only half the income of two-parent families. Recall from table 2 that the cost of raising children for a single-parent family is very high as a fraction of their income. Additionally, table 1 showed that child-care expenses absorbed a significant fraction of a working mother's income. Children living with a single mother are much more likely to live in poverty than those living within a married household, as the right-hand panel of figure 13 shows. A much greater percentage of female-headed families receive welfare than married ones do—see the left panel of figure 13.

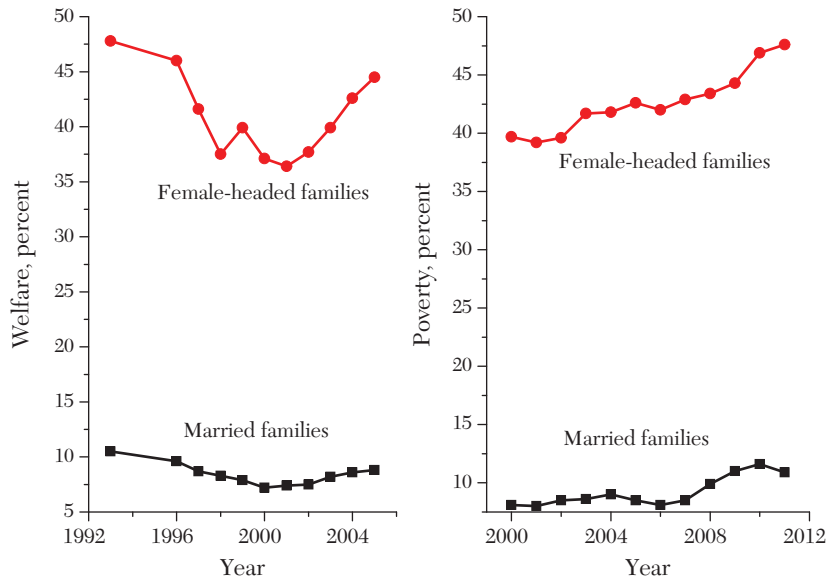


Figure 13. Households on Welfare (1993–2005) and Child Poverty (2000–2011) in the United States

Sources: Indicators of Welfare Dependence, Annual Report to Congress, 2008, US Department of Health and Human Services; and US Bureau of the Census, Current Population Survey, Annual Social and Economic Supplements.

Children in single-parent households also enjoy less time with their parents. Table 6 shows how much time mothers spend on child care per week. The upper panel shows total hours per week, while the lower panel reports hours per child. The total time spent on child care increased in recent decades for all groups. As documented in section 2, total nonmarket hours (household work) of women declined significantly during this period. The numbers in table 6 suggest that time saved by better household technologies was partly used for spending more time with children. More educated mothers spend more time with their children than less educated mothers. Finally, child-care time per child in single-parent families is about

an hour less per week than in two-parent families.

4.1 *A Model with Female-Headed Households*

Imagine a world where there are two types of men and women, those with a high level of productivity, $\phi_2/2$, and those with a low level, $\phi_1/2$, so that $\phi_1 < \phi_2$. A person earns a wage based on her/his productivity level on the labor market. The fraction of each sex that is the low type is ν . Men and women meet in a marriage market. They sort according to their productivity so that a paired man and woman are from the same socioeconomic class. The couple draws a marital bliss shock, $\bar{b} \in \{-b, b\}$. The realized value of

TABLE 6
US CHILD-CARE TIME BY MOTHERS, HOURS PER WEEK

Year	< College	College	Unmarried	Married
<i>Total</i>				
1965	8.56	10.81	6.02	10.31
1975	7.73	7.38	6.80	7.94
1985	7.65	8.82	5.76	8.77
2003	12.51	16.39	11.14	14.41
2013	13.00	15.63	11.51	14.5
<i>Per child</i>				
1965	3.38	5.09	2.82	4.71
1975	4.02	3.63	3.98	3.90
1985	4.67	5.38	3.72	5.27
2003	7.24	9.63	7.38	8.04
2013	7.76	9.39	7.42	8.47

Notes: The numbers in the upper panel are hours spent on total child care (primary, educational, and play) by women. The lower panel reports hours per week per child in the household. The sample is restricted to households with a child present.

Source: Aguiar and Hurst (2007, table II) updated with the 2004–13 American Time Use Survey (ATUS)

the shock will take the low value, $-b$, with probability ε . The couple decides whether or not to marry based upon the realized value of this shock. Regardless of whether they marry or not, each woman has two children, a girl and a boy. If the couple separates, the children always live with their mother. An adult's productivity level is influenced by the amount of human capital investment (education) that s/he received when young. In particular, if a household invests η in their children's education, then both children will draw the high level of productivity, $\phi_2/2$, with probability π . If they do not, then both children will draw the high level with probability $\lambda < \pi$.

Let a married household have preferences of the form

$$\ln(c) + 2\phi' + \tilde{b},$$

where c is consumption, $\phi' \in \{\phi_1/2, \phi_2/2\}$ represents the productivity level of the

children, and $\tilde{b} \in \{-b, b\}$ denotes the level of marital bliss. Parents care about the success of their children, as measured by the kids' productivity. Utility for a single mother is given by

$$\ln(c) + 2\phi'.$$

She does not realize any bliss from marriage. The preferences for a single man have the form

$$\ln(c).$$

Since the children do not live with single fathers, it is assumed that the latter does not enjoy a benefit from the former. Therefore, an estranged father will not willingly invest in his children because he does not care about them.

Should a household invest or not in educating their children? If a single mother of type i does not invest in her children,

her consumption will be $\phi_i/2$. The expected utility from her children is then $(1 - \lambda)\phi'_1 + \lambda\phi'_2$. Alternatively, if she does invest in her children, she will realize expected utility from them in the amount $(1 - \pi)\phi'_1 + \pi\phi'_2 > (1 - \lambda)\phi'_1 + \lambda\phi'_2$, but her consumption will be reduced to $\phi_i/2 - \eta < \phi_i/2$. Let $e = 0$ indicate if a household does not invest in its children (or does not educate them) and $e = 1$ denote that it does. The single mother's decision to invest in her children is summarized by

$$e = \begin{cases} 1 & \text{if } \ln(\phi_i/2) - \ln(\phi_i/2 - \eta) \\ & \leq (\pi - \lambda)(\phi'_2 - \phi'_1), \\ 0 & \text{otherwise.} \end{cases}$$

A single mother will only invest in her children if the gain in utility from improved child quality, $(\pi - \lambda)(\phi'_2 - \phi'_1)$, exceeds the loss from the drop in her consumption, $\ln(\phi_i/2) - \ln(\phi_i/2 - \eta)$. The gain in utility from improved child quality derives from the higher odds, $\pi > \lambda$, of getting a good draw, $\phi'_2 > \phi'_1$.

The decision for a married household is very similar:

$$e = \begin{cases} 1 & \text{if } \ln(\phi_i) - \ln(\phi_i - \eta) \\ & \leq (\pi - \lambda)(\phi'_2 - \phi'_1), \\ 0 & \text{otherwise.} \end{cases}$$

Note that $\ln(\phi_i) - \ln(\phi_i - \eta)$ is decreasing in ϕ_i . So, a married household is more likely to invest in their children because they are wealthier, $\phi_i > \phi_i/2$.

4.2 An Equilibrium with Single Mothers

It is easy to construct the following situation where:

- (i) Some children will grow up with a single mother.

- (ii) Children who grow up with single mothers will have, on average, lower levels of human capital than those who don't.

- (iii) Girls who grow up living with a single mother are more likely to become single mothers than girls who grow up in a two-parent family.

To construct such an equilibrium, assume the following conditions.

CONDITION 3: (*All married couples educate their children*)

$$\ln(\phi_1) - \ln(\phi_1 - \eta) < (\pi - \lambda)(\phi'_2 - \phi'_1).$$

Observe that if the above equation holds for a type-1 household, then it certainly holds for a type-2 one, because $\ln(\phi_i) - \ln(\phi_i - \eta)$ is decreasing in ϕ_i .

CONDITION 4: (*Single mothers do not educate their children*)

$$\ln(\phi_2/2) - \ln(\phi_2/2 - \eta)$$

$$> (\pi - \lambda)(\phi'_2 - \phi'_1).$$

If a high-type single mother chooses not to educate her child, then a low-type one won't either, since $\ln(\phi_i/2) - \ln(\phi_i/2 - \eta)$ falls with increases in ϕ_i .

CONDITION 5: (*Couples with the high bliss shock, b , always marry*)

$$\ln(\phi_1 - \eta) + (1 - \pi)\phi'_1 + \pi\phi'_2 + b$$

$$> \ln(\phi_1/2) + (1 - \lambda)\phi'_1 + \lambda\phi'_2.$$

The left side gives the value of marriage for a type-1 couple, which draws the high value for the bliss shock, b . The right side gives the value of single life for the woman. The

man will also want to marry if this condition holds, because the value of single life for him is $\ln(\phi_1/2)$. Again, $\ln(\phi_i - \eta) - \ln(\phi_1)$ is increasing in ϕ_i , so that if the above condition holds for ϕ_1 , then it also holds for ϕ_2 .

CONDITION 6: (*High-productivity couples with a bad bliss shock, $-b$, choose to marry while low-productivity ones do not*)

$$\begin{aligned} \ln(\phi_2 - \eta) + (1 - \pi)\phi'_2 + \pi\phi'_2 - b \\ > \ln(\phi_2/2) + (1 - \lambda)\phi'_1 + \lambda\phi'_2, \end{aligned}$$

and

$$\begin{aligned} \ln(\phi_1 - \eta) + (1 - \pi)\phi'_1 + \pi\phi'_2 - b \\ < \ln(\phi_1/2) + (1 - \lambda)\phi'_1 + \lambda\phi'_2. \end{aligned}$$

When the first condition holds for a type-2 woman, then it will also hold for a type-2 man. Whether the second condition holds for a type-2 man is irrelevant, since if the woman doesn't want to marry, then a union won't form.

4.2.1 The Steady State with Single Mothers

Now think of an overlapping-generations model where a person lives two periods, the first as a child, the second as an adult. Given the above conditions, type-1 matches that draw the low value for the bliss shock, $-b$, will not result in a marriage. The single woman will then pick not to educate her children—condition (4). Let ν denote the steady-state fraction of low-productivity adults. By assumption, then, ν will not change over time. It is determined by

$$\begin{aligned} \nu = \nu\varepsilon(1 - \lambda) + \nu(1 - \varepsilon)(1 - \pi) \\ + (1 - \nu)(1 - \pi). \end{aligned}$$

To understand this equation, focus on the right-hand side. There are exactly three ways

a person can become a low-productivity adult. First, s/he may have had a low-productivity mother. There are ν low-productivity mothers in the population. A fraction ε of these will remain unmarried because they draw the low value for the bliss shock. Their children will be a low type with probability $(1 - \lambda)$. Thus, the number of low-type children spawned from low-type single mothers is $\nu\varepsilon(1 - \lambda)$. This explains the first term. Second, a low-productivity mother marries with probability $1 - \varepsilon$. Even though the children in her household will be educated, they may still turn out to be a low type with probability $1 - \pi$. This gives the second term. Third, there are $1 - \nu$ high-type marriages in the population. A fraction $(1 - \pi)$ of these marriages will have low-type children. Therefore, the number of low-type children arising from high-type marriages is $(1 - \nu)(1 - \pi)$. Consequently, the right-hand side gives the total number of low-type children, which in a steady state must equal ν , or the left-hand side. Solving gives

$$0 < \nu = \frac{1 - \pi}{1 - \varepsilon[(1 - \lambda) - (1 - \pi)]} < 1.$$

Taking stock of the situation:

- (i) The fraction of mothers who are single is $\nu\varepsilon$. Not surprisingly, this fraction is increasing in the odds that a low-type mother will not get married, ε . This fraction also rises in the probabilities that children will turn out to be low types, $1 - \lambda$ and $1 - \pi$.
- (ii) The expected level of human capital for a child growing up with a single mother is $(1 - \lambda)\phi_1/2 + \lambda\phi_2/2$, which is less than the expected level of human capital for a child growing up in a two-parent family, $(1 - \pi)\phi_1/2 + \pi\phi_2/2$.

- (iii) The odds of a girl who grew up with a single mother becoming a single mother herself are $\varepsilon(1 - \lambda)$, while the probability of a girl who grew up in a two-parent family becoming a single mother is $\varepsilon(1 - \pi)$, where $\varepsilon(1 - \lambda) > \varepsilon(1 - \pi)$.

4.3 Discussion

Economic status is highly correlated across generations. Corak (2013) and Durlauf and Shaorshadze (forthcoming) review the extensive literature on intergenerational mobility. Measuring intergenerational mobility is a challenging task that requires panel data that link the economic status of children to that of their parents. Guell, Rodriguez Mora, and Telmer (2015) and Clark (2014) propose a new methodology for measuring intergenerational mobility that exploits the cross-sectional data on the joint distribution of surnames and economic outcomes. Clark (2014) documents that there has been very little change in social mobility over the past few centuries.

A growing body of literature in economics and other social sciences emphasizes the importance of initial conditions for intergenerational mobility. Carneiro and Heckman (2003) and Cunha et al. (2006), among others, show that differences between children, both in their cognitive and noncognitive skills, appear at very early ages and that the family environment plays a significant role in generating these differences. Furthermore, Cunha and Heckman (2007) and Cunha, Heckman, and Schennach (2010) emphasize the importance of early childhood investment for the effectiveness of investment at later ages. Huggett, Ventura, and Yaron (2011) calculate that the initial conditions at labor market entry, ages twenty to twenty-five, can account for about 60 percent of the variation in lifetime earnings—considerably more than shocks received during the working lifetime. Keane and Wolpin

(1997) find an even larger role for initial conditions.

Becker and Tomes (1979, 1986) and Loury (1981) constitute the main building blocks of theoretical research on intergenerational mobility. In the Becker and Tomes model, altruistic parents, given their preferences and constraints, decide how much to invest in their children. Lee and Seshadri (2014) integrate the Becker and Tomes framework into a standard life-cycle economy. Their results suggest that investment in children and parents' human capital, rather than the persistence of innate abilities, have the largest impact on intergenerational mobility. Restuccia and Urrutia (2004) and Caucutt and Lochner (2012) also study models with multi-period human capital investment and show that public policies that target early ages are more successful than later interventions.

The plight of children living with a single mother was documented in a classic book by McLanahan and Sandefur (1994). Aiyagari, Greenwood, and Guner (2000) build an overlapping-generations model with endogenous marriage and divorce to analyze the impact of single parenthood on intergenerational mobility. Their analysis combines the Becker and Tomes model of human capital investment on children with a search model of marriage. The analysis is extended to include endogenous fertility in Greenwood, Guner, and Knowles (2003). Suppose that single women are eligible for a lump-sum welfare payment in the amount ω . From examining condition (6), it can be seen that this payment makes it more likely for a low-productivity woman to choose single life because the value of not marrying increases to $\ln(\phi_1/2 + \omega) + (1 - \lambda)\phi'_1 + \lambda\phi'_2$. Still, if the payment were large enough perhaps a single mother would invest in her children's education, since now the cost of doing this is $\ln(\phi_1/2 + \omega) - \ln(\phi_1/2 + \omega - \eta)$, which is decreasing in ω —condition (4). In a similar vein, suppose that a single father has to pay

TABLE 7
SELECTED US WOMEN'S RIGHTS MILESTONES

Year	Event
1920	19th Amendment
1920	Women's Bureau
1936	<i>United States v. One Package</i>
1960	Birth Control Pill
1963	Equal Pay Act
1964	Civil Rights Act, Title VII
1964	EEOC
1965	<i>Griswold v. Connecticut</i>
1969	No Fault Divorce, California
1970	<i>Shultz v. Wheaton Glass Co.</i>
1972	Title IX, Education Amendments
1973	<i>Roe v. Wade</i>
1978	Pregnancy Discrimination Act
1986	<i>Meritor Savings Bank v. Vinson</i>

child support in the amount ξ . This reduces the value of single life for a low-productivity man to $\ln(\phi_1/2 - \xi)$, so he would more likely remain married—his analogue to condition (6). This raises the value of single life for a low-productivity woman and also gives her more wherewithal to educate her kids. The impact of welfare payments on single motherhood is modeled in Greenwood, Guner, and Knowles (2000). In situations where the household solves a bargaining problem, as discussed in section 2.2, the relative weight on the female's utility in the household optimization problem may shift as a result of changes in public policy—see Greenwood, Guner, and Knowles (2003).

5. Social Change

Just as families have changed over time, so have culture, social norms, and social institutions. Clearly, these societal changes influence what happens within families. Likewise, widespread shifts in the structure of families will have an impact on culture, social norms, and social institutions. The approach here

models both family structure, and culture and social institutions, as functions of the economic environment. As the economy changes, so do family structure, culture, and social institutions. The induced changes in family structure, culture, and social institutions interact with each other.

5.1 Women's Rights in the Workplace

Milestones for women's rights in the United States are presented in table 7. This list is far from complete. It emphasizes topics discussed here, to wit: married female labor-force participation, marriage and divorce, and reproduction. In 1920 an amendment to the US constitution granted women the right to vote. In the same year, the US Department of Labor established the Women's Bureau. Its purpose was to collect information about women in the workforce and to improve their working conditions. The Equal Pay Act was passed in 1963. This act made it illegal to pay a woman less than what a man would earn for the same job. Around the same time, the Civil Rights Act, Title VII, prohibited employment

discrimination on the basis of sex. The act established the Equal Employment Opportunity Commission (EEOC), charged with investigating complaints and imposing penalties. In 1970, the US Court of Appeals for the Third Circuit ruled that Title VII applied to jobs with “substantially equal” task requirements for men and women, although not necessarily in title or job description. Title IX of the Education Amendments, passed in 1972, banned discrimination against women in education. This facilitated entry into professional schools, among other things. Employment discrimination against pregnant women was prohibited in 1978. Last, in 1986 the Supreme Court found that sexual harassment on the job violated Title VII of the Civil Rights Act.

5.2 *A Model of Women’s Rights*

A stylized model of the process whereby women gain rights in the workplace is now formulated. Imagine an economy populated by married households, each with two children. Both the husband and wife have one unit of time. The husband spends all of his time working in the market at the wage w . The wife has three potential uses of her time: working at home, spending time with her children, and working in the market at the wage rate rw , where $r \in \{0, \phi\}$ reflects women’s rights in the workplace. When $r = 0$ (no rights), married women are prohibited from working, while when $r = \phi \leq 1$ (equal rights) they can work at the wage ϕw .² Here, ϕ represents the gender gap. Even in a world without discrimination, women may be paid differently than men at a particular point in time. This could occur because jobs

in the past required more brawn than brains. Labor is indivisible. A mother must spend the *fixed* amount of time $\mathfrak{h}$ on housework. If the woman works in the market, then she must work the *fixed* amount $\mathfrak{l}$ there.

The household’s utility function is given by

$$c + \lambda q + \eta \mathbf{q},$$

where c is the household’s consumption, q is the quality of the two children, and $\mathbf{q}$ is the *average* quality of children in society. The constant η measures how households in society care about other people’s children, as opposed to their own, which is reflected by λ . The variable λ is distributed across households according to a uniform distribution on the interval $[0, 1]$. Thus, some households care more (less) about the quality of their offspring (consumption) than others. The quality of children is specified by

$$q = \ln(t),$$

where t is the mother’s time spent with them. Households also care about the average quality of children in society, $\mathbf{q}$. Low-quality children in society may lead to social problems such as crime, unwanted pregnancies, unemployment, and the like.

The household must decide whether or not the woman should work. The woman’s unit of time is split between housework, $\mathfrak{h}$, working in the market, $l \in \{0, \mathfrak{l}\}$, and improving the quality of the household’s children, t ; thus, $t = 1 - \mathfrak{h} - l$. The household’s decision amounts to solving the maximization problem

$$\max_{l \in \{0, \mathfrak{l}\}} \{w + rwl + \lambda \ln(1 - \mathfrak{h} - l) + \eta \mathbf{q}\},$$

where l indicates whether the wife works in the market ($l = \mathfrak{l}$) or not ($l = 0$). With $\hat{\lambda}$ defined as the threshold value where the household is indifferent between the woman

²Think about the situation corresponding with $r = 0$ as reflecting a world where there is severe discrimination against women, making it undesirable for them to work. This equilibrium could be supported by a trigger strategy mechanism, whereby people won’t truck with other people who hire women or other people who do business with people that hire women.

working in the market or not, it is easy to deduce that

$$l = \begin{cases} 1 & \text{if } \lambda < \hat{\lambda} = \Lambda(r) \\ & \equiv rw\mathfrak{l}/\ln[(1-\mathfrak{h})/(1-\mathfrak{l}-\mathfrak{h})], \\ 0 & \text{otherwise.} \end{cases}$$

The fraction of women working in society is just $\hat{\lambda} = \Lambda(r)$, which is a function of women's rights, r . When $r = 0$ (an absence of rights) then $\hat{\lambda} = \Lambda(0) = 0$ and no women will work.

LEMMA 4: (Married female labor-force participation) *The fraction of women working in society, $\hat{\lambda} = \Lambda(r)$, is increasing in women's rights, r , and decreasing in the amount of housework required, $\mathfrak{h}$. It is not affected by the average quality of children in society, $\mathbf{q}$.*

Observe that the average quality of children in society, $\mathbf{q}$, does not impact whether or not the woman in a household works in the market, at least given the level of women's rights, r . It turns out that the level of women's rights is influenced by the quality of children, though, and vice versa. As will be seen, this is one channel through which the economic environment affects women's rights.

Goldin (1990) discusses marriage bars that prevented married women from working. These were regulations in the first half of the twentieth century that barred women from working after marriage and prohibited employers from hiring married ones. In 1928, for example, 62 percent of school districts would not hire married women. This figure rose to 77 percent in 1942. Private firms had similar proscriptions.

Suppose that households can vote on a law that allows married women to work or not. When women can work, then $r = \phi$, and when they cannot, $r = 0$. For a law

to pass, 50 percent of households must vote for it. The median voter in society is the household with $\lambda = 0.5$. The law will pass if and only if the median household is in favor of it. When women can work, the average quality of children will be $\mathbf{q} = \hat{\lambda} \ln(1 - \mathfrak{l} - \mathfrak{h}) + (1 - \hat{\lambda}) \ln(1 - \mathfrak{h})$, while when they can't it is $\mathbf{q} = \ln(1 - \mathfrak{h})$, where again $\hat{\lambda} = \Lambda(r)$ is the fraction of women who work. A household will only vote for the law if the wife will work. It is easy to understand why. If the wife in the household will not work, then that household will not gain any income from voting for the law. But, the average quality of children in society will fall. So, the household will vote no. If the median household ($\lambda = 0.5$) votes for the law, then so will all households with a $\lambda < 0.5$.

The criteria underlying the median household's vote is straightforward to formalize. If women can work (a yes vote), the median household's utility will be

$$\begin{aligned} & w(1 + \phi\mathfrak{l}) + 0.5 \ln(1 - \mathfrak{l} - \mathfrak{h}) \\ & + \eta\{\Lambda(\phi)\ln(1 - \mathfrak{l} - \mathfrak{h}) \\ & + [1 - \Lambda(\phi)] \ln(1 - \mathfrak{h})\}, \end{aligned}$$

while if they can't [a no vote, which implies $\hat{\lambda} = \Lambda(0) = 0$], it is

$$w + 0.5 \ln(1 - \mathfrak{h}) + \eta \ln(1 - \mathfrak{h}).$$

Notice that the median household cares about the average quality of children in society, which is a function of the aggregate level of married female labor-force participation, $\hat{\lambda} = \Lambda(\phi)$. So, the median household will vote yes if $w\phi\mathfrak{l} - \eta\Lambda(\phi)\ln[(1-\mathfrak{h})/(1-\mathfrak{l}-\mathfrak{h})] > 0.5 \ln[(1-\mathfrak{h})/(1-\mathfrak{l}-\mathfrak{h})]$ and no otherwise. By solving out for $\hat{\lambda} = \Lambda(\phi)$, the decision to vote yes ($r = \phi$) or no ($r = 0$)

to grant women the right to work can be written as

$$r = \begin{cases} \phi & \text{if } w\phi\mathfrak{I}(1 - \eta) \\ & > 0.5 \ln[(1 - \mathfrak{h})/(1 - \mathfrak{I} - \mathfrak{h})], \\ 0 & \text{otherwise.} \end{cases}$$

PROPOSITION 5: (Women's rights) *Women's rights are more likely in societies where the requirement for housework, $\mathfrak{h}$, is low and the value of a woman's work in the market place, ϕw , is high. Women are likely to have fewer rights in society's that place more emphasis on children or where η is high.*

Therefore, technological progress in the home, or a reduction in $\mathfrak{h}$, is conducive to the development of women's rights. So is technological progress in the market. The latter can occur due to either an increase in the general level of wages, w , or from an increase in the value of jobs that women are suited for, as reflected by ϕ —say because of a shift away from brawn to brain.

5.2.1 Discussion

Fernandez, Fogli, and Olivetti (2004) model shifts in culture as changes in tastes over time. They build a dynamic model of culture in which mothers' employment affects their sons' preferences toward their wife working or not. In their model, sons with a working mother become less biased against their wives working. As more women enter the labor force, a new culture is transmitted from mothers to sons, which leads to even more women entering the labor force in the next generation. Empirically, the authors find that the employment status of a man's mother has a significant impact on the likelihood that his wife works. This is true even after controlling for various characteristics of husbands and wives, such as education, income, and religion. In follow-up work, Fernandez and Fogli (2009) find that second-generation

American women whose ancestors worked in countries where married female labor-force participation is higher tend themselves to have higher rates of labor-force participation. Likewise, second-generation American women tend to have higher rates of fertility when their ancestors originated from countries where fertility is higher.

The model of women's rights in the workplace is inspired by Doepke and Tertilt (2009). They model the empowerment of women over time. In their framework, at the start of time, only the husband's preferences matter for household decisions. The world then transits to a situation where both the husband's and wife's preferences are weighted equally. Women in their framework value the welfare of descendants more than men. A man faces a trade-off. On the one hand, he would like to run his own household. On the other, he would like his daughter to be empowered in her household. As the return to education increases, it pays for men to empower women in the household. This result arises because women care more about children than men. Consequently, women's empowerment spurs human capital investment and increases the welfare of the lineage. Eventually, this effect is strong enough so that men favor empowerment. Doepke, Tertilt, and Voena (2012) survey the literature on women's rights. They note that the extent of female empowerment in a nation is highly correlated with its income (see Doepke, Tertilt, and Voena 2012, figure 1). This fact suggests that culture, social norms, and social institutions are likely to be functions of the economic environment. Miller (2008) discusses how suffrage rights for women in the United States led to an expansion of public-health spending that resulted in a reduction in child mortality. In a similar vein, Cavalcanti and Tavares (2011) find that increased labor-force participation for women is associated with larger government. Still, culture may change slowly.

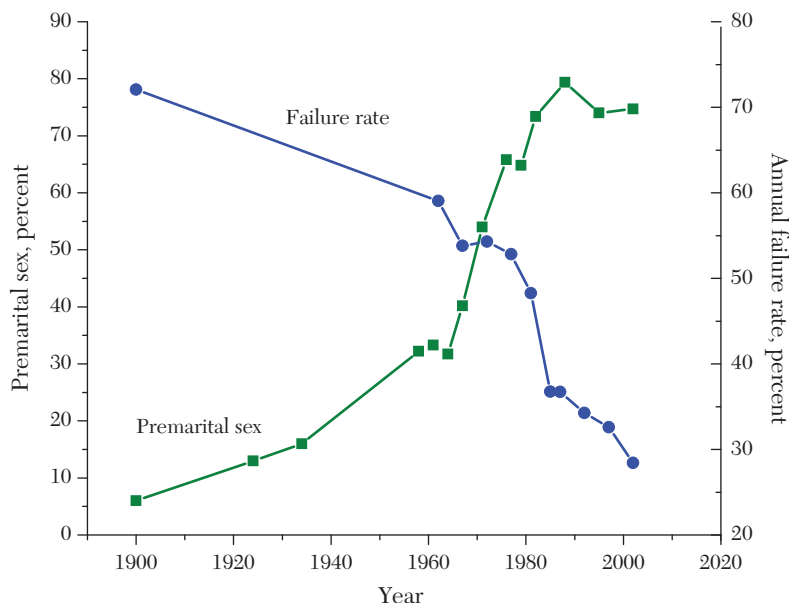


Figure 14. Premarital Sex and the Failure Rate for Contraception, the United States

Notes: The graph shows the fraction of nineteen-year-old women who had premarital sex as well as the annual failure rate for contraception.

Source: Fernandez-Villaverde, Greenwood, and Guner (2014).

Alesina, Giuliano, and Nunn (2013) show that historical differences in agricultural technology across countries (labor-intensive shifting cultivation done by women versus capital-intensive plough agriculture done by men) have left their marks on the status of women even today.

5.3 The Sexual Revolution, 1900–2000

There may be no better illustration of social change than the sexual revolution that occurred during the twentieth century. It is an excellent example of how technological progress, in this case contraception, can lead to a dramatic change in culture. At the beginning of the past century, only a paltry number of unmarried teenage girls engaged

in premarital sex; 6 percent in 1900 (see figure 14). By the end of the century, a large majority did, roughly 75 percent in 2002. What caused this increase? The contraception revolution.

Both the technology for contraception and education about its use changed dramatically over the course of the past century. Condoms and diaphragms became more effective, the birth control pill and IUDs arrived, and contraception became widely available due to the establishment of birth control clinics and changes in laws. Some of these laws are documented in table 7. In 1936, in the *United States v. One Package*, the US Court of Appeals for the Second Circuit established Margaret Sanger's right to order a pessary

TABLE 8
THE PERCENTAGE OF US TEENAGE GIRLS HAVING PREMARITAL
SEX AS A FUNCTION OF GUILT, MOTHER'S FEELINGS, AND
RELIGION

<i>Feel guilty from sex?</i>	%
Strongly agree	17
Agree	37
Neither agree or disagree	57
Disagree	75
Strongly disagree	77
 <i>How would your mother feel?</i>	 %
Strongly disapprove	31
Disapprove	50
Neither approve or disapprove	81
Approve	76
 <i>Role of Religion</i>	 %
Very important	38
Fairly important	53
Fairly unimportant	60
Not important at all	49

Source: Fernandez-Villaverde, Greenwood, and Guner (2014)

(a diaphragm) through the mail from a doctor in Japan. Similarly, in 1965 the Supreme Court in *Griswold v. Connecticut* struck down a Connecticut law that banned people from using “any drug, medicinal article or instrument for the purpose of preventing conception.” Estelle Griswold had opened a birth control clinic. In 1972, a women’s right to an abortion was affirmed in *Roe v. Wade*. These rulings resulted in a dramatic drop in the cost of premarital sex. In addition, the annual failure rate for contraception fell precipitously. In 1900, following the conventional birth control practices at the time, a teenage girl would have had a 72 percent chance of becoming pregnant if she engaged in premarital sex for the full course of a year. By 2002, this dropped to 28 percent.

How did societies in yesteryear prevent out-of-wedlock births? Providing unwed mothers with material support was a great

financial burden for parents, churches, and governments of the time. Societal real incomes were very low. So, society worked hard to punish young men and women who engaged in sex outside of marriage. Often, this involved shaming the people involved. For example, in 1601 England, the Lancashire Quarter Sessions ordered an unmarried father and mother of a child to be publicly whipped. The two then had to sit in the stocks naked from the waist up with a placard on their heads that read “These persons are punished for fornication.” Even today, the shame an unmarried teenage girl associates with sex is related to her propensity to engage in it. A girl who feels premarital sex is shameful is much less likely to engage in it, as table 8 makes clear. In 2002, only 17 percent of teenage girls (fifteen to nineteen year olds) who thought that premarital sex would cause shame had sexual intercourse, compared with 77 percent who

felt that it would not cause shame. In a similar vein, just 31 percent of girls who felt that their mothers would strongly disapprove if they had premarital sex had coitus, compared with 81 percent who thought that their mothers would be agnostic. People's attitudes are an input into culture. So, modeling how they are formed and evolve is important for understanding social change.

5.4 A Model of Premarital Sex

Imagine a problem facing a teenage girl. She is deciding whether or not to engage in premarital sex. When making the decision, she rationally calculates the costs and benefits from this activity. The benefit is joy from sex. Suppose this is governed by the level of the girl's libido, denoted by l (for this subsection). Let libido be distributed across girls according to a uniform distribution on $[0, 1]$. The cost of the activity is the chance that the girl becomes pregnant. A teenage pregnancy reduces the odds that a girl will attain a good education, work in a fulfilling job, or find a desirable partner in the marriage market. Represent the utility that a girl with an out-of-wedlock birth will have when she is an adult by A^o , and the utility that a girl without one will realize by A^n . Presume, of course, that $A^n > A^o$. Think about $A^n - A^o$ as representing the economic cost of pregnancy. Additionally, assume that a girl who becomes pregnant will feel shame in the amount s . The determination of the level of shame will be discussed later. Finally, even if the girl engages in premarital sex, she may not become pregnant. The level of π reflects the state of society's contraceptive technology. Let π give the odds of safe sex, or not becoming pregnant. Therefore, $1 - \pi$ is the probability of becoming pregnant, or the failure rate.

5.4.1 A Teenage Girl's Decision Making

Direct attention now to a teenage girl's decision about whether or not to engage

in premarital sex. On the one hand, if the girl is abstinent, then she will realize an expected lifetime utility level of A^n . On the other hand, if she engages in premarital sex, she will realize the enjoyment l , but will become pregnant with probability $1 - \pi$. Her expected lifetime utility level will be $l + \pi A^n + (1 - \pi)A^o$. Additionally, if she becomes pregnant she will feel shame in the amount s , which must be netted out of A^o . The teenage girl will pick the option that generates the highest level of expected lifetime utility. Her decision can be summarized as follows:

$$\begin{cases} \text{Abstinent} & \text{if } A^n \geq l + \pi A^n + (1 - \pi)(A^o - s), \\ \text{Sexually active} & \text{otherwise.} \end{cases}$$

5.4.2 Premarital Sexual Activity

How much premarital sexual activity will there be in society? The threshold level of libido, l^* , at which a girl is indifferent between having premarital sex or not is given by³

$$l^* = (1 - \pi)(A^n + s - A^o).$$

This has a nice interpretation. The expression equates the utility of sex, given by l^* , with its expected cost, the difference in future utilities induced by an out-of-wedlock birth, multiplied by the probability of pregnancy. All girls with an $l > l^*$ will engage in premarital sex while those with $l < l^*$ will not. The number of girls experiencing premarital sex, p , is given by

$$(18) p = 1 - l^* = 1 - (1 - \pi)(A^n + s - A^o).$$

³The threshold level of libido must lie between zero and one. Clearly, it will always be positive since $A^n > A^o$. Set $l = 1$ whenever $(1 - \pi)(A^n + s - A^o) > 1$.

LEMMA 5: (Premarital sex). *The fraction of teenage girls engaging in premarital sex, p , is decreasing in the failure rate of contraception, $1 - \pi$; the amount of shame associated with an out-of-wedlock birth, s ; and the economic cost of an out-of-wedlock birth, $A^n - A^o$.*

One might expect that girls from a higher socioeconomic background will have more to lose from having an out-of-wedlock birth. That is, $A^n - A^o$ will be larger for girls from a higher socioeconomic background than for the ones from a lower one. In fact, in the United States, the odds of a girl having premarital sex decline with family income. In the 2002 National Survey of Family Growth, for instance, 70 percent of girls between the ages of fifteen and nineteen in the bottom income decile had experienced it, versus 47 percent in the top one.

5.4.3 Socialization by Parents

How is the level of shame determined? Suppose that it is determined by parental socialization. In particular, assume that parents try to mold their daughters' psyches when they are young so that they will be less likely to engage in premarital sex when they are older. Parents do this recognizing that their teenage daughters will do whatever is in their own best interest. In economic terms, daughters will maximize their own utilities subject to the constraints that they face. However, parents can influence the paths that their children will follow as adults by molding their offsprings' utility functions when they are young. Hence, the actions a daughter takes as a teen will be partially based upon the socialization she received when young. To operationalize this idea, let parents choose the level of shame, s , to minimize the odds of an out-of-wedlock birth for their daughter. Shaming is a costly socialization process. It requires parental effort in terms of instilling sexual mores in a

daughter. The cost function for this is given by

$$\frac{s^{1+1/\gamma}}{1 + 1/\gamma}, \quad \text{where } 0 < \gamma.$$

Assume that parents do not know the level of libido that their teenage daughter will have. The odds of an out-of-wedlock birth are given by $(1 - \pi)[1 - (1 - \pi)(A^n + s - A^o)]$. The parents' decision problem can be formulated as

$$\min_s \delta(1 - \pi)[1 - (1 - \pi)(A^n + s - A^o)] + \frac{s^{1+1/\gamma}}{1 + 1/\gamma},$$

where the term δ governs the disutility parents realize if their teenage daughter becomes pregnant. The solution to this minimization problem is

$$(19) \quad s = [\delta(1 - \pi)^2]^\gamma.$$

LEMMA 6: (Parental socialization). *The amount of parental socialization, s , that a daughter receives is increasing in the failure rate for contraception, $1 - \pi$, and in the parents' disutility from an out-of-wedlock birth to their teenage daughter, δ .*

5.4.4 Socialization by Church and State

Other participants in the economy may also have an interest in socializing children. Churches and governments desire to curtail premarital sex by teenagers. They do this for both economic and moral reasons. Governments provide welfare for unwed mothers and their children. Historically, this was a large expense for the church. Social institutions desire to minimize this expense. Teenage girls who are religious are less likely

to participate in premarital sex. A girl who says religion is very important is much less likely to have premarital sex than one who does not think so. In 1994, only 38 percent of fifteen- to nineteen-year-old teenage girls who said religion was “very important” to them had premarital sex versus the 60 percent who said that religion was “fairly unimportant”—see table 8.

Now suppose that the church can influence the disutility that parents will experience if their daughters become pregnant. Recall that δ reflects the disutility that parents place on a daughter having an out-of-wedlock birth. Let

$$(20) \quad \delta = \frac{r^{\kappa/\gamma}}{\kappa^{1/\gamma}}, \quad 0 < \kappa < 1,$$

where r is the amount of religious indoctrination that the church undertakes. Thus, the parents will incur a higher level of disutility the bigger is the amount of religious indoctrination. Indoctrination is costly and undertaken according to the cost function

$$r\psi.$$

The church wants to minimize the number of out-of-wedlock births, while taking into account the cost of religious indoctrination. The number of out-of-wedlock births can be calculated using (18), (19), and (20) to be

$$(1 - \pi)[1 - (1 - \pi)(A^n - A^o) - \frac{r^\kappa}{\kappa}(1 - \pi)^{1+2\gamma}].$$

Suppose that the church's objective is to minimize the sum of out-of-wedlock births and the cost of religious indoctrination. The church's minimization problem will appear as

$$\min_p (1 - \pi) \left[1 - (1 - \pi)(A^n - A^o) - \frac{r^\kappa}{\kappa}(1 - \pi)^{1+2\gamma} \right] + r\psi,$$

which has the solution

$$r = \left[\frac{(1 - \pi)^{2+2\gamma}}{\psi} \right]^{1/(1-\kappa)}.$$

LEMMA 7: (Socialization by the church). *The amount of religious indoctrination, r , is increasing in the failure rate for contraception, $1 - \pi$, and is decreasing in cost of religious education, ψ .*

5.4.5 Social Change

It is now easy to deduce that as contraception becomes more effective, there will be less socialization by parents, the church, and the state against the perils of premarital sex. This reduction happens because as the odds of a sexually active unmarried teenage girl becoming pregnant drop, there is less need for socialization and socialization is a costly process. When the failure rate, $1 - \pi$, falls, so does the amount of religious indoctrination, r , by lemma 7. Parents will care less if their teenage daughters become pregnant, since δ will now be smaller. Thus, by lemma 6 parents too will engage in less socialization. Therefore, s will drop. All of this causes a rise in premarital sex, according to lemma 5. A sociologist would observe a simultaneous rise in premarital sex and a decrease in proscription against it. He may then interpret this as social change causing the rise in premarital sex.

PROPOSITION 6: (Social change). *A drop in the failure rate for contraception, $1 - \pi$, will cause a rise in premarital sex, p , and a drop in parental, s , and religious, r , proscription against it.*

5.4.6 Discussion

Parents want the best for their children. They spend a great deal of time and effort

educating their kids. Children must learn many things in order to function effectively as adults. Not all of the lessons taught by parents are of a formal nature. Parents may tell their children to study and/or work hard, not drink excessively or do drugs, be wise with their money, and not be dishonest, etc. Becker (1993) started the modern analysis of how a child's preferences can be molded by parental investments. He explored how parents may predispose children's preferences toward providing them with old-age support. Becker and Mulligan (1997) analyze how parents can manipulate a child's rate of time preference. This idea is extended in Doepke and Zilibotti (2008), who study the decline of the aristocracy that accompanied the British Industrial Revolution. They argue that parents who thought their children might enter the class of skilled workers instilled in their offspring a patience that allowed the children to sacrifice today to acquire the human capital necessary so that they would earn more tomorrow. Doepke and Zilibotti (2014) study the effects of parenting styles. They show that a lower level of household inequality results in more permissive parenting because the stakes are lower, a prediction supported by the empirical evidence. The model of premarital sex developed here is a simplified version of Fernandez-Villaverde, Greenwood, and Guner (2014). It has obvious similarities with the above work, but extends the notion to show how other players in society, such as churches or governments, may also try to shape preferences. Their work illustrates how shifts in the economic environment can lead to changes in culture.

A different perspective on preference formation is taken in the well-known work of Bisin and Verdier (2001). They assume that parents want their children to behave like them. The cultural transmission of corruption is modelled by Hauk and Saez-Marti (2002) using a variant of the Bisin and Verdier (2001) framework. They assume

that an honest person will suffer by behaving dishonestly. Honest parents educate their children, at a cost, with the hope that the latter will inherit (in a probabilistic manner) this notion of guilt associated with dishonest behavior. This is similar to the concept of shame analyzed here.

6. Fertility

The birth of a child radically changes a family. It obviously increases its size, but it also changes its age and sex composition. The birth of a child is also costly. It requires resources such as time and goods, often for many years. Finally, birth rates have first-order effects on population growth and on the age structure of a population. For all these reasons, economists have developed models to better understand the determinants and effects of fertility decisions.⁴

The Demographic Transition.—A “demographic transition” was experienced by countries throughout the world, starting in Europe and North America around 1800. The transition was from the high mortality and fertility rates that characterized the preindustrial era toward the low mortality and fertility rates of the modern era. This can be seen in figure 15, which plots crude birth rates for four regions of the world. Besides the crude birth rate, which measures contemporaneous fertility, lifetime fertility also plummeted in countries experiencing a demographic transition. In the United States, for example, the total fertility rate (the average births over a woman's life) for a white woman was seven in 1800 but had fallen to two by 1980—see Hernandez (1993). Before the demographic transition, some areas of the world experienced flat or increasing birth rates.

⁴There exist a variety of ways to measure fertility, which are reviewed in appendix A.10.

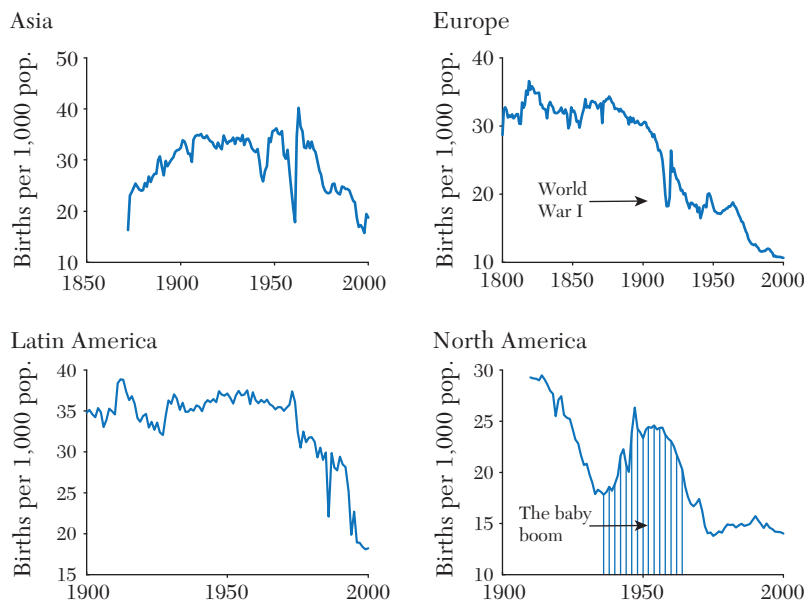


Figure 15. Demographic Transitions by Regions

Note: The graph shows population-weighted crude birth rates. North America: Canada and the United States; Europe: Belgium, Denmark, France, Italy, Norway, Spain, Sweden, and the United Kingdom; Latin America: Argentina, Chile, and Mexico; Asia: China, India, and Japan.

Source: Mitchell (2003)

The Income—Fertility Correlation.—There is a negative cross-sectional correlation between fertility and income. Figure 16 shows this relationship for various cohorts in the United States. The negative correlation between fertility and income not only holds in cross sections of individuals but also across countries, as documented by Manuelli and Seshadri (2009). In the case of the US economy, Jones and Tertilt (2008) compute an overall elasticity of children ever born to income of -0.38 across various cohorts. They also document that although this correlation varies from one cohort to another, it remains significantly negative between -0.45 (for the 1888 cohort) and -0.16 (for the 1933 cohort).

The Baby Boom.—Many developed countries experienced a baby boom about the middle of the twentieth century. This baby boom is most clearly seen in the North America panel of figure 15 and spans the years 1936 to 1965. In the United States, the number of children ever born went from 2.3 for women born in 1907 (whose average child was born in 1934) to 3.1 for women born in 1932 (whose average child was born in 1959).⁵ But the baby boom was not just in the United States. Australia, New Zealand, Canada, France, Norway, and Sweden, to cite only a few, also experienced baby booms.

⁵See Hernandez (1993, table 2.2, p. 24).

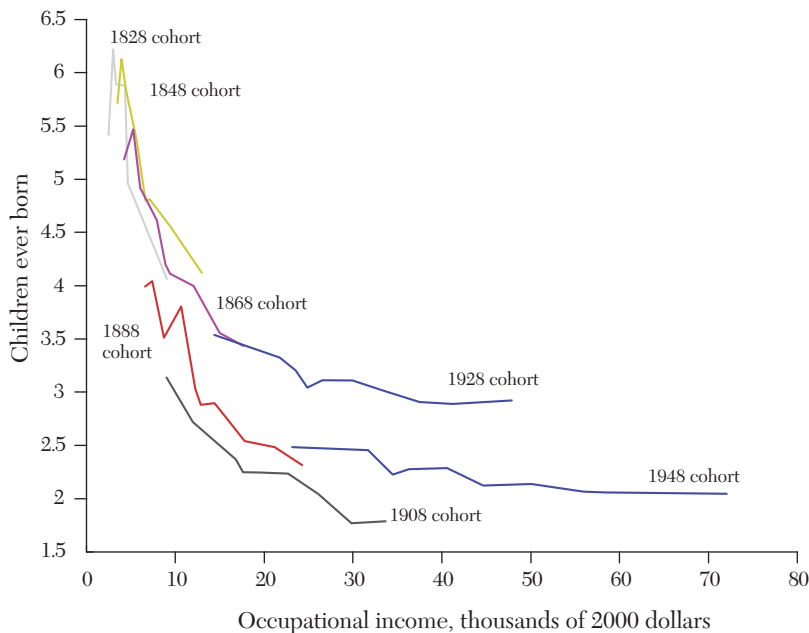


Figure 16. Children Ever Born by Occupational Income in the United States

Note: The graph shows children ever born by occupational income for select cohorts of women in the United States.

Source: Jones and Tertilt (2008)

Fertility and Social Upheaval.—Periods of wars, civil wars, revolutions, bouts of occupation by foreign invaders, etc., typically lead to lower fertility, as documented by Caldwell (2004). In Spain, for instance, births declined during the civil war years (1935–42) just as much as during the preceding thirty-five years. In Europe, World War I was associated with a very large decline in fertility, about 50 percent below trend—see figure 15. The missing births during the war were at least as large as, and sometimes larger than, the military casualties on the battlefield. The dent in the age composition of European demography caused by World War I was felt long after the war and throughout the twentieth century.

6.1 A Model of Fertility

A model is now presented to describe some of the key trade-offs associated with fertility choices, as emphasized in the literature. The economy is populated by households formed by two adults, a husband and a wife. Assume that the husband and wife are of the same age and live for two periods. They each have one unit of time per period. Each period, the husband spends all of his time working in the market at the wage w_m . The wife has two potential uses of her time: working and spending time taking care of children. Denote the productivity of the wife when she works by w_f . Productivity, w_f , can

be interpreted either as a market wage or as productivity in home production.

The preferences of the household are represented by the utility function

$$(21) \quad \mathbf{U}(c, k, q) + \beta \mathbf{U}(c', k', q).$$

In this specification, $\beta \in (0, 1)$ is the subjective discount factor. The function $\mathbf{U}$ is a utility index with standard properties. The variables c and c' represent household consumption in the first and second periods, respectively. Similarly, the variables k and k' represent the number of kids in the household during these periods. Assume that k' may differ from k because some children may die between the first and second periods. Let $s \in (0, 1)$ represent the fraction of children that are still present in the household during the second period:

$$(22) \quad k' = sk.$$

The variable k (as opposed to k') measures fertility. The variable q represents the average quality of children.

Children are costly; recall table 2, which shows their monetary cost, and table 6, which shows their time cost. Let the function $\mathbf{C}(k, q; w_f)$ measure the period cost of raising k children with quality q when the wife's opportunity cost of time is w_f . Depending on its specific form, this function regulates the cost of children in goods and/or time. The function $\mathbf{C}$ is a description of the technology linking inputs, goods and/or time, to the number of children produced. The budget constraint of the household is

$$(23) \quad c + \frac{c'}{r} + \mathbf{C}(k, q; w_f) + \frac{\mathbf{C}(sk, q; w_f')}{r} \\ = w_m + w_f + \frac{w'_m + w'_f}{r},$$

where r is the gross rate of interest.⁶ The household's optimization problem is to choose consumption, c and c' , the number of children, k , and their quality, q , to maximize (21) subject to (23). At an optimum, the following first-order condition must be satisfied for k :

$$(24) \quad \mathbf{U}_2(c, k, q) + \beta s \mathbf{U}_2(c', sk, q) \\ = \mathbf{U}_1(c, k, q) \left[\mathbf{C}_1(k, q; w_f) + \frac{s \mathbf{C}_1(sk, q; w_f')}{r} \right].$$

The marginal (lifetime) utility of children is on the left-hand side of this equation; that is, the increase in lifetime utility resulting from an additional birth. The right-hand side shows the marginal (lifetime) cost of children. The term in brackets is the marginal goods cost of children; that is, the increase in the present value of costs resulting from an additional birth. This is multiplied by the marginal utility of current consumption, so as to convert the cost into utils.

A second optimality condition must be satisfied for q :

$$(25) \quad \mathbf{U}_3(c, k, q) + \beta \mathbf{U}_3(c', sk, q) \\ = \mathbf{U}_1(c, k, q) \left[\mathbf{C}_2(k, q; w_f) + \frac{\mathbf{C}_2(sk, q; w_f')}{r} \right].$$

The left-hand side is the marginal benefit of increasing children's quality while the right-hand side is the marginal cost. Finally, an Euler equation must hold to equalize the marginal cost and marginal benefit of savings:

$$(26) \quad \mathbf{U}_1(c, k, q) = \beta r \mathbf{U}_1(c', sk, q).$$

⁶Even though the term w_f appears on the right-hand side of the budget constraint, the disposable income of the household in the first period is not $w_m + w_f$. The function $\mathbf{C}(k, q; w_f)$ represents the purchase of all the inputs needed to raise children, including potentially the wife's time at price w_f .

In what follows, simplified versions of this generic model are studied to analyze how productivity, mortality, and changes in the cost and/or demand for child quality affect fertility.

6.2 The Effects of Technology

Consider the following form for the utility function:

$$U(c, k, q) = U(c) + \gamma V(k).$$

This version of the model abstracts from child quality to simplify the discussion. A child may require θ units of the wife's time and/or e units of goods. The cost function is then

$$C(k, q; w_f) = (e + \theta w_f) k.$$

Different cases can be represented using this formulation. There is a general case where a child requires time and goods, that is $\theta > 0$ and $e > 0$. There are also two special cases. One case is when a child requires only time, that is $\theta > 0$ and $e = 0$, and another case is when a child requires only goods, that is $\theta = 0$ and $e > 0$. Wages grow at the common (gross) rate $g > 1$: $w_f' = gw_f$ and $w_m' = gw_m$. The first-order condition for fertility, equation (24), now reads

$$(27) \quad \gamma V_1(k) + \beta \gamma s V_1(sk) \\ = U_1(c) \left[e \left(1 + \frac{s}{r} \right) + \theta w_f \left(1 + \frac{gs}{r} \right) \right].$$

Observe that the left-hand side of equation (27), which represents the marginal benefit of children, is decreasing in k while the right-hand side, or the marginal cost, is

increasing. The solution of this equation is depicted in figure 17.⁷

Technology affects the household's decisions via the following channels. First, it can change the productivity of the husband and/or that of the wife, that is w_m and/or w_f . Second, it can affect the resources needed to produce children, that is e and/or θ . Accordingly, the effects of changes in technology are now developed in four propositions.

PROPOSITION 7: (Husband's earnings and fertility). *Fertility, k , is increasing in the productivity of the husband, w_m .*

To understand proposition 7, contemplate equation (27). It reveals that an increase in w_m , the productivity of the husband, reduces the marginal cost of children and, therefore, raises fertility. To understand why, hold fertility, k , constant at any arbitrary level. An increase in w_m unambiguously raises consumption and, thus, reduces its marginal utility. This implies that the cost of diverting resources away from consumption and into more kids is reduced. Thus, the right-hand side of equation (27) decreases, as represented in figure 18, implying an increase in the optimal fertility choice. This is an income effect.

REMARK 1: (Fertility and wars). *Proposition 7 can be used to deduce the effect of a war on fertility. Recall that wars are associated with large decreases in fertility rates. This is because male-specific wartime mortality is akin to a decrease in expected household income. That is, when making its fertility choice, the household takes into account that the woman may have to raise the children on just her own income, if her husband becomes a casualty of war. This implies a decrease in*

⁷In figures 17 and 18, the quantity displayed on the vertical axis is marginal utility.

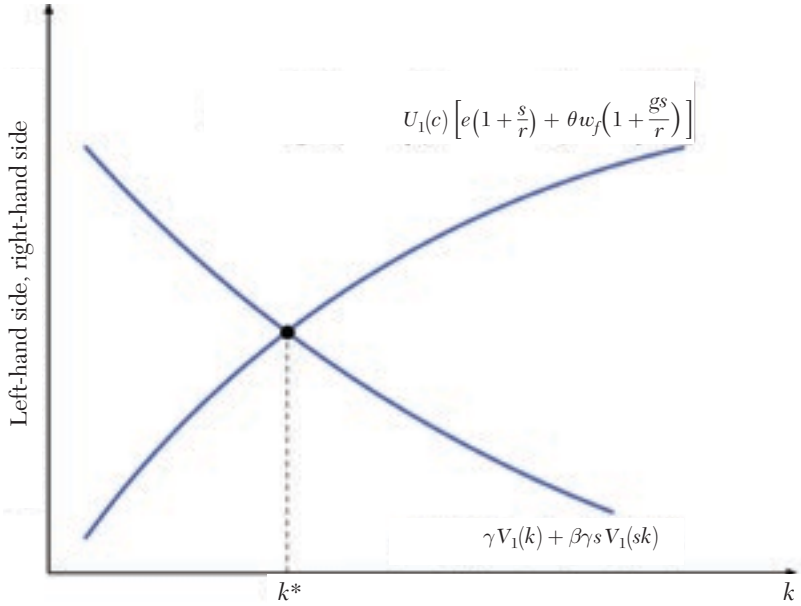


Figure 17. The Determination of Fertility

fertility via the income effect. Death benefits and the possibility of remarriage would mitigate this effect.

PROPOSITION 8: (Wife's earnings and fertility).

- A. *Children cost goods* ($\theta = 0, e > 0$)—Fertility, k , is increasing in the productivity of the wife, w_f .
- B. *Children cost time* ($\theta > 0, e = 0$)—An increase in w_f causes fertility, k , to
- (i) decrease if $U_1(c)$ is nondecreasing;
 - (ii) either increase or decrease if $U_1(c)$ is decreasing.

Part A of proposition 8 can be explained with the same argument as for proposition 7. When a child requires no time, an increase

in the wife's productivity acts as an income effect. It raises consumption and, therefore, decreases the marginal cost of a child.

Part B of proposition 8 deals with the case where an increase in the wife's productivity implies both an income and a substitution effect. These effects may act in the same or opposite directions, depending upon the shape of preferences. To see this, it pays to rewrite equation (27) as

$$\begin{aligned}
 (28) \quad & [\gamma V_1(k) + \beta \gamma s V_1(sk)] \\
 & \times \left[\frac{1}{\theta} \left(1 + \left(\frac{w_f}{w_m} \right)^{-1} \right) \frac{g+r}{gs+r} - k \right] \\
 & = U_1(c)c + \beta U_1(c')c'.
 \end{aligned}$$

(See Appendix A.5 for a derivation of this equation.) First, when w_f increases, there is an increase in the opportunity cost of the wife's time. This is represented by a decrease

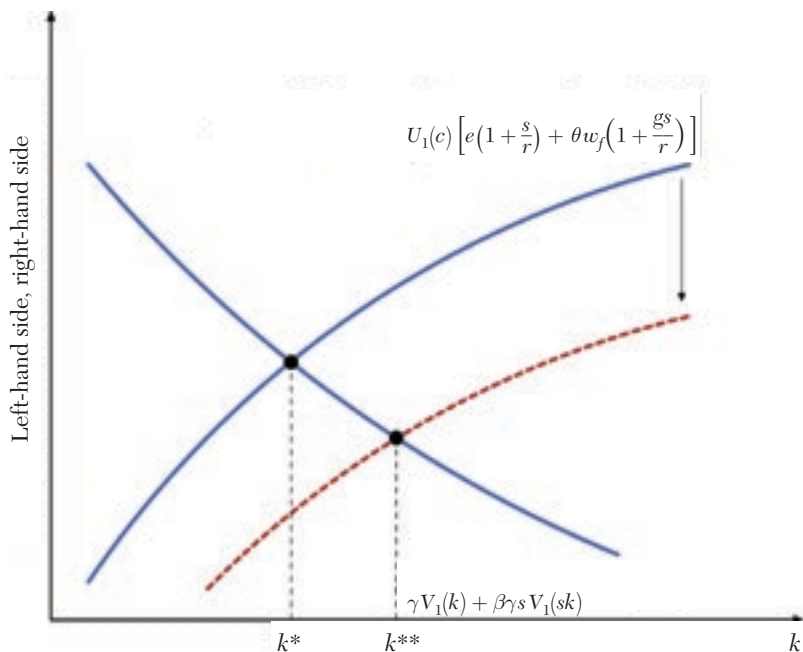


Figure 18. The Effect of a Decrease in the Marginal Cost of Children on Fertility

of the left-hand side of equation (28) for each value of k . Second, an increase in w_f raises consumption. This reduces the marginal cost of a child since $U_1(c)$ drops, and hence, so does the cost of taking resources away from consumption. To understand what will happen to the left-hand side of equation (28) for each value of k , consider the case of logarithmic utility, $U(c) = \ln(c)$. The term $U_1(c)c$ is constant and so will be the right-hand side of equation (28). In this case, fertility decreases from k^* to k^{**} , as represented in figure 19. If the utility index U is such that $U_1(c)c$ is increasing, the right-hand side of equation (28) rises and fertility further decreases to k^{***} as represented in figure 19. Therefore, when $U_1(c)c$ is non-decreasing, fertility falls as w_f increases, as stated in part B(i) of proposition 8. When

$U_1(c)c$ is decreasing, however, the reduction in the marginal utility of consumption may be strong enough to offset a part, or all, of the rising opportunity cost of time for the wife. In this case (which is not represented in figure 19), the final value of fertility would be greater than k^{**} . Its position relative to the initial fertility, k^* , is ambiguous, however, without additional restrictions on the shape of preferences.

The result in proposition 8 deserves a few additional remarks. First, the shape of the utility index V is immaterial to the conclusion. Note, however, that it is relevant for the magnitude of the effect of productivity on fertility. Second, the term $U_1(c)c$ is informative because it measures the pace at which the marginal utility of consumption, which is a component of the cost of children,

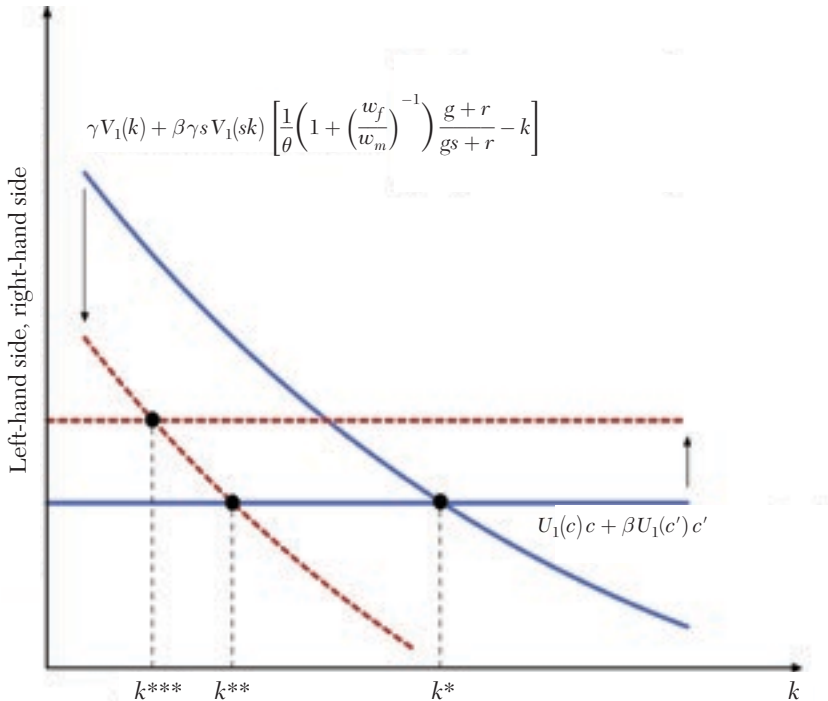


Figure 19. The Effect of Women's Productivity on Fertility

decreases relative to the rise in consumption induced by productivity increases. A decreasing $U_1(c)c$ means that the marginal utility of consumption decreases “fast” as consumption increases, implying the marginal cost of a child may eventually decrease. An increasing $U_1(c)c$ has the opposite interpretation.

REMARK 2: (Gender wage gap and fertility). *Proposition 8 can be used to deduce the effect of changes in the ratio w_f/w_m , which in the context of this model measures the gender gap in earnings. Specifically, proposition 8 is equivalent to studying changes in the gender gap, w_f/w_m , holding w_m fixed.*

PROPOSITION 9: (Earnings and fertility).

- A. *Children cost goods* ($\theta = 0, e > 0$)—A proportional increase in w_m and w_f causes fertility, k , to increase.
- B. *Children cost time* ($\theta > 0, e = 0$)—A proportional increase in w_m and w_f causes fertility, k , to
 - (i) remain constant if $U_1(c)c$ is constant (income and substitution effects cancel out);
 - (ii) increase if $U_1(c)c$ is decreasing (dominating income effect);

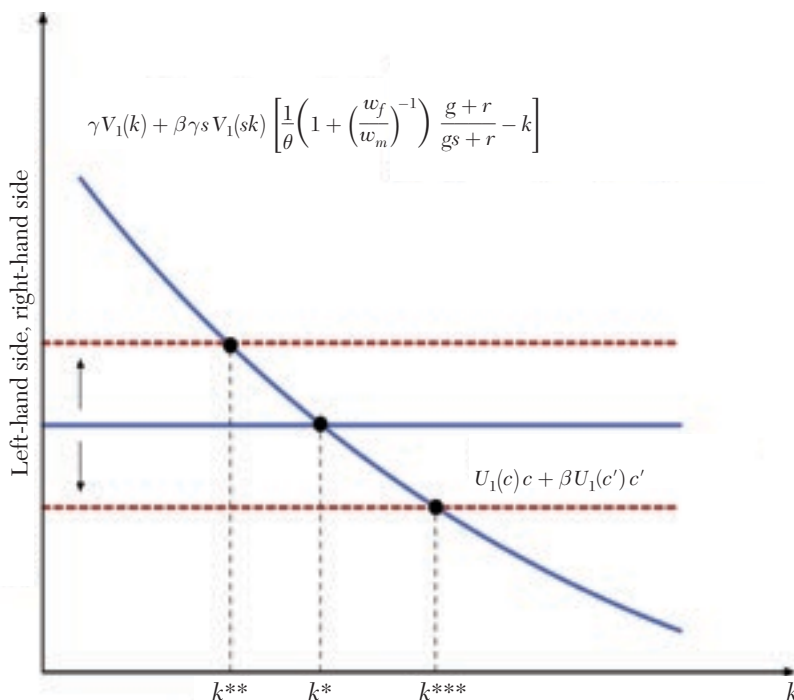


Figure 20. The Effect of a Proportional Increase in the Productivity of Men and Women on Fertility

- (iii) *decrease if $U_1(c)c$ is increasing (dominating substitution effect).*

Proposition 9 describes the effect of proportional growth in w_m and w_f . Think, for example, of two generations: the first one faces low values for w_m and w_f while the second one faces high values because w_m and w_f grow. Note that the two generations face the same growth rate, g . Part A of proposition 9 follows from an inspection of equation (27): a proportional increase in w_m and w_f raises consumption, which unambiguously reduces the marginal cost of a child when $\theta = 0$. Part B of the proposition deals with the case when there is a time cost for children. In this case there is both an income and a

substitution effect. Consider equation (28) again. A proportional change in w_m and w_f leaves its left-hand side unchanged. Thus, the result depends only upon the property of the right-hand side of the equation as consumption rises. Three cases are discussed and illustrated by figure 20. In the case of a logarithmic utility index U , the term $U_1(c)c$ is constant. Fertility remains constant at k^* when both w_m and w_f increase proportionately. This is the well-known case where the income and substitution effects cancel each other out. If $U_1(c)c$ is increasing, the substitution effect dominates and fertility decreases. If $U_1(c)c$ is decreasing, the opposite result prevails.

PROPOSITION 10: (Household technology and fertility). *A decrease in the time cost of raising a child, θ , and/or in the good cost, e , increases fertility.*

The result in proposition 10 derives directly from an inspection of equation (27), since a decrease in θ and/or e leads to a decrease in the marginal cost of raising a child. The effect of such change is represented graphically in figure 18.

6.3 The Effect of Infant and Child Mortality

To discuss the effects of a change in child mortality, that is, a change in the survival parameter, s , return to equation (27). Note that the survival probability s affects both the left- and the right-hand sides of the equation. On the right-hand side of equation (27), an increase in s raises the expected cost of children. On the left-hand side, the marginal (lifetime) utility of children depends upon the survival probability via two channels. Focus on the term $sV_1(sk)$. Again, hold k fixed at any arbitrary level. On the one hand, an increase in s raises the marginal (lifetime) utility of children because it raises the expected flow of utils derived from children. On the other hand, an increase in s implies that more children are present in the household in the second period. This reduces the (period) marginal utility, $V_1(sk)$, and, therefore, reduces the marginal lifetime utility of children. Which effect dominates depends upon the specific functional form chosen for V .

The following proposition gives a condition under which the effect of s on the number of children born, and the number of surviving children, is unambiguous.

PROPOSITION 11: (Effect of child mortality). *When $sV_1(sk)$ is nonincreasing in s , an increase in the survival probability, s , reduces fertility, k , so that $dk/ds < 0$, and*

raises the number of surviving children, implying $d(sk)/ds > 0$.

The proof of this result is given in Appendix A.6. The condition that $sV_1(sk)$ is nonincreasing in s implies that $V_1(sk) + skV_{11}(sk) \leq 0$. It is therefore a condition on the curvature of the utility function V . Under this condition, the effect of an increase in s can be represented, graphically, by an upward shift of the increasing (marginal cost) curve in figure 17, and a downward shift of the decreasing (marginal benefit) curve. Note that in the case of logarithmic utility, $V(k) = \ln(k)$, the function $sV_1(sk)$ is a constant. The two effects of s on the marginal utility of a child exactly offset each other. In this case the sole effect of an increase in s is to raise the marginal cost of a child. Proposition 11 also states that the number of surviving children, sk , increases with the survival rate, s . That is, when the survival rate, s , increases by 1 percent, fertility, k , decreases by less than 1 percent. As a result the number of surviving children, sk , increases.

When $sV_1(sk)$ is increasing in s , the marginal (lifetime) utility of children is increasing in their survival. Graphically, the effect of an increase in s would be represented by upward shifts of both curves in figure 17, so the effect on k is ambiguous without further specifying functional forms and parameters. Examples 1 and 2 below use common forms for the utility function that are found in the literature.

EXAMPLE 1: (CRRA preferences for fertility). *Let $V(k) = k^{1-\rho}/(1-\rho)$ with $\rho \geq 1$. Then $sV_1(sk) = s^{1-\rho}k^{-\rho}$ is nonincreasing in s . An increase in s reduces fertility and increases the number of surviving children.*

EXAMPLE 2: (Logarithmic preferences for fertility with a minimum number of children). *Let $V(k) = \ln(k - \mathfrak{k})$ where $\mathfrak{k} > 0$ is*

a minimum requirement for the number of children. Then $sV_1(sk) = 1/(k - \mathfrak{f}/s)$ is decreasing in s . An increase in s reduces fertility and increases the number of surviving children.

6.4 Quality and Human Capital

6.4.1 The Quality–Quantity Tradeoff

To simplify the discussion, consider a static version of the model; that is, let $\beta = 0$. Preferences are represented by the utility function

$$(29) \quad U(c) + \gamma V(k) + \eta H(q).$$

To specify the cost function, suppose that a child requires θ units of time and that he can be endowed with quality q through goods spending, e . Specifically, imagine that there is a production function

$$q = Q(e),$$

describing the relationship between expenditure per child and the quality of children. Assume that the household has a total of one unit of time and that the market wage is w . That is, for simplicity, abstract from the distinction between husband and wife. The budget constraint now reads

$$(30) \quad c + k(e + \theta w) = w.$$

The first-order conditions for fertility, k , and expenditures, e , are

$$(31) \quad \gamma V_1(k) = U_1(c)(e + \theta w)$$

and

$$(32) \quad \eta J_1(e) = U_1(c)k,$$

where $J(e) \equiv H(Q(e))$. The right-hand side of equation (31) is the marginal cost of a child. It depends upon the marginal utility of

consumption, $U_1(c)$, as well as the wage rate, w , and the level of expenditures on quality, e . This is because (1) the higher the marginal utility of consumption, the more costly it is to use resources for childrearing instead of consumption; (2) the higher the wage rate, the higher the opportunity cost of time spent with a child; and (3) the higher the expenditures per child, the more costly the marginal child. The right-hand side of equation (32) is the marginal cost of goods spending on quality. Imagine increasing expenditure on child quality by one unit. This will cost k units of consumption because the household has k kids. To convert this into a cost in utility terms, multiply this by the marginal utility of consumption, to capture the loss from cutting consumption by k units. Note that the marginal cost of investing in child quality is increasing in the number of children.

To discuss the effect of productivity on fertility, it is useful to define the elasticity of substitution between fertility and consumption, σ_{ck} , and between fertility and expenditures on quality, σ_{ek} :

$$\sigma_{ck} \equiv \frac{d \ln(c/k)}{d \ln(V_1(k)/U_1(c))}$$

and

$$\sigma_{ek} \equiv \frac{d \ln(e/k)}{d \ln(V_1(k)/J_1(e))},$$

where $\sigma_{ck}, \sigma_{ek} > 0$. The elasticity σ_{ck} shows how the c/k ratio changes along an indifference curve as the relative price of consumption and fertility changes. The latter, at an optimum, is equal to the marginal rate of substitution; hence, the term $d \ln(V_1(k)/U_1(c))$ in the denominator for σ_{ck} . Similarly, the elasticity σ_{ek} shows how the e/k ratio changes along an indifference curve as the relative price of expenditures and fertility changes. Note, here, that the marginal rate of substitution between

expenditures and fertility, $V_1(k)/J_1(e)$, depends on the function J , which reflects a composition of the utility and production functions for child quality. As the price of, say, fertility relative to consumption rises—that is, as the marginal rate of substitution $V_1(k)/U_1(c)$ rises—the optimal c/k ratio rises in a proportion given by σ_{ck} . A high elasticity of substitution means, therefore, a high willingness to substitute away from the relatively more expensive commodity. Thus, these elasticities control how the household will adjust its spending on c , k , and e as wages, w , change. These elasticities play an important role in the proposition below. Appendix A.7 derives their exact formulas.

PROPOSITION 12: (Productivity and the quality–quantity trade-off). *The effect of productivity, w , on fertility, k , and expenditure on child quality (in intensive form), e/w , is described by*

$$(33) \quad \frac{d \ln(k)}{d \ln(w)} = \frac{1}{\Delta} \left(\frac{c}{w-c} \right) \left(\frac{1}{\sigma_{ck}} - 1 \right) \left(\frac{1}{\sigma_{ek}} - \frac{ek}{w-c} \right) + \frac{1}{\Delta} \left(\frac{ek}{w-c} \right) \left(\frac{1}{\sigma_{ek}} - 1 \right) \left(\frac{1}{\sigma_{ck}} + \frac{c}{w-c} \right),$$

and

$$(34) \quad \frac{d \ln(e/w)}{d \ln(w)} = -\Delta^{-1} (1/\sigma_{ek} - 1) \times \left(1/\sigma_{ck} - 1 + \frac{1}{1-c/w} \right),$$

where $\Delta > 0$ is defined in Appendix A.8.

The proof of this result is given in Appendix A.8.

Case 1: $\sigma_{ek} = \sigma_{ck} = 1$. It follows from equation (33) that $d \ln(k)/d \ln(w) = 0$; i.e., fertility remains constant when productivity increases. This is the well-known case where income and substitution effects cancel out. Specifically, an increase in productivity makes the time spent taking care of a child relatively more expensive. This is a force toward a reduction in fertility. But, there is also an income effect: the household can afford more children. This is a force toward an increase in fertility. In the case at hand, these two effects exactly offset each other. It is obvious from equation (34) that e/w remains constant. This implies that expenditure per child, e , rises in lockstep with wages.

Case 2: $\sigma_{ek} > 1$ and $\sigma_{ck} = 1$. Equation (33) implies $d \ln(k)/d \ln(w) < 0$; i.e., fertility decreases when productivity increases. Here, fertility declines because it is more substitutable with expenditures than in case 1. Equation (34) reveals that e/w rises. Thus, expenditure per child, e , moves up at a faster clip than wages, w . Thus, due to the high degree of substitutability between k and e , the household reduces the quantity of children in favor of higher quality per child, as productivity rises and expenditures on quality become relatively cheaper. (Equation (A16) in Appendix A.8 establishes that c/w remains constant.)

Case 3: $\sigma_{ek} = 1$ and $\sigma_{ck} > 1$. Equation (33) implies $d \ln(k)/d \ln(w) < 0$; i.e., fertility decreases when productivity increases. In this case, the fertility decline occurs for a different reason than in the previous case: fertility declines because it is more substitutable with consumption than in case 1. The household substitutes the quantity of children for more consumption as productivity rises and expenditures on consumption become relatively cheaper. As can be seen from equation (34), e/w remains constant. Therefore, it is obvious from the

budget constraint that c/w must move up with productivity—the doubter can check equation (A16) in Appendix A.8. So, the high substitutability of the quantity of children with consumption induces the household to increase its spending on consumption at a faster rate than the increase in productivity, while maintaining the budget share of quality spending constant.

So, there are at least two distinct mechanisms through which increases in productivity lead to decreases in fertility: the substitution of fertility for quality as in case 2, or the substitution for consumption as in case 3. Note that the two mechanisms are not observationally equivalent. The behavior of c/w and e/w can help one tell, using data, which case is at hand.

It is important to note that σ_{ek} , the elasticity of substitution between the quantity of children and their average quality, depends upon the properties of the function $J(e)$ (see Appendix A.7), which is identical to $H(Q(e))$. This function combines, therefore, characteristics of preferences, via H , as well as technology, via Q . The examples below use some common forms for tastes and technology that are found in the literature.

EXAMPLE 3: (CRRA preferences for consumption, logarithmic preferences over quantity and quality, and an isoelastic production technology). Let $U(c) = c^{1-\rho}/(1-\rho)$, with $0 < \rho < 1$, $V(k) = \ln(k)$, $H(q) = \ln(q)$, and $Q(e) = \chi_0 e^{\chi_1}$. Then, $\sigma_{ck} > 1$ and $\sigma_{ek} = 1$ —see the elasticity formulas in Appendix A.7. It is immediate that case 3 will transpire. Fertility will fall with wages while expenditure per child will increase in proportion with them.

EXAMPLE 4: (Logarithmic preferences for consumption and quantity, CRRA preferences over quality, and an isoelastic production technology). Let $U(c) = \ln(c)$,

$V(k) = \ln(k)$, $H(q) = q^{1-\rho}/(1-\rho)$, with $0 < \rho < 1$, and $Q(e) = \chi_0 e^{\chi_1}$. Then, $\sigma_{ck} = 1$ and $\sigma_{ek} > 1$ —see Appendix A.7. Here case 2 will obtain. Again, fertility will fall with wages, while expenditure per child will increase faster than them.

6.4.2 Human Capital (as Child Quality)

In the previous setup, parents derive utility from the quality of their children. It is straightforward to turn this into a model where they care about their offspring's earnings. To achieve this, interpret $Q(e)$ as human capital that can be rented on the labor market in exchange for a wage. Suppose that children leave their parents' home and become adults after one period. Let the parents' preferences be represented by a slightly modified version of equation (29):

$$U(c) + \gamma V(k) + \eta H(gwQ(e)).$$

The term $gwQ(e)$ represents the future earnings of a child supplying human capital, $Q(e)$, at the wage rate per efficiency unit of labor prevailing next period, gw . The human capital of each parent is normalized to 1. Thus, in this setup, parents' preferences are defined over the future earnings of their children, which they can influence via investment in education. The budget constraint and first-order condition for k are as described by equations (30) and (31), respectively. The first-order condition for e now reads

$$(35) \quad \eta H_1(gwQ(e)) gw Q_1(e) = U_1(c)k.$$

A difference between the quality–quantity model of the previous section and the human capital model can be seen by comparing the first-order conditions (32) and (35). In equation (32), a change in the wage, w , affects the optimal decision because it changes the marginal cost of expenditures on quality, on the right-hand side, via a change in the marginal utility of consumption. This effect

is also present in the human capital model. But, there is another effect: a change in productivity also changes the marginal benefit of expenditures, on the left-hand side of equation (35), because it changes the future earnings of a child.

Note that the growth rate of productivity, g , also matters for the determination of fertility. This is another difference with the quality–quantity model of the previous section. A change in the growth rate of productivity has two opposing effects. On the one hand, an increase in g lowers the marginal benefit of a child's future earnings because, all else equal, it raises these earnings. This is represented by the term $\eta H_1(gwQ(e))$ on the left-hand side of equation (35). On the other hand, an increase in g raises the marginal benefit of investing in the quality of the child since each unit of human capital gets paid at a higher price. This is represented by the term $gwQ_1(e)$ on the left-hand side of equation (35). Which one of these two effects dominates depends on the shape of preferences. The following result is established in Appendix A.9.

PROPOSITION 13: (The effect of productivity growth on fertility and human capital). *Assume that $U(c) = \ln(c)$, $V(k) = \ln(k)$, and $H(q) = q^{1-\rho}/(1-\rho)$, with $\rho > 0$. An increase in the growth rate of productivity, g , causes fertility, k , to*

- (i) *decrease if $\rho \in (0, 1)$;*
- (ii) *remain constant if $\rho = 1$;*
- (iii) *increase if $\rho > 1$.*

Human capital, $Q(e)$, moves oppositely with fertility, k .

6.5 Discussion

The modern approach to fertility in macroeconomics stems from the work of Razin

and Ben-Zion (1975), Barro and Becker (1989), and Becker and Barro (1988). These authors are among the first to model fertility decisions in versions of the optimal growth model, which has proven useful in the analysis of growth-related phenomena.

Economists have argued that the demographic transition may be, among other things, a response of fertility to rising income and/or productivity. Galor and Weil (1996), for instance, propose a model where the accumulation of physical capital leads to an increase in women's relative wages because capital is more complementary to women's input than to men's. This increase in the relative wages of women raises the cost of childbearing and, thus, leads to a reduction in fertility. In the language of the model developed above, the mechanism proposed by Galor and Weil (1996) works as described in proposition 8B(i). Galor and Weil (2000) present a theory of the long-run relationship between population growth, technology, and economic growth. In their theoretical model, an economy can stagnate, then enter a regime where both population growth and technological progress increase and, finally, experience a demographic transition jointly with sustained economic growth. Galor (2012) presents a similar, simplified model. Galor (2005, 2011) presents a unified view of the transition from stagnation to growth that, in the data, coincided with the demographic transition.

Galor (2005, 2011) also insists on the importance of human capital and education to understand the demographic transition. A key aspect of his analysis is that the production of human capital depends on expenditures, as well as on the future growth rate of productivity. In section 6.4, this was modeled by letting the quality of a child be measured directly by his future earnings: $gwQ(e)$. As shown in proposition 13, this mechanism can lead to fewer children being born, but now they are of higher quality. On the quality–quantity

trade-off, Cavalcanti, Kocharkov, and Santos (2017) observe that the total fertility rate in Kenya for 2008 was 4.6. Out of this, unwanted children corresponded to 1.2 kids. They analyze the impact of lowering the cost of contraception (including abortion). They find that this can lead to a significant increase in per capita GDP, because it results in fewer children of higher quality.

In Greenwood and Seshadri (2002) and Greenwood, Seshadri, and Vandenbroucke (2005), the driving force behind the secular decline in fertility is rising productivity. Both of the models are fit to very long-run US fertility data. In the former, an education decision is modeled so that the analysis is consistent with the rise in education and the reallocation of labor from agriculture to industry. In the latter, the rise in productivity generates an increase in the opportunity cost of raising children, following the logic explained in proposition 9 of section 6.2.

The decline in infant and child mortality has also been discussed as a potential factor explaining the demographic transition. Many authors have adopted versions of the perfect foresight model presented in here, where there is no uncertainty over the number of surviving children. Eckstein, Mira, and Wolpin (1999) is an example. They posit a period utility function for children of the form presented in example 2, that is $V(k) = \ln(k - \mathfrak{k})$, where $\mathfrak{k} > 0$ acts as a minimum number of children needed in the household. They estimate their model using Swedish time series data, and find that the reduction in infant and child mortality rates is the most important factor driving the fertility decline. Productivity also reduced fertility, but the reduction in adult mortality had no effect by itself. Reductions in infant and child mortality rates, alone, cannot explain the decline in the number of *surviving* children, as shown by proposition 11.

Bar and Leukhina (2010) propose a model of the transition from stagnation to growth,

along the lines of Barro and Becker (1989) and Hansen and Prescott (2002). The mechanism through which infant and child mortality reduces fertility, in their model, is similar to that described in proposition 11. Bar and Leukhina (2010) use their model to assess the contribution of productivity growth versus that of declining mortality in explaining the time series of fertility in England between 1650 and 1950. They find that changes in young-age mortality account for close to 60 percent of the decline in fertility.

Boldrin and Jones (2002) present a version of the Barro and Becker model where children care about their parents' old-age consumption. In this model, fertility is an investment into future consumption and a decrease in infant mortality reduces fertility. Doepke (2005) also presents versions of the Barro and Becker model to analyze the effect of infant and child mortality. He distinguishes two classes of models, one where there is no uncertainty over the number of surviving children and another where there is. In the model developed in this section, there is no uncertainty. Theoretical results for models with uncertainty on the number of surviving children are also presented in Sah (1991) and Kalemli-Ozcan (2003). In a quantitative exercise Doepke (2005) finds that calibrated versions of his models yield similar implications: they all predict a decline in fertility rates as infant mortality declines, but none predicts a decline in the number of surviving children as seen in the data. These are the results of proposition 11.

Besides technological progress, the demand for human capital, and mortality, other factors have been contemplated in the economic literature discussing the demographic transition. Doepke (2004) analyzes differences in the timing and pace of the demographic transition across countries (see figure 15), and ascribes them to differences in child labor laws. Boldrin, de Nardi, and Jones (2015) argue that government-provided

old-age pensions are strongly associated with a low fertility.

Becker and Barro (1988) ascribe the baby boom to a catch-up effect associated with low fertility during World War II. Greenwood, Seshadri, and Vandenbroucke (2005) present a list of arguments to dispute the notion that the baby boom was a response to World War II. They propose a theory where the main impetus for the baby boom comes from improvements in household productivity that made it possible for women to spend less time raising children. They argue that such improvements were made available by the diffusion of electricity and associated appliances, the introduction of new goods such as frozen foods, the growing availability of running water and gas heaters, etc., as well as the rationalization of the home following scientific management principles discovered on factory floors. In the framework of the model presented here, their theory can be viewed as technological advance causing a decrease in θ , the time cost of raising a child. Suppose, as they do, that $U_1(c)c$ is increasing. Then, in the long run, proportional increases to w_m and w_f lead to a downward trend in fertility. When a new technology is introduced to save time in the household in the form of a lower θ , the right-hand side of equation (28) increases, leading to an increase in fertility—see proposition 10 above. As time passes, the continued growth of productivity eventually takes over, and fertility resumes its declining trend: the baby boom is over.

Albanesi and Olivetti (2014) also propose an explanation of the baby boom based on technological improvements. They document that there has been a remarkable decline in maternal mortality during the twentieth century: in 1900, the maternal mortality rate was above eighty deaths (per 10,000 live births), while it was below ten by 1950. Thus, most of the gain occurred during the first half of the century, namely, during the 1930s. They conduct an empirical

analysis to assess the effect of maternal mortality on fertility. They find that a decline in maternal mortality of ten deaths (per 10,000 live births) yields an increase in completed fertility of 0.27 children per married woman. One can again associate improvements in medical technology with a decline in θ , in the context of the model presented here. The idea behind this association is that θ may be composed of both the time spent recovering from the delivery of the child and the time spent taking care of the child. Improvements in medical technology reduce the time spent recovering from the delivery and therefore reduce the cost of a child.

Another attempt at explaining the baby boom using modern macroeconomic tools is by Doepke, Hazan, and Maoz (2015). Their theory links the baby boom to the job market conditions faced by young women immediately after World War II. These conditions were worse than usual. This was because older women, who started to work during the war and gained experience, retained these jobs after the war. This increased labor-force participation of older women lowered market wages for women relative to men and induced younger women to stay home and raise children. Doepke, Hazan, and Maoz (2015) present evidence of the decline in the relative wage of women and the increased labor-force participation of older women after the war. They build an equilibrium model of fertility and labor-force participation, which they calibrate to US data. In their main experiment, they find that their model accounts for 80 percent of the increase in fertility during the baby boom as a consequence of World War II, which they model as a shock to government spending, and male and female labor-force participation. In the framework of the model presented here, equation (28) shows that a decline in w_f/w_m raises fertility. This, in a nutshell, summarizes part of the argument in Doepke, Hazan, and Maoz (2015): the

poor labor-market conditions for women following World War II led to an increase in fertility.

Zhao (2014) associates the baby boom with changes in the marginal tax rate induced by World War II. Specifically, Zhao (2014) shows that the marginal income tax rate for an average American increased from 4 to 25 percent during the war. He proposes a model where children cost time and where, using the language of the simple model described above, preferences are such that $U_1(c)c$ is increasing. Thus, an increase in the marginal tax rate reduces the opportunity cost of time and the marginal cost of a child, as per the result of proposition 9. On the basis of a computational experiment, Zhao (2014) concludes that this change in tax policy was an important cause of the baby boom.

Manuelli and Seshadri (2009) document, and seek to explain, the negative correlation between fertility and income across countries. They build a model of fertility choice and investment in both a child's human and health capitals. In a quantitative exercise, they find that differences in productivity across countries account for a large fraction of cross-country differences in fertility. Jones, Schoonbroodt, and Tertilt (2011) review a variety of mechanisms that may generate the negative income–fertility correlation. They emphasize that almost all theories rely on the existence of a time cost of raising a child. In the language of the model presented here, $\theta > 0$ is critical. They also emphasize that models that distinguish between husbands and wives need to rely on positive assortative matching of spouses for fertility to be declining in the husband's income. Positive assortative mating is discussed in section 3.2. Finally, they consider models with endogenous investment in child quality. They find that modeling the choice of a child's quality does not, per se, lead to a negative income–fertility correlation—in line with proposition 12.

Caldwell (2004) documents how fertility typically collapses during periods of war and social unrest. Becker and Barro (1988) argue that the increase in military spending in the United States during World War II reduced wealth and, therefore, lowered fertility. Similarly, Vandenbroucke (2014) presents a quantitative analysis where the outbreak of a war is akin to an expected income shock; i.e., the expected income of the household decreases when the husband is more likely to die. To the extent that children are normal goods, the model then predicts that fertility should decline during periods of war.

7. Quantitative Theory

This section presents a simple example illustrating how macroeconomic models of family-related decisions can be used to answer quantitative questions. The question at hand regards the baby boom in the United States. Specifically, how did technological innovation in the household sector contribute to the baby boom?

7.1 A Structural Model of the Baby Boom

A one-period version of the model presented in the section on fertility is laid out. This amounts to setting $\beta = 0$ in equation (21). There is only one adult in each family, so the model abstracts from differences between men and women. Preferences are represented by

$$\mathbf{U}(c, k, q) = \phi \frac{c^{1-\eta}}{1-\eta} + (1-\phi)k,$$

implying that there is no quality dimension to children. The cost of children is specified by

$$\mathbf{C}(k, q; w) = \theta wk,$$

so that only the time spent on children matters. There are two technologies available to help a family raise children. The

technologies are differentiated by their time cost, θ , and their price, p . The first technology (old) has a high time cost, θ_{high} , while the second technology (new) has a low cost, θ_{low} , where $\theta_{low} < \theta_{high}$. The new technology has a price of $p = e$, while the old one is free so that $p = 0$. Thus, technology is represented by the bundle (θ, p) .

The decision problem for a family who has chosen technology (θ, p) is

$$V(w, \theta, p) = \max_{c, k} \left\{ \phi \frac{c^{1-\eta}}{1-\eta} + (1-\phi)k \right\},$$

subject to

$$c + \theta wk = w - p.$$

The technology adoption decision for a family is given by

$$(\theta, p) = \begin{cases} (\theta_{high}, 0) \text{—old} & \text{if } V(w, \theta_{high}, 0) > V(w, \theta_{low}, e), \\ (\theta_{low}, e) \text{—new} & \text{if } V(w, \theta_{high}, 0) < V(w, \theta_{low}, e). \end{cases}$$

One might expect that a high-wage family will be more inclined to adopt the second technology. To address this, in what follows, families will be differentiated by their wage rate, w .

7.2 Empirical Estimation

The model will now be matched with the US data. Depending on the question at hand, researchers may use cross-sectional data, time-series data, or a combination of both. Models can be matched to various types of stylized facts. These facts may be key moments in the data, such as means, standard deviations, and correlations. Models may also be matched with the results of regression equations, such as the estimated coefficients on certain variables. The matching may be done informally (the eyeball metric),

or formally using various distance estimators, such as minimum distance or simulated method of moments. Often when regression coefficients are matched in the analysis, an indirect inference procedure is incorporated into the estimation.

For the purpose of this illustration, the model will be matched to data from cross sections of US counties: data at the household level is not available. First, the model must be readied to generate within- and across-county heterogeneity. Let $\ln(w_{ij}) \sim N(w_j, \sigma_A)$ denote the distribution of wages for individual i in county j , and let w_j be distributed in line with $\ln(w_j) \sim N(0, \sigma_B)$. Second, two periods must be considered. The first period will correspond to the lowest total fertility rate in the United States before the baby boom, the mid-1930s. The second period will correspond to the peak of the baby boom, the late 1950s. To be precise, the total fertility rate of women born in 1932 (whose average child was born in 1959) was 28 percent higher than that of women born in 1907 (whose average child was born in 1934). This increase in fertility will be targeted in the empirical estimation. The differences between the first and second period will be engineered by assuming that all wages grow by 2 percent per year and that the cost of the new technology, e , decreases by 2 percent per year. There are twenty-five years between the two periods. Assume that $\theta_{low} = 0.5 \times \theta_{high}$. The parameters of the model, $\omega \equiv (\eta, \phi, e, \theta_{high}, \sigma_A, \sigma_B)$, are estimated via the following, unweighted, minimum-distance estimation:

$$\min_{\omega} M(\omega)'M(\omega)$$

where $M(\omega)$ is a 6×1 vector of the *differences* between the moments implied by the model and their empirical counterparts. An indirect inference procedure is incorporated

TABLE 9
TARGETED MOMENTS: DATA AND MODEL

Moments	Model	Data
Variance of county income	0.17	0.17
Variance of household income	0.37	0.35
Adoption, 1940	0.38	0.35
TFR ratio	1.25	1.28
Coefficient, fertility on adoption	−0.32	−0.35
Coefficient, Δ fertility on Δ adoption	−0.17	−0.17

into the estimation. Specifically, the moments used are the following:

- (i) The variance of log income across US counties. This statistic is obtained from Haines (2004). The earliest year in his dataset is 1950, and the figure for the variance of log income across counties in that year is 0.17.
- (ii) The variance of log income across US households. The value 0.35 is obtained from US census data for 1940.
- (iii) The fraction of households that had adopted a modern technology in 1940. The value of this statistic is 0.35 and is taken from Bailey and Collins (2011). This measure is built from census data reporting the proportion of housing units equipped with refrigerators, washing machines, electric stoves, and electric services.
- (iv) The ratio of the total fertility rate of the late 1950s to that of the mid 1930s. The value of this statistic is 1.28.
- (v) The coefficient from a regression of fertility on adoption, conditional on income, in the first period using county-level data. The value of this

statistic is −0.35 and is estimated in Baily and Collins (2011).

- (vi) The coefficient from a regression of the change in fertility on the change in adoption, conditional on the change in income, using county-level data. The value of this statistic is −0.17, and again the source is Bailey and Collins (2011).

7.2.1 *Results*

The estimated parameters are

$$\eta = 0.10, \phi = 0.49, e = 1.60,$$

$$\theta_{high} = 15.62, \sigma_A = 0.66, \sigma_B = 0.51.$$

The fit to the targeted moments is reported in table 9. The model is able to replicate the observed baby boom that took place between the mid 1930s and the late 1950s. The engine behind the baby boom is technological progress in the household sector, which implied a drop in the price of the time-saving technology. The adoption of the time-saving technology promotes fertility—recall proposition 10. The baby boom occurred despite the fact that at the same time wages rose, which would have operated to curtail fertility—proposition 9B(iii).

7.2.2 Koopmans (1947)

At first glance, the above regression coefficients would appear to contradict the result in proposition 10, that a fall in the time cost of raising children will lead to a rise in fertility. This may lead an empirical researcher to reject the model. Yet, the model mimics these results almost perfectly. Koopmans (1947) noted some time ago that care must be exercised when interpreting the coefficients on nonstructural regression equations. First, the theory outlined is specified at the level of the household, while the regressions are run using county-level data. Second, adoption, fertility, and income are all endogenous variables. These factors do not invalidate the use of the above regression coefficients in the indirect inference exercise. They affect their interpretation, though. Recall that households differ by their wage rates. High-wage households will have lower fertility. Now, high-wage households are more likely to adopt the expensive new technology than are low-wage ones. So, if high-wage households adopt the new technology first, then a regression of fertility on adoption at the county-wide level may show that technology adoption is associated with a drop in fertility; i.e., even though the fertility rate of those who have adopted the new technology may have risen, their rate will still be lower than for those who did not adopt the technology.

7.3 Discussion

The model of the baby boom estimated here is a simplified version of the Greenwood, Seshadri, and Vandenbroucke (2005) model that is presented in Greenwood, Seshadri, and Vandenbroucke (2015). For more detail on the estimation procedure, see the latter paper. This type of estimation procedure is common in modern family economics. For example, Greenwood et al. (2016) use minimum-distance estimation to match a dynamic stochastic model of

marriage, divorce, educational attainment, and married female labor-force participation to the post-World War II US data. Eckstein, Keane, and Lifshitz (2016) do a similar exercise using simulated method of moments.

Bailey and Collins (2011) misconstrue, for the reasons discussed earlier, the regression coefficients presented above as rejecting the Greenwood, Seshadri, and Vandenbroucke (2005) hypothesis. As is shown, their regressions results can easily be appended to an estimation of the Greenwood, Seshadri, and Vandenbroucke (2015) model by using an indirect inference procedure; the model is fully capable of matching them in addition to other stylized facts. This is an example of how care must be taken when interpreting nonstructural regression equations. For an introduction to indirect inference, see Smith (2008).

Examples of papers (previously discussed) in the genre of quantitative theory are numerous. A short and nonexhaustive list covering various topics and quantitative methods is as follows: Benhabib, Rogerson, and Wright (1991) study household production; Greenwood, Seshadri, and Yorukoglu (2005) research married female labor-force participation; Albanesi and Olivetti (2016) investigate the decisions of women to work and bear children; Regalia and Rios-Rull (2001) examine the rise in the number of single mothers; Fernandez and Wong (2017) and Voena (2015) probe the marriage market; Fernandez-Villaverde, Greenwood, and Guner (2014) explore premarital sex; Eckstein, Mira, and Wolpin (1999) and Greenwood, Seshadri, and Vandenbroucke (2005) analyze fertility.

8. Conclusions

This review surveys macroeconomic models of married female labor-force participation, marriage and divorce, social change,

and fertility. The modern macroeconomic literature dealing with family-related decisions can be read through the lenses of these models. This literature has not answered all questions, but much progress has been made, thanks to the use of economic modeling, in explaining (1) the huge rise in married female labor-force participation; (2) the large drop in marriage; (3) the uptick in positive assortative mating; (4) the issue of children living in poverty with a lone parent; and (5) baby booms and busts.

An example is provided illustrating how macroeconomic models of the family can be matched with stylized facts from the data using minimum-distance estimation. Theories must be quantified before they can be used for any serious policy analysis. The set of moments to be targeted can include facts from nonstructural regression analysis. So, nonstructural regression may be used as an input into quantitative theory. Sometimes care must be taken when interpreting the results from nonstructural regressions. The connection between economic theory and nonstructural regressions may at times prove to be elusive.

To date, a general equilibrium model of fertility, marriage and divorce, and married female labor-force participation has not been developed. It seems likely that the secular decline in fertility is connected with the rise in married female labor-force participation. Matching these long-run facts, in addition to the cross-sectional facts on female labor-force participation and fertility, would be an important thing to do. The development of such a macroeconomic model is essential for understanding a host of policy questions surrounding the family.

Some sample policy questions include: first, should child care be subsidized? Such policies may help working women manage jobs and families. This is important because disruptions to working women's careers, due to child birth, may account for a large

portion of the gender wage gap. Additionally, in a world with declining fertility, where the ratio of the elderly to the young population is high, such policies might be important for publicly financing retirement. This leads to the second question: should fertility be promoted? Third, should the tax system be designed to promote marriage? As was discussed, children growing up in single-parent families appear to do worse in life than those growing up in two-parent families. So, in a world where more and more children are born out of wedlock, promoting marriage could be important. Additionally, the movement toward more single-parent households is connected to the recent rise in income inequality. Should aid be given to single parents? On the one hand, this helps to lift many single-parent households out of poverty and aids children, while on the other hand it might promote an increase in the number of single households with children. This leads to the last question. How much insurance is provided within a family when their members are hit by earnings or health shocks? While traditionally taxes and transfers have been the key tools of social insurance, making female labor supply more flexible as a result of, for example, better child care arrangements, could be an effective policy tool as well. These are difficult questions for which modern macroeconomic models are well suited.

Another avenue for future research is to study further how children's preferences are formed. In particular, young adults' attitudes about leisure, work, and time preference are important. Economic theory predicts that individuals who place a low weight on leisure and who have a low rate of time preference will work more, accumulate more human capital and wealth, and invest more in their health. Thus, these factors could play a role in determining poverty and its intergenerational transmission. Rich parents instill a different set of ideals in their kids than poor

parents. How parents, peer groups, and social institutions, such as schools, affect children's preferences is important to study. As was discussed, research shows that the amount of shame a child feels about engaging in risky behavior has an impact on their actions. So, such channels matter.

The growing availability of data for the developing world also offers directions for future research. Existing models of family-related behavior have, to a large extent, been constructed to understand today's developed countries and their past demographic and economic histories. Can

these models confront the data from developing countries? For example, the decline in fertility in developing countries is much more rapid than what occurred in developed countries, as can be seen from figure 15. Is this due to more rapid growth in income or are these transitions fundamentally different? Similar questions also arise for theories of women's labor-force participation, marriage, divorce, sexual behavior, etc. Can one use models built for developed countries or are new ones needed? An answer to this question is important for guiding development policies around the world.

MATHEMATICAL APPENDIX

A1. Proof of Proposition 1 (Married Female Labor Supply)

Consider an interior solution; i.e., assume that $l_f > 0$. To begin with, the budget constraint can be written as

$$c = w_f \left(\frac{1}{\phi} \mathfrak{I}_m + l_f - qd \right).$$

Substituting the budget constraint and

$$d = \left(\frac{(1 - \kappa)q}{\kappa} \right)^{\frac{1}{\sigma-1}} (2 - \mathfrak{I}_m - l_f) \equiv R(q)(2 - \mathfrak{I}_m - l_f)$$

into the first-order condition for l_f yields

$$\begin{aligned} \frac{\alpha}{\left(\frac{1}{\phi} \mathfrak{I}_m + l_f - R(q)(2 - \mathfrak{I}_m - l_f)q \right)} &= \frac{(1 - \alpha)(1 - \kappa)(2 - \mathfrak{I}_m - l_f)^{\sigma-1}}{[\kappa [R(q)(2 - \mathfrak{I}_m - l_f)]^\sigma + (1 - \kappa)(2 - \mathfrak{I}_m - l_f)^\sigma]} \\ &= \frac{(1 - \alpha)(1 - \kappa)}{[\kappa R(q)^\sigma + (1 - \kappa)](2 - \mathfrak{I}_m - l_f)}, \end{aligned}$$

which can be rearranged to arrive at

$$(A1) \quad \alpha [\kappa R(q)^\sigma + (1 - \kappa)](2 - \mathfrak{I}_m - l_f) = (1 - \alpha)(1 - \kappa) \left(\frac{1}{\phi} \mathfrak{I}_m + l_f - R(q)q(2 - \mathfrak{I}_m - l_f) \right).$$

Using the above formula for d ,

$$\frac{\partial R(q)}{\partial q} = -\frac{1}{1-\sigma} \left(\frac{1-\kappa}{\kappa} \right)^{\frac{1}{\sigma-1}} q^{\frac{2-\sigma}{\sigma-1}} < 0,$$

and

$$\frac{\partial [R(q)q]}{\partial q} = -\frac{\sigma}{1-\sigma} \left(\frac{1-\kappa}{\kappa} \right)^{\frac{1}{\sigma-1}} q^{\frac{1}{\sigma-1}} < 0,$$

since $0 < \sigma < 1$. Hence, $-R(q)q$ is increasing in q . As a result, a lower q is associated with an upward shift in the left-hand side of equation (A1). The right-hand side of equation (A1) is increasing in q . Hence, a lower value of q is connected with a downward shift in the right-hand side of equation (A1). Thus, the female labor supply must go up as a result of a lower q . It is easy to deduce that l_f is increasing in ϕ .

A2. Nash Bargaining Solution

First consider the problem where the household maximizes a weighted sum of utilities,

$$\max_{c, l_m, l_f} \mu [\alpha \ln(c) + (1-\alpha) \ln(1-l_m)] + (1-\mu) [\alpha \ln(c) + (1-\alpha) \ln(1-l_f)],$$

subject to

$$c = w_m l_m + w_f l_f.$$

Focus on the first-order condition for l_f . It is given by

$$(A2) \quad \alpha \frac{w_f}{c} = (1-\alpha)(1-\mu) \frac{1}{1-l_f}.$$

Consider now the Nash bargaining problem with B and G as outside options for the man and woman. It is given by

$$\max_{c, l_m, l_f} \{ [\alpha \ln(c) + (1-\alpha) \ln(1-l_m) - B] \times [\alpha \ln(c) + (1-\alpha) \ln(1-l_f) - G] \}.$$

The first-order condition for l_f is now given by

$$\begin{aligned} & 2\alpha^2 \ln(c) \frac{1}{c} w_f + \alpha(1-\alpha) \frac{1}{c} \ln(1-l_f) w_f + \alpha(1-\alpha) \frac{1}{c} \ln(1-l_m) w_f - \alpha \frac{1}{c} w_f G - \alpha \frac{1}{c} w_f B \\ & = \alpha(1-\alpha) \ln(c) \frac{1}{1-l_f} + (1-\alpha)^2 \ln(1-l_m) \frac{1}{1-l_f} - (1-\alpha) \frac{1}{1-l_f} B, \end{aligned}$$

which can be written as

$$\begin{aligned}
 \text{(A3)} \quad & \frac{\alpha w_f}{c} [2\alpha \ln(c) + (1 - \alpha) \ln(1 - l_f) + (1 - \alpha) \ln(1 - l_m) - B - G] \\
 & = (1 - \alpha) \frac{1}{1 - l_f} [\alpha \ln(c) + (1 - \alpha)(1 - l_m) - B].
 \end{aligned}$$

Let

$$\begin{aligned}
 1 - \mu &= \frac{[\alpha \ln(c) + (1 - \alpha)(1 - l_m) - B]}{[2\alpha \ln(c) + (1 - \alpha) \ln(1 - l_f) + (1 - \alpha) \ln(1 - l_m) - B - G]} \\
 &= \frac{H - B}{(H - B) + (W - G)},
 \end{aligned}$$

where $W \equiv \alpha \ln(c) + (1 - \alpha)(1 - l_f)$ and $H \equiv \alpha \ln(c) + (1 - \alpha)(1 - l_m)$. It is immediate that the first-order conditions in equations (A2) and (A3) are identical. Similar steps can be followed to show that the first-order conditions for l_m are also identical for these two problems when $\mu = (W - B) / [(H - B) + (W - G)]$.

A3. Proof of Lemma 1 (Household Allocations)

To start, define the size variable z as taking the value 1 for a single household and 2 for a married one. Substitute out for $c - \mathfrak{c}$ and n in the objective functions for the single and married households. Then maximize with respect to d and h to obtain the following two first-order conditions (for $z = 1, 2$):

$$\text{(A4)} \quad \frac{\alpha}{c - \mathfrak{c}} wq = (1 - \alpha) z^{-\zeta} [\kappa d^\sigma + (1 - \kappa) h^\sigma]^{\zeta/\sigma - 1} \kappa d^{\sigma - 1},$$

and

$$\text{(A5)} \quad \frac{\alpha}{c - \mathfrak{c}} w = (1 - \alpha) z^{-\zeta} [\kappa d^\sigma + (1 - \kappa) h^\sigma]^{\zeta/\sigma - 1} (1 - \kappa) h^{\sigma - 1}.$$

(The z subscript has been dropped on the endogenous variables to conserve on notation.) Next, divide (A5) into (A4) and simplify to get

$$\text{(A6)} \quad d = h \left[\frac{(1 - \kappa) q}{\kappa} \right]^{1/(\sigma - 1)} \equiv R(q) h.$$

Finally, by substituting (A6) and the household's budget constraint into equation (A5), a single equation in h arises (for $z = 1, 2$):

$$\text{(A7)} \quad \alpha [\kappa R(q)^\sigma + (1 - \kappa)]^{1 - \zeta/\sigma} h^{1 - \zeta} = (1 - \alpha) (1 - \kappa) z^{-\zeta} \left[\left(z - \frac{\mathfrak{c}}{w} \right) - h - q R(q) h \right].$$

PROOF OF THE LEMMA: Rewrite equation (A7) as

$$\begin{aligned} & \alpha [\kappa R(q)^\sigma + (1 - \kappa)]^{1-\zeta/\sigma} h^{-\zeta} h + (1 - \alpha)(1 - \kappa) [1 + qR(q)] z^{-\zeta} h \\ & = (1 - \alpha)(1 - \kappa) z^{-\zeta} (z - \mathbf{c}/w). \end{aligned}$$

Now begin with part (iii). If z increases by the factor $\lambda > 1$, then $z - \mathbf{c}/w$ rises by the factor $\rho \equiv (\lambda z - \mathbf{c}/w)/(z - \mathbf{c}/w) > \lambda$. Now the right-hand side rises by the factor $\lambda^{-\zeta} \rho$. Observe that if h rises by the factor ρ , then the left-hand side will increase by more than the factor $\lambda^{-\zeta} \rho$, because $\rho^{-\zeta} > \lambda^{-\zeta}$ when $\zeta < 0$. Therefore, to restore equality between the left-hand and right-hand sides of the above equation, h must rise by less than the factor ρ . Next, part (ii) is implied by equation (A6) and part (iii). By using parts (ii) and (iii), in conjunction with the household's budget constraint, part (i) arises. Part (iv) follows from part (i) by noting that $(c_2 - \mathbf{c}) \geq 2(c_1 - \mathbf{c})$, since $[(2 - \mathbf{c}/w)/(1 - \mathbf{c}/w)] \geq 2$. Thus, $(c_2 - \mathbf{c})/2 \geq (c_1 - \mathbf{c})$, which implies $c_2 > c_1$. Now, it is feasible for a married household to choose $c_2 = 2c_1$, $d_2 = 2d_1$, and $h_2 = 2h_1$. The fact they didn't implies that $M(w, q) > S(w, q)$. Hence, part (v) holds. ■

A.4 The Rise in Assortative Mating

In the setup used in section 3.2, the married household's first-order conditions are

$$\frac{\alpha}{\epsilon} w q = (1 - \alpha) [\kappa d^\sigma + (1 - \kappa)(\eta_h h)^\sigma]^{-1} \kappa d^{\sigma-1},$$

and

$$(A8) \quad \frac{\alpha}{\epsilon} w \phi_j = (1 - \alpha) [\kappa d^\sigma + (1 - \kappa)(\eta_h h)^\sigma]^{-1} (1 - \kappa) \eta_h^\sigma h^{\sigma-1}.$$

This now implies that

$$(A9) \quad d = \left[\frac{(1 - \kappa) \eta_h^\sigma q}{\kappa \phi_j} \right]^{1/(\sigma-1)} h \equiv R(q) \left(\frac{\phi_j}{\eta_h} \right)^{1/(1-\sigma)} \eta_h h.$$

Finally, by substituting equation (A9) and the household's budget constraint into equation (A8), a single equation in h arises:

$$\begin{aligned} & \alpha \phi_j \left[\kappa R(q)^\sigma \left(\frac{\phi_j}{\eta_h} \right)^{\sigma/(1-\sigma)} + (1 - \kappa) \right] h \\ & = (1 - \alpha)(1 - \kappa) \left[\mu_i + \phi_j - \phi_j h - q R(q) \left(\frac{\phi_j}{\eta_h} \right)^{1/(1-\sigma)} \eta_h h \right]. \end{aligned}$$

This equation can be rearranged to achieve

$$h = \frac{(1 - \alpha)(1 - \kappa)(1 + \phi_j / \mu_i)}{\alpha(\phi_j / \mu_i) \left[\kappa R(q)^\sigma \left(\frac{\phi_j}{\eta_h} \right)^{\sigma/(1-\sigma)} + (1 - \kappa) \right] + (1 - \alpha)(1 - \kappa)(\phi_j / \mu_i) \left[1 + qR(q) \left(\frac{\phi_j}{\eta_h} \right)^{\sigma/(1-\sigma)} \right]}.$$

Housework, h , is decreasing in the ratio of the wife's productivity in the market to her productivity at home, ϕ_j / η_h , and in the ratio of her productivity in the market to her husband's, ϕ_j / μ_i .

A5. Derivation of Equation (28)

Consider equation (27) and divide through by $e(1 + s/r) + \theta w_f(1 + gs/r)$:

$$\left[\gamma V_1(k) + \beta \gamma s V_1(sk) \right] \frac{1}{e(1 + \frac{s}{r}) + \theta w_f(1 + \frac{gs}{r})} = U_1(c).$$

Multiply by lifetime income, which is given by the intertemporal budget constraint

$$c + \frac{c'}{r} = (w_m + w_f) \left(1 + \frac{g}{r} \right) - k \left[e \left(1 + \frac{s}{r} \right) + \theta w_f \left(1 + \frac{gs}{r} \right) \right]$$

to obtain

$$\left[\gamma V_1(k) + \beta \gamma s V_1(sk) \right] \left[\frac{(w_m + w_f) \left(1 + \frac{g}{r} \right)}{e \left(1 + \frac{s}{r} \right) + \theta w_f \left(1 + \frac{gs}{r} \right)} - k \right] = U_1(c) \left[c + \frac{c'}{r} \right].$$

Using the Euler equation, $U_1(c) = \beta r U_1(c')$, to rewrite the right-hand side as $U_1(c)c + \beta U_1(c')c'$, assuming $e = 0$, and rearranging the left-hand side yields

$$\left[\gamma V_1(k) + \beta \gamma s V_1(sk) \right] \left[\frac{1}{\theta} \left(1 + \left(\frac{w_f}{w_m} \right)^{-1} \right) \frac{g + r}{gs + r} - k \right] = U_1(c)c + \beta U_1(c')c'.$$

A6. Proof of Proposition 11

The optimization problem can be written

$$\max_{k, c} \{ F(k, c) = U(c) + \beta U(c') + \gamma V(k) + \gamma \beta V(sk) \},$$

subject to

$$c' = rW - rc - k(rx + x's),$$

where $W \equiv w_m + w_f + (w'_m + w'_f)/r$ is the household's total wealth, and $x \equiv e + \theta w_f$ and $x' \equiv e + \theta w'_f$ represent the current and future cost of a child. Assume that $sV_1(sk)$ is a nonincreasing function of s . This implies

$$V_1(sk) + sk V_{11}(sk) \leq 0.$$

At a maximum for F , the following first- and second-order conditions must hold:

$$F_k, F_c = 0,$$

$$F_{kk}, F_{cc} < 0,$$

$$F_{kk}F_{cc} - F_{kc}^2 \equiv \Delta > 0.$$

These derivatives are

$$F_k = -\beta U_1(c')(rx + sx') + \gamma V_1(k) + \beta \gamma s V_{11}(sk),$$

$$F_c = U_1(c) - \beta r U_1(c'),$$

$$F_{kk} = \beta U_{11}(c')(rx + sx')^2 + \gamma V_{11}(k) + \beta \gamma s^2 V_{11}(sk),$$

$$F_{cc} = U_{11}(c) + \beta r^2 U_{11}(c'),$$

$$F_{kc} = \beta r U_{11}(c')(rx + sx').$$

The discriminant, Δ , is given by

$$\begin{aligned} \Delta = & \left[\beta U_{11}(c')(rx + sx')^2 + \gamma V_{11}(k) + \beta \gamma s^2 V_{11}(sk) \right] \left[U_{11}(c) + \beta r^2 U_{11}(c') \right] \\ & - (\beta r)^2 U_{11}(c')^2 (rx + sx')^2, \end{aligned}$$

which simplifies to

$$\Delta = \left[\gamma V_{11}(k) + \beta \gamma s^2 V_{11}(sk) \right] \left[U_{11}(c) + \beta r^2 U_{11}(c') \right] + \beta U_{11}(c')(rx + sx')^2 U_{11}(c).$$

Note also the following derivatives:

$$F_{ks} = \beta U_{11}(c')(rx + sx')kx' - \beta U_1(c')x' + \beta \gamma V_1(sk) + \beta \gamma sk V_{11}(sk),$$

$$F_{cs} = \beta r U_{11}(c')kx'.$$

A6.1 The Sign of dk/ds

Total differentiation of the first-order conditions with respect to c , k , and s yields

$$\begin{bmatrix} F_{kk} & F_{ck} \\ F_{ck} & F_{cc} \end{bmatrix} \begin{bmatrix} dk/ds \\ dc/ds \end{bmatrix} = \begin{bmatrix} -F_{ks} \\ -F_{cs} \end{bmatrix}.$$

Cramer's rule states that

$$\frac{dk}{ds} = \Delta^{-1} \begin{vmatrix} -F_{ks} & F_{ck} \\ -F_{cs} & F_{cc} \end{vmatrix}.$$

Let X represent the determinant in the above expression, which can be written as

$$\begin{aligned} X = & -\left[\beta U_{11}(c')(rx + sx')kx' - \beta U_1(c')x' + \beta\gamma V_1(sk) + \beta\gamma sk V_{11}(sk)\right] \left[U_{11}(c) + \beta r^2 U_{11}(c')\right] \\ & + (\beta r)^2 U_{11}(c')^2 kx'(rx + sx'), \end{aligned}$$

or

$$\begin{aligned} X = & -\left[-\beta U_1(c')x' + \beta\gamma V_1(sk) + \beta\gamma sk V_{11}(sk)\right] \left[U_{11}(c) + \beta r^2 U_{11}(c')\right] \\ & - \beta U_{11}(c')(rx + sx')kx' U_{11}(c). \end{aligned}$$

Recall that $V_1(sk) + sk V_{11}(sk) \leq 0$ by assumption. It follows that the first bracketed term is negative. Hence $X < 0$, and $dk/ds < 0$.

6.4.2 The Sign of $d(sk)/ds$

Establishing that $d(sk)/ds > 0$ is equivalent to showing that $-dk/ds \times s/k < 1$. Now, the elasticity of k with respect to s is

$$\frac{dk}{ds} \frac{s}{k} = -\frac{\Phi s}{\Delta k},$$

where

$$\begin{aligned} \Phi = -X = & \left[-\beta U_1(c')x' + \beta\gamma V_1(sk) + \beta\gamma sk V_{11}(sk)\right] \left[U_{11}(c) + \beta r^2 U_{11}(c')\right] \\ & + \beta U_{11}(c')(rx + sx')kx' U_{11}(c). \end{aligned}$$

Note that $\Phi = -X > 0$. The objective is to prove that $|dk/ds \times s/k| < 1$, which amounts to demonstrating that $\Phi_s - \Delta k < 0$. Consider the difference $\Phi_s - \Delta k$:

$$\begin{aligned}\Phi_s - \Delta k &= [-\beta U_1(c')x' + \beta\gamma V_1(sk) + \beta\gamma sk V_{11}(sk)] [U_{11}(c) + \beta r^2 U_{11}(c')] s \\ &\quad + \beta U_{11}(c')(rx + sx') k x' U_{11}(c) s \\ &\quad - [\gamma V_{11}(k) + \beta\gamma s^2 V_{11}(sk)] [U_{11}(c) + \beta r^2 U_{11}(c')] k - \beta U_{11}(c')(rx + sx')^2 U_{11}(c) k,\end{aligned}$$

or

$$\begin{aligned}\Phi_s - \Delta k &= [-\beta U_1(c')x's + \beta\gamma s V_1(sk) - \gamma k V_{11}(k)] [U_{11}(c) + \beta r^2 U_{11}(c')] \\ &\quad - rx\beta U_{11}(c')(rx + sx') U_{11}(c) k.\end{aligned}$$

The first-order condition $F_k = 0$ implies $-\beta U_1(c')sx' = \beta U_1(c')rx - \gamma V_1(k) - \beta\gamma s V_1(sk)$. The difference $\Phi_s - \Delta k$ then becomes

$$\begin{aligned}\Phi_s - \Delta k &= [\beta U_1(c')rx - \gamma V_1(k) - \gamma k V_{11}(k)] [U_{11}(c) + \beta r^2 U_{11}(c')] \\ &\quad - rx\beta U_{11}(c')(rx + sx') U_{11}(c) k.\end{aligned}$$

Recall that $V_1(k) + k V_{11}(k) \leq 0$ by assumption. It follows that the first-bracketed term is positive. Hence $\Phi_s - \Delta k < 0$ and

$$-\frac{dk}{ds} \frac{s}{k} < 1.$$

A7. Elasticities of Substitution, σ_{ck} and σ_{ek}

Consider the preferences described in equation (29). The elasticity of substitution between the number of children, k , and consumption, c , is determined below, as well as the elasticity of substitution between k and expenditures on quality, e .

A7.1 Consumption and Fertility, σ_{ck}

Holding expenditures, e , constant, an indifference curve in the (c, k) plane is represented by

$$d \ln(c) = -\gamma \frac{k}{c} \frac{V_1(k)}{U_1(c)} d \ln(k).$$

This implies

$$d \ln(c/k) = - \left(1 + \gamma \frac{k}{c} \frac{V_1(k)}{U_1(c)} \right) d \ln(k).$$

Consider the marginal rate of substitution, $\gamma V_1(k)/U_1(c)$. Taking derivatives implies

$$\begin{aligned} d \ln \left(\frac{V_1(k)}{U_1(c)} \right) &= \frac{V_{11}(k)}{V_1(k)} k d \ln(k) - \frac{U_{11}(c)}{U_1(c)} c d \ln(c) \\ &= \frac{V_{11}(k)}{V_1(k)} k d \ln(k) + \gamma \frac{U_{11}(c)}{U_1(c)} k \frac{V_1(k)}{U_1(c)} d \ln(k). \end{aligned}$$

Therefore,

$$\begin{aligned} \sigma_{ck} &= \frac{d \ln(c/k)}{d \ln(V_1(k)/U_1(c))} \\ &= - \frac{1 + \gamma \frac{k}{c} \frac{V_1(k)}{U_1(c)}}{\frac{V_{11}(k)}{V_1(k)} k + \gamma \frac{U_{11}(c)}{U_1(c)} \frac{V_1(k)}{U_1(c)} k}. \end{aligned}$$

A7.2 Expenditures on Quality and Fertility, σ_{ek}

Holding expenditures, c , constant, an indifference curve in the (e, k) plane is represented by

$$d \ln(e) = - \frac{\gamma k}{\eta e} \frac{V_1(k)}{J_1(e)} d \ln(k),$$

implying

$$d \ln(e/k) = - \left(1 + \frac{\gamma k}{\eta e} \frac{V_1(k)}{J_1(e)} \right) d \ln(k).$$

Consider the marginal rate of substitution, $(\gamma/\eta) V_1(k)/J_1(e)$. Taking derivatives implies

$$\begin{aligned} d \ln \left(\frac{V_1(k)}{J_1(e)} \right) &= \frac{V_{11}(k)}{V_1(k)} k d \ln(k) - \frac{J_{11}(e)}{J_1(e)} e d \ln(e) \\ &= \frac{V_{11}(k)}{V_1(k)} k d \ln(k) + \frac{\gamma J_{11}(e)}{\eta J_1(e)} k \frac{V_1(k)}{J_1(e)} d \ln(k). \end{aligned}$$

Hence,

$$\begin{aligned}\sigma_{ek} &= \frac{d \ln(e/k)}{d \ln(V_1(k)/J_1(e))} \\ &= -\frac{1 + \frac{\gamma k}{\eta e} \frac{V_1(k)}{J_1(e)}}{\frac{V_{11}(k)}{V_1(k)}k + \frac{\gamma J_{11}(e)}{\eta J_1(e)}k \frac{V_1(k)}{J_1(e)}}.\end{aligned}$$

A8. Proof of Proposition 12 (Productivity and the Quality–Quantity Trade-off)

Rearranging the first-order conditions (31) and (32), while making use of (30), gives

$$\frac{\gamma V_1(k)/J_1(e)}{\eta \frac{e}{k}} = 1 + \theta w/e \quad \text{and} \quad \gamma \frac{V_1(k)/U_1(c)}{c/k} = w/c - 1.$$

Taking logs and then taking the total differential of the two expressions yields

$$\begin{aligned}(1/\sigma_{ek} - 1)d \ln(e/k) &= \frac{\theta}{1 + \theta w/e} d(w/e), \\ (1/\sigma_{ek} - 1)d \ln(c/k) &= \frac{1}{w/c - 1} d(w/c).\end{aligned}$$

This can be rearranged into

$$(A10) \quad (1/\sigma_{ek} - 1)d \ln(e/k) = -\frac{\theta}{e/w + \theta} d \ln(e/w),$$

$$(A11) \quad (1/\sigma_{ek} - 1)d \ln(c/k) = -\frac{1}{1 - c/w} d \ln(c/w).$$

Note also that

$$d \ln(e/k) = d \ln(e/w) + d \ln(w) - d \ln(k),$$

$$d \ln(c/k) = d \ln(c/w) + d \ln(w) - d \ln(k).$$

Thus, the system (A10)–(A11) can be rewritten as

$$(A12) \quad \left(1/\sigma_{ek} - 1 + \frac{\theta}{e/w + \theta}\right) d \ln(e/w) - (1/\sigma_{ek} - 1)d \ln(k) = -(1/\sigma_{ek} - 1)d \ln(w),$$

$$(A13) \quad \left(1/\sigma_{ek} - 1 + \frac{1}{1 - c/w}\right) d \ln(c/w) - (1/\sigma_{ek} - 1)d \ln(k) = -(1/\sigma_{ek} - 1)d \ln(w).$$

The budget constraint (30) can be used to write

$$(A14) \quad d \ln(k) = -\frac{c/w}{1-c/w} d \ln(c/w) - \frac{e/w}{e/w+\theta} d \ln(e/w).$$

Substituting equation (A14) into the system (A12)–(A13) leads to the following formulation:

$$\begin{bmatrix} 1/\sigma_{ek} - 1 + \frac{\theta}{e/w+\theta} + (1/\sigma_{ek} - 1)\frac{e/w}{e/w+\theta} & (1/\sigma_{ek} - 1)\frac{c/w}{1-c/w} \\ (1/\sigma_{ek} - 1)\frac{e/w}{e/w+\theta} & 1/\sigma_{ek} - 1 + \frac{1}{1-c/w} + (1/\sigma_{ek} - 1)\frac{c/w}{1-c/w} \end{bmatrix} \\ \times \begin{bmatrix} d \ln(e/w)/d \ln(w) \\ d \ln(c/w)/d \ln(w) \end{bmatrix} = \begin{bmatrix} -(1/\sigma_{ek} - 1) \\ -(1/\sigma_{ek} - 1) \end{bmatrix}.$$

Let Δ denote the discriminant of this system. The second-order conditions of the optimization problem imply that, at a maximum, $\Delta > 0$. Applying Cramer's rule to this system, one finds

$$(A15) \quad \frac{d \ln(e/w)}{d \ln(w)} = -\Delta^{-1}(1/\sigma_{ek} - 1) \left(1/\sigma_{ek} - 1 + \frac{1}{1-c/w} \right),$$

and

$$(A16) \quad \frac{d \ln(c/w)}{d \ln(w)} = -\Delta^{-1}(1/\sigma_{ek} - 1) \left(1/\sigma_{ek} - 1 + \frac{\theta}{e/w+\theta} \right).$$

Substituting these equations into (A14) yields

$$\begin{aligned} \frac{d \ln(k)}{d \ln(w)} &= \frac{1}{\Delta} \frac{c}{w-c} (1/\sigma_{ek} - 1) \left(1/\sigma_{ek} - \frac{ek}{w-c} \right) \\ &+ \frac{1}{\Delta} \frac{ek}{w-c} (1/\sigma_{ek} - 1) \left(1/\sigma_{ek} + \frac{c}{w-c} \right). \blacksquare \end{aligned}$$

A9. Proof of Proposition 13 (The Effect of Productivity Growth on Fertility)

Preferences are represented by $\ln(c) + \alpha \ln(k) + \eta H(gwQ(e))$. The first-order conditions for c and k imply $\alpha/k = (e + \theta w)/c$. It follows, then, from the budget constraint (30) that $c = w/(1 + \alpha)$ and

$$(A17) \quad k = \frac{\alpha}{1 + \alpha} \frac{w}{e + \theta w}.$$

The optimization problem then becomes equivalent to $\max_e \eta H(gwQ(e)) - \alpha \ln(e + \theta w)$, with the following first-order condition:

$$(A18) \quad \eta(gw)^{1-\rho} Q(e)^{-\rho} Q_1(e) = \frac{\alpha}{e + \theta w}.$$

Write this first-order condition as $A(e, g) - B(e) = 0$, and note that the second-order condition for a maximum implies $A_1(e, g) - B_1(e) < 0$. Implicitly differentiating with respect to e and g yields

$$\frac{de}{dg} = -\frac{A_2(e, g)}{A_1(e, g) - B_1(e)},$$

where $A_2(e, g) = (1 - \rho)g^{-\rho}\eta w^{1-\rho}Q(e)^{-\rho}Q_1(e)$. Thus, the sign of de/dg , and hence $dQ(e)/dg = Q_1(e)de/dg$, is the same as the sign of $A_2(e, g)$. The result for fertility follows from the observation that k is decreasing in e , as indicated by equation (A17). ■

A10. Measures of Fertility

There are many measures of fertility. To review them, define $k_{i,t}$ as the average number of children born to a woman of age i during year t :

$$k_{i,t} = \frac{\text{Number of births to women of age } i}{\text{Number of women of age } i}.$$

This is a first measure known as the *age-specific fertility rate*. Let $p_{i,t}^f$ denote the denominator of this ratio, that is, the total number of women of age i during year t . Similarly, let $p_{i,t}^m$ denote the total number of men. Another measure of fertility is the *crude birth rate*,

$$\text{Crude Birth Rate}_t = \frac{\sum_i k_{i,t} p_{i,t}^f}{\sum_i (p_{i,t}^f + p_{i,t}^m)},$$

which is simply the number of birth per population during a particular period of time. A third measure, the *general fertility rate*, reports births per women in their child-bearing period during a particular time:

$$\text{General Fertility Rate}_t = \frac{\sum_{i=15}^{49} k_{i,t} p_{i,t}^f}{\sum_{i=15}^{49} p_{i,t}^f}.$$

These are measures of contemporaneous fertility. Two additional measures deal with lifetime fertility. The *total fertility rate* measures the number of children a woman would have over the

course of her fertile life if she experienced the *current* age-specific fertility rates throughout her life. For women 15 years of age at date t , the total fertility rate is

$$\text{Total Fertility Rate}_t = \sum_{i=15}^{49} k_{i,t}.$$

Note, again, that the total fertility rate at time t is built from the age-specific fertility rate observed at time t . A fifteen-year-old woman at time t may not experience at age sixteen—that is, at date $t + 1$ —the same age-specific fertility rate that a sixteen year old experiences at time t . That is, $k_{i+1,t+1}$ may differ from $k_{i+1,t}$. Thus, the Total Fertility Rate measures the fertility of an “imaginary” woman who would experience throughout the rest of her life the current age-specific fertility rates of each age group. The total fertility rate is useful to build projections about the lifetime fertility of a young cohort, assuming that age-specific fertility rates will remain constant. For women past their childbearing years, it is possible to construct a final measure, the completed fertility rate for a woman born at date t :

$$\text{Completed Fertility Rate}_t = \sum_{i=15}^{49} k_{i,t+i}.$$

This is a measure of realized lifetime fertility.

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